CHAPTER

ATA-25

EQUIPMENT AND FURNISHINGS

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

Contents

Equipment and Furnishings, General

Introduction

The flight compartment has seating for the flight crew and a flight observer, it also has readily accessible normal and emergency equipment.

The passenger compartment furnishings are provided to ensure the comfort and safety of the passengers and cabin crew.

The emergency equipment located in the flight compartment and the passenger compartment is used by the flight crew during emergency situations.

The galleys on board the aircraft supply a place, to store and prepare food and beverages for the passengers and flight crew.

The lavatory compartment supplies the crew and passengers with washroom facilities.

The forward and aft baggage compartments have sufficient space for the safe storage of the passengers' baggage and cargo.

On aircraft with the ModSum 4−458912 incorporated, the forward baggage compartment is removed to accommodate additional passenger seats.

On aircraft with ModSum 4−459363 incorporated, the G1 galley is removed to increase the volume of the aft baggage compartment to 828 ft^3 (23.45 m^3).

On aircraft with ModSum 4−459364 incorporated, the G2 galley is removed to increase the volume of the aft baggage compartment to 828 ft^3 (23.45 m^3).

On aircraft with ModSum 4−459141 incorporated, the aft baggage compartment 411 ft^3 (11.63 m^3) is removed to increase the volume of the aft cargo compartment to 828 ft^3 (23.45 m^3).

General Description

The Equipment and Furnishings chapter is divided into the following sub−chapters:

- Flight Compartment Furnishings
- Passenger Compartment Furnishings
- Galleys
- Lavatories
- Baggage Compartments
- Emergency Equipment
- Insulation

Flight Compartment Furnishings

Introduction

The flight compartment has seating for the flight crew and a flight observer. It also has readily accessible normal and safety equipment. The flight compartment panels give access to different equipment and controls.

General Description

The flight compartment has seats for the pilot, the copilot and for an observer. Crew aids are supplied to help the crew in different operations.

Insulation blankets are installed where necessary in the flight compartment, and can be removed for structural inspection.

Access panels in the compartment give access to different equipment. The floor area is covered with a foam backed vinyl–coated covering.

The flight compartment furnishings include the components that follow:

- Pilot and Copilot Seats
- Observer Seat
- Crew Storage
- Floor Cover
- Sidewall Panels and Covering, Flight Compartment
- Ceiling Panels and Covering, Flight Compartment

Pilot and Co-pilot Seats

Refer Figure 25-1

The pilot's and copilot's seats are mechanically adjustable (fore/aft and vertically) and are installed on track rails in the flight compartment. The seats have an aluminium alloy frame structure equipped with reclining backs and fold up padded armrests. Each seat has a removable seat cushion, contoured back cushion with lumbar support, coat hanger, head rest and a document holder.

The base frame of each seat is attached to four plate assemblies which roll on two machined tracks attached to the flight compartment floor. A roller is installed in each plate assembly to allow the seat to move freely along the tracks. Each seat is adjustable fore and aft along the floor tracks by means of a lock release lever located on the inboard side of each seat. The seat is held in place by four locking pins which engage in holes located along the length of the floor tracks. Pulling the lock release lever up unlocks the seat by disengaging the locking pins allowing the seat to move freely along the tracks.

The height control lock lever, located on the outboard side of each seat, raises or lowers the seat with the help of a gas spring. The recline adjustment lever, located on the right side of each seat, reclines the back rest from a range of 7 to 34 degrees.

The seat also has controls for the adjustment of the lumbar support. Turning the control knob on the right side of the seat adjusts the support IN and OUT. Turning the knob on the left side adjusts the support UP and DOWN.

The padded armrests fold up to make it easier to enter and exit the seat. The armrests can be individually adjusted UP or DOWN by a knob located at the forward end of each armrest.

The seats have a safety harness with a crotch strap, lap belts and a shoulder harness. An inertial reel controls the movement of the shoulder harness which is spring loaded to the retracted position. The inertia reel may be operated in a manual or automatic mode. Selection of the inertia reel to lock, locks the harness in position.

Selection of the inertia reel to unlock lets the harness move freely until a force of 2 G's or greater is applied automatically locking the inertia reel. The inertia reel lock levers are located on the inboard side of the seats. This allows each flight crew member to access to the other crew member's lock lever in case of incapacitation.

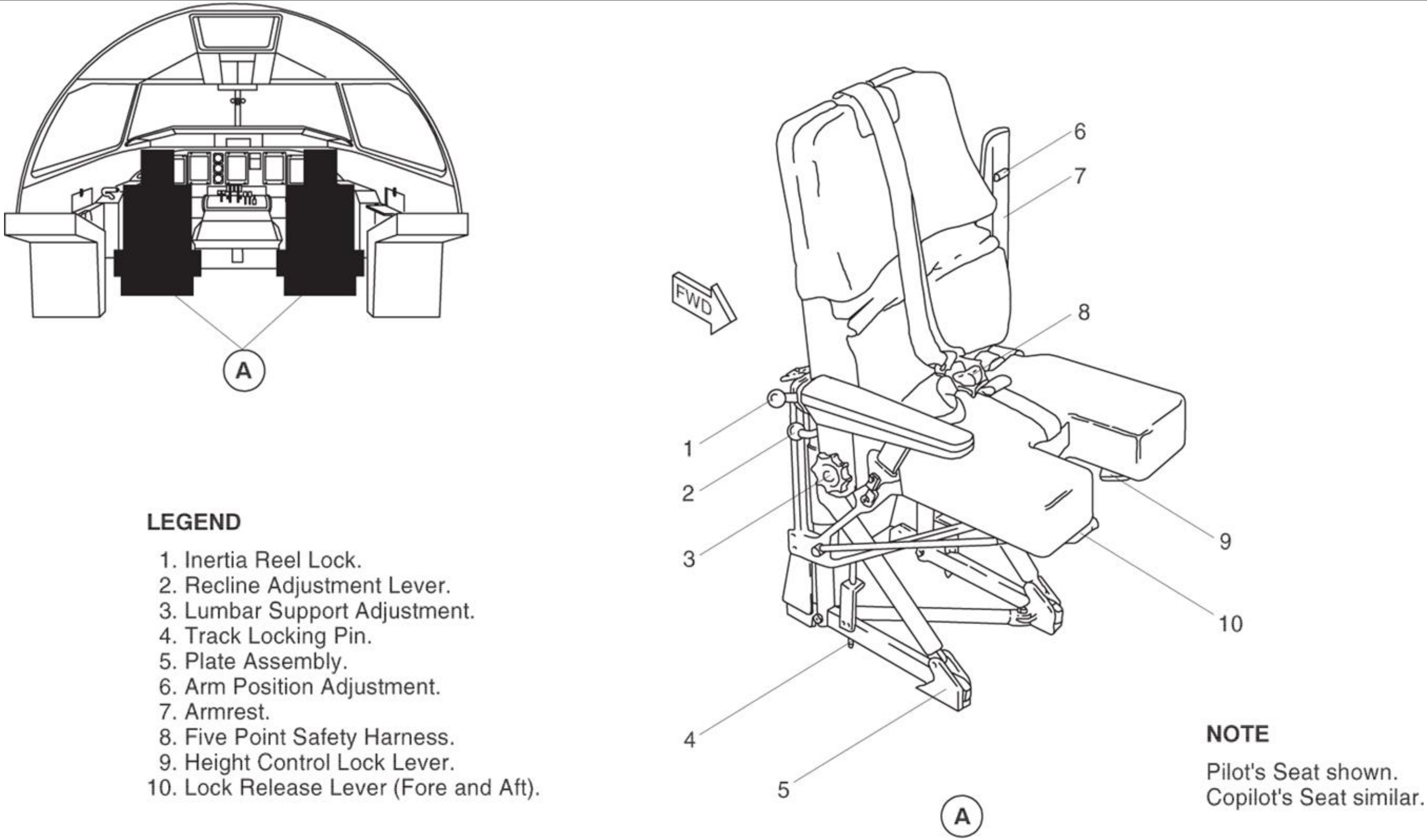


Figure 25-1 Pilot and Co-Pilot Seats

Observer Seat

Refer Figure 25-2

The observer's seat has a machined steel frame, with a nylon covered padded seat cushion and a restraint system. The seat is located forward of the flight compartment door and is attached to the avionics rack by two hinge assemblies.

The seat frame (with the bottom cushion) can be folded up and attached to the left side wall by a strap when the seat is not in use. When folded down, the seat is supported and held in place by latches on the inboard side of the lavatory wall. A padded back and head rest cushion is installed on the flight compartment door.

The seat has a single shoulder harness installed on an inertia reel and a lap belt with a metal to metal latching device. The flight compartment door can open when the observer's seat is in use.

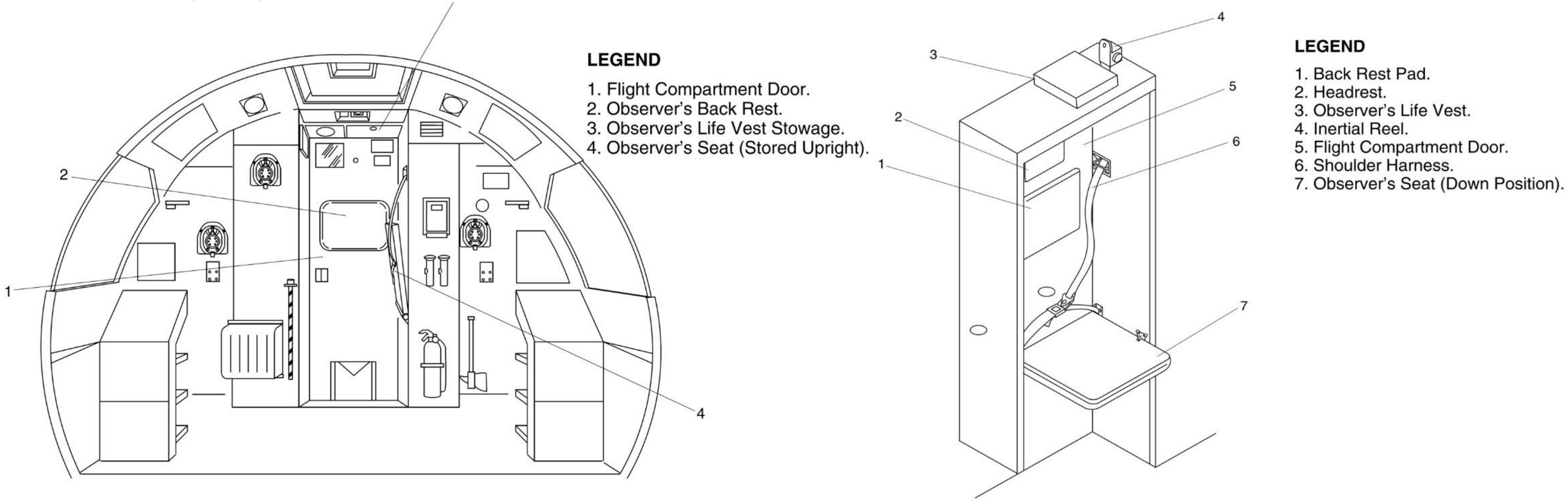


Figure 25-2 Observer's Seat

Crew Storage

Refer Figure 25-3 & Refer Figure 25-4

The flight compartment has the crew aids that follow:

- Cup holders
- Map pouch
- Foot rests
- Assist handles
- Microphone
- Chart holder
- Individual air gasper
- Air conditioning grilles
- Console map light
- Dome light
- Reading light
- Flash light
- Life vest
- Smoke goggles
- Oxygen mask.

Floor Covering

The flight compartment floor area is covered with a foam backed vinyl–coated covering. There are cut–outs in the floor covering for various access panels.

The entire floor area of the flight compartment is covered with a serviceable vinyl–coated floor covering.

Access panels in the floor provide access to the nose landing gear emergency release and main landing gear alternate extension hand pump.

Sidewall Panels and Covering, Flight Compartment

Access panels in the floor give access to the nose landing gear emergency release, and main landing gear alternate extension hand pump. There are access covers for the life vests and the crew escape rope.

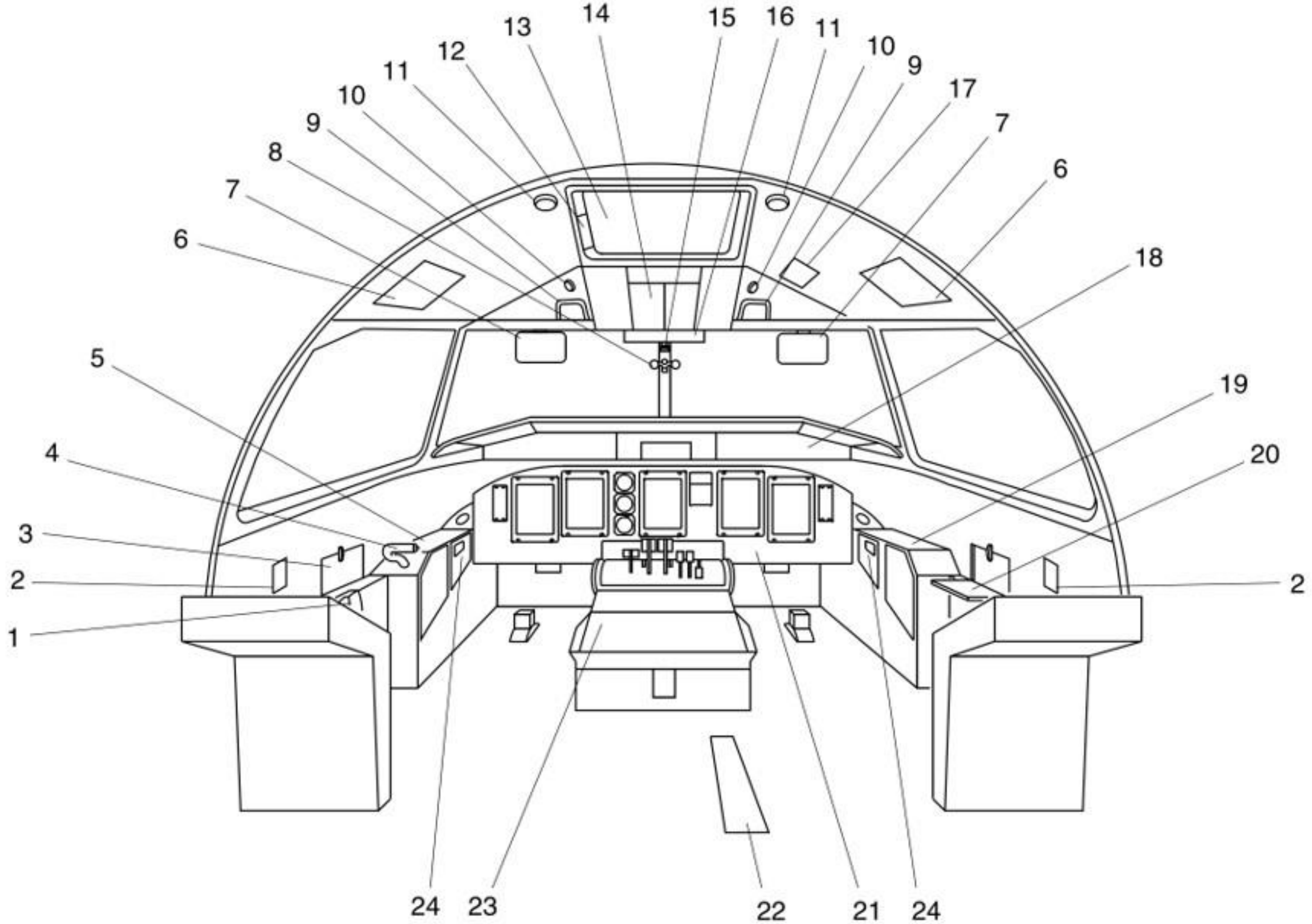
Ceiling Panels and Covering, Flight Compartment

Refer Figure 25-5

The ceiling panels in the flight compartment are as follows:

- Left hand headliner panel
- Right hand headliner panel
- Escape hatch surround panel
- Escape hatch trim panel
- Ceiling threshold panel

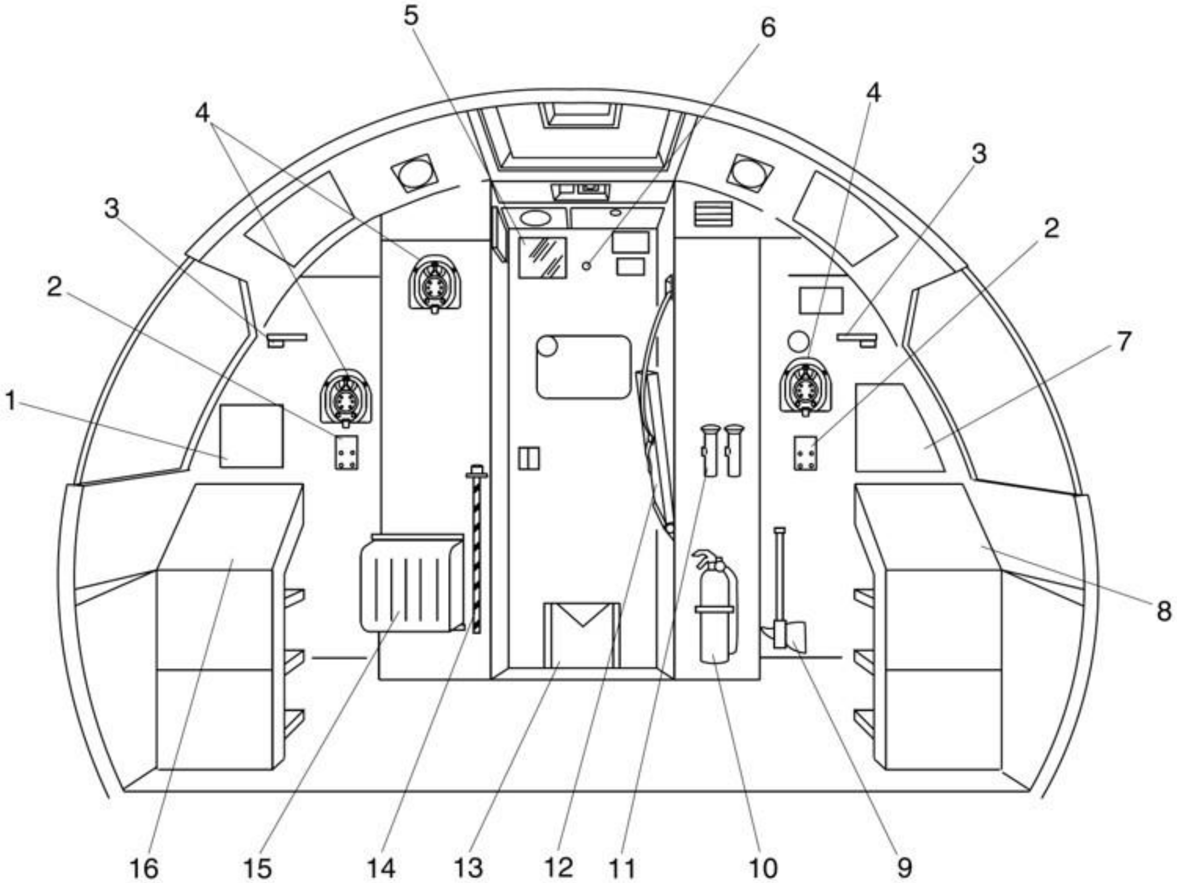
The flight compartment ceiling panels are made of molded polycarbonate material. The panels are secured to the structure by velcro tape, screws, bolts and panel mounted components. The ceiling panels are coloured when molded.



LEGEND

1. Chart Holder.
2. Flow Control Levers.
3. Pilot's Map Table (Closed).
4. Steering Hand Control.
5. Pilot's Side Panel.
6. Life Vest Stowage.
7. Sun Visor.
8. Eye Level Indicator.
9. Assist Handle.
10. Utility Light.
11. Dome Light.
12. Emergency Escape Rope Storage.
13. Emergency Exit.
14. Overhead Console.
15. Standby Compass.
16. Caution & Warning Panel.
17. Landing Gear Alternate Release Door.
18. Glareshield.
19. Copilot's Side Panel.
20. Copilot's Map Table (Open).
21. Instrument Panel.
22. Landing Gear Alternate Extend Door.
23. Centre Console.
24. Smoke Goggles.

Figure 25-3 Flight Compartment Equipment – Detailed 1 of 2



LEGEND

1. Variable Frequency AC Circuit Breaker Panel.
2. Headset Jacks.
3. Circuit Panel Light.
4. Oxygen Mask.
5. Mirror.
6. Viewer.
7. Avionics Circuit Breaker Panel.
8. Left DC Circuit Breaker Panel.
9. Fire Axe.
10. Fire Extinguisher.
11. Flashlights.
12. Observer's Seat.
13. Weight and Balance Manual.
14. Landing Gear Emergency Extension Handpump Handle.
15. Protective Breathing Equipment (PBE).
16. Right DC Circuit Breaker Panel.

Figure 25-4 Flight Compartment Equipment – Detailed 2 of 2

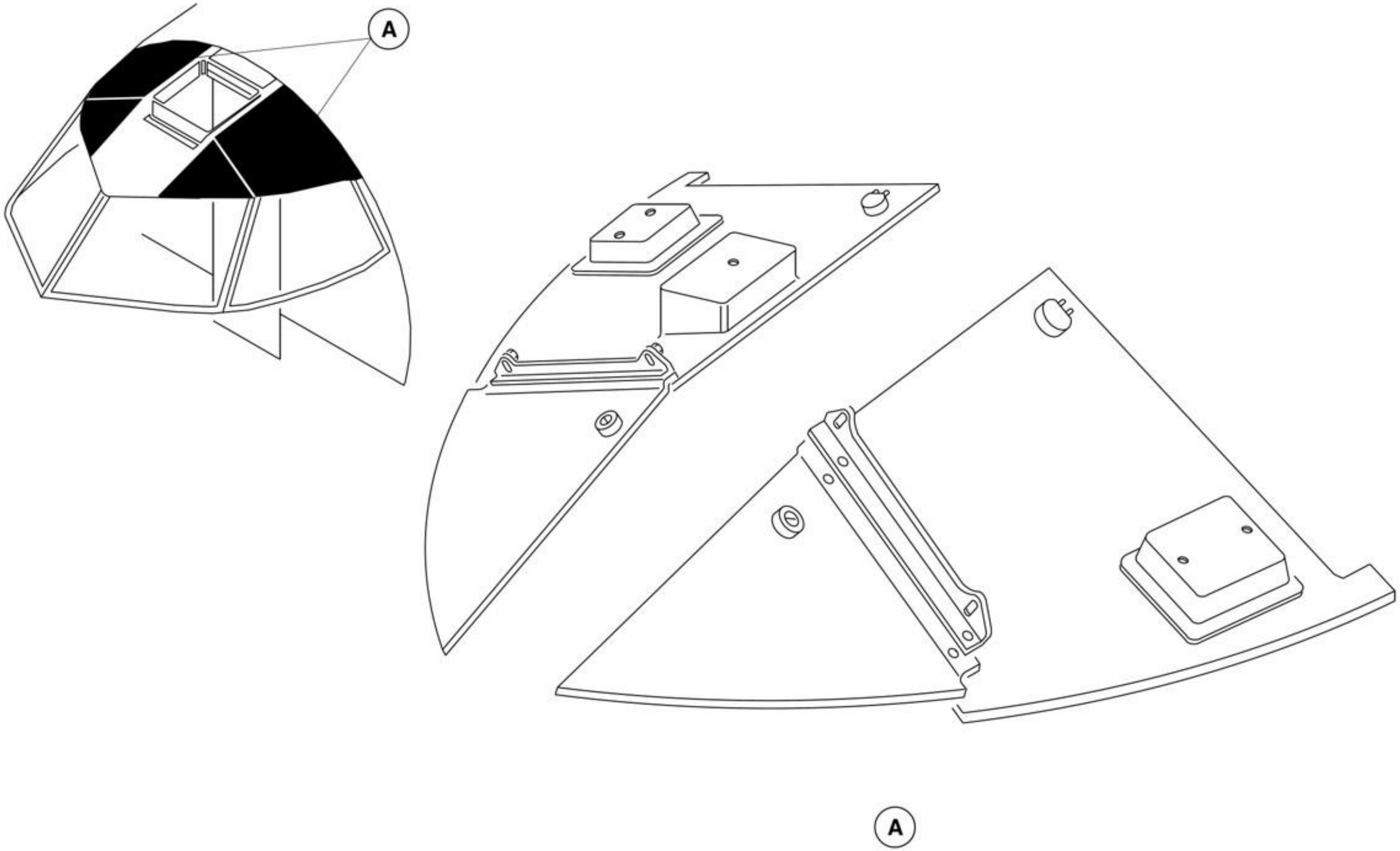


Figure 25-5 Flight Compartment Head Liner

Passenger Compartment Furnishings

Introduction

The passenger compartment furnishings are provided to ensure the comfort and safety of the passengers and cabin crew.

On aircraft with ModSum 4−459134 incorporated, the aft flight attendant seat is removed to increase the volume of the aft baggage compartment to 828 ft^3 (23.45 m^3).

On aircraft with ModSum 4−459146 incorporated, the aft galley area curtain is removed to increase the volume of the aft baggage compartment to 828 ft^3 (23.45 m^3).

General Description

The passenger compartment furnishings include the components that follow:

- Passenger Seats
- Flight Attendant Seats
- Overhead Storage
- Wardrobes and Partitions
- Floor Panels and Coverings
- Sidewall Panels and Covering, Passenger Compartment
- Ceiling Panels and Covering, Passenger Compartment
- Passenger Service Units

Passenger Compartment Furnishings - Component Description

Passenger Seats

Refer Figure 25-6

The passenger cabin is designed to accommodate different seating configurations. The baseline configuration has up to 68 passengers, seated four abreast, at a mixed pitch of 33, 34, 35 and 36 in. (838, 864, 889 and 914 mm). The other configurations 78 seats at 30 in. (762 mm) pitch.

The standard passenger seats for the aircraft are reclining or non−reclining, and have fold−down meal trays, armrests with an ashtray or armrests without an ashtray and under−seat baggage restraints. The seats without a fold−down meal tray in front of them have plug−in meal trays.

The passenger seats are installed with its own seat belt and metal to metal buckle. The pads are put on the top of each seat backrest to make it smooth for the safety of the passengers. Optionally, the coat hooks on the latch of the fold−down trays and the literature pockets on the backrest aft surface of the seats are installed for the passengers.

The passenger seats are made of an anodized aluminum frame, with plastic foam flotation cushions with dress covers. The bottom cushions for all the seats can be removed and used as a flotation device in the event of the aircraft ditching in water. The cushions are attached to the seats by Velcro tape. The foam cushions, liners and seat dress covers satisfy the FAA and CAA flammability requirements.

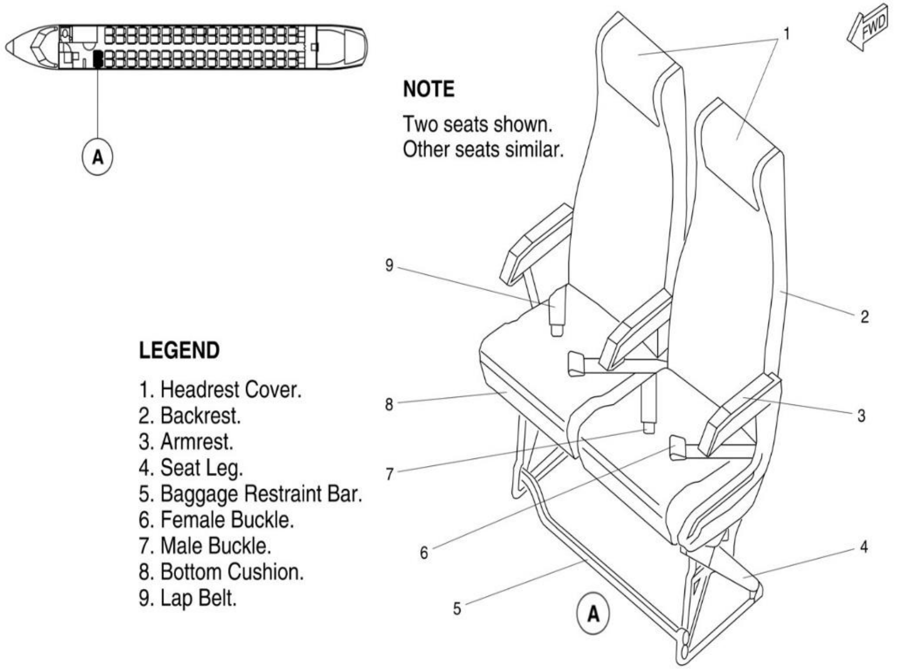
The economy class consists of standard passenger seats with recline control, armrests without ashtrays, under seat baggage restraint bars and life vest stowage pockets. The first row of seats on the left and right sides have in−arm meal tray provisions and the remaining seats are without in−arm meal tray provisions.

All the seats in the economy class have a back tray behind each seat. Each passenger seat in business class and economy class has its own seat belt assembly. The bottom cushions for all the seats can be removed and used as a flotation device in the event of the aircraft ditching in water. The foam cushions, liners and seat dress covers satisfy the FAA and CAA flammability requirements.

On aircraft with ModSum 4−459733 incorporated, the economy class consists of 90 standard passenger seats with no recline control, armrests without ashtrays, life vest stowage pockets, seat belts and literature pockets. Each seat has a back mounted meal tray and under seat baggage restraint bars except the last row seats. The foam cushions, liners and seat dress covers satisfy the FAA and CAA flammability requirements.

The seats are installed on inboard and outboard seat tracks and are attached to the tracks by fittings and studs with anti−rattle locks. The seat leg studs fit into their related notches in the floor tracks. The seat frame structure, seat legs and track attachments can withstand a force of 16Gs. The seats can be moved along the tracks in one inch increments to give spacing flexibility. The fittings which attach the seats to the tracks are quick disconnect to allow for easy installation and removal. Seat track covers are installed to protect the tracks from dirt and water spillage.

A two section lap belt is installed on each passenger seat and is connected to the seat frame by bolts. Each belt has a male and female buckle and a locking device which allows adjustment of the seat belt length.



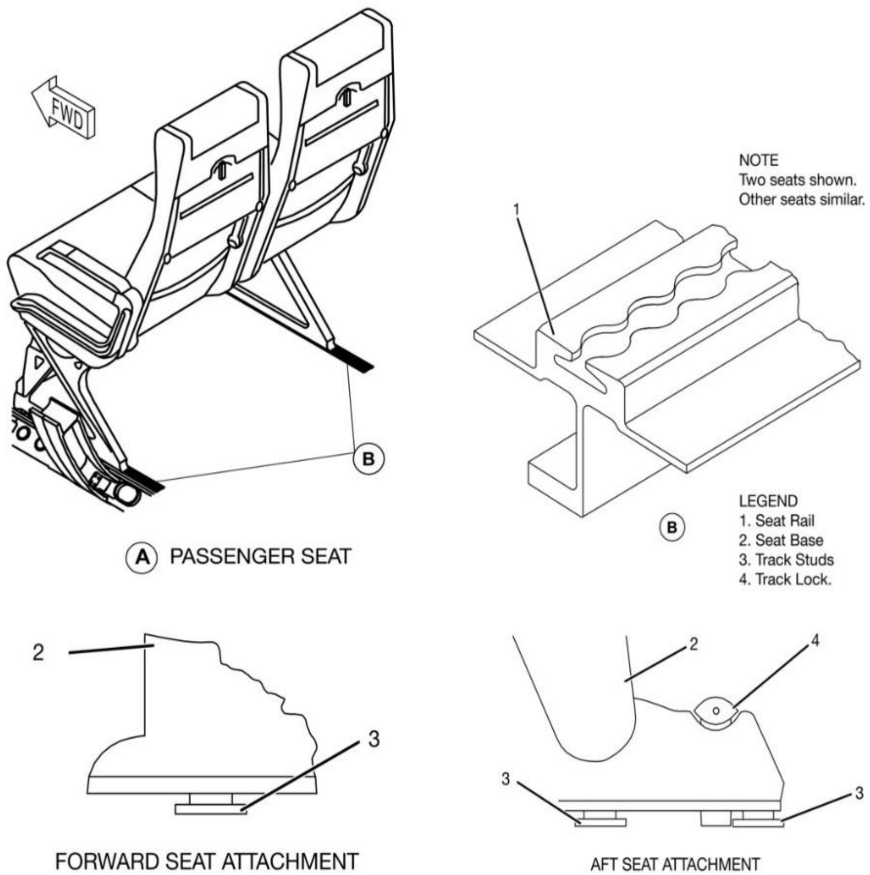


Figure 25-6 Passenger Seat

Portable Meal Tray

Refer Figure 25-7

Fold away meal trays are installed in the seat backrest, except for the front row of seats, where provision is made for a portable tray to be plugged into the front section of the arm rest.

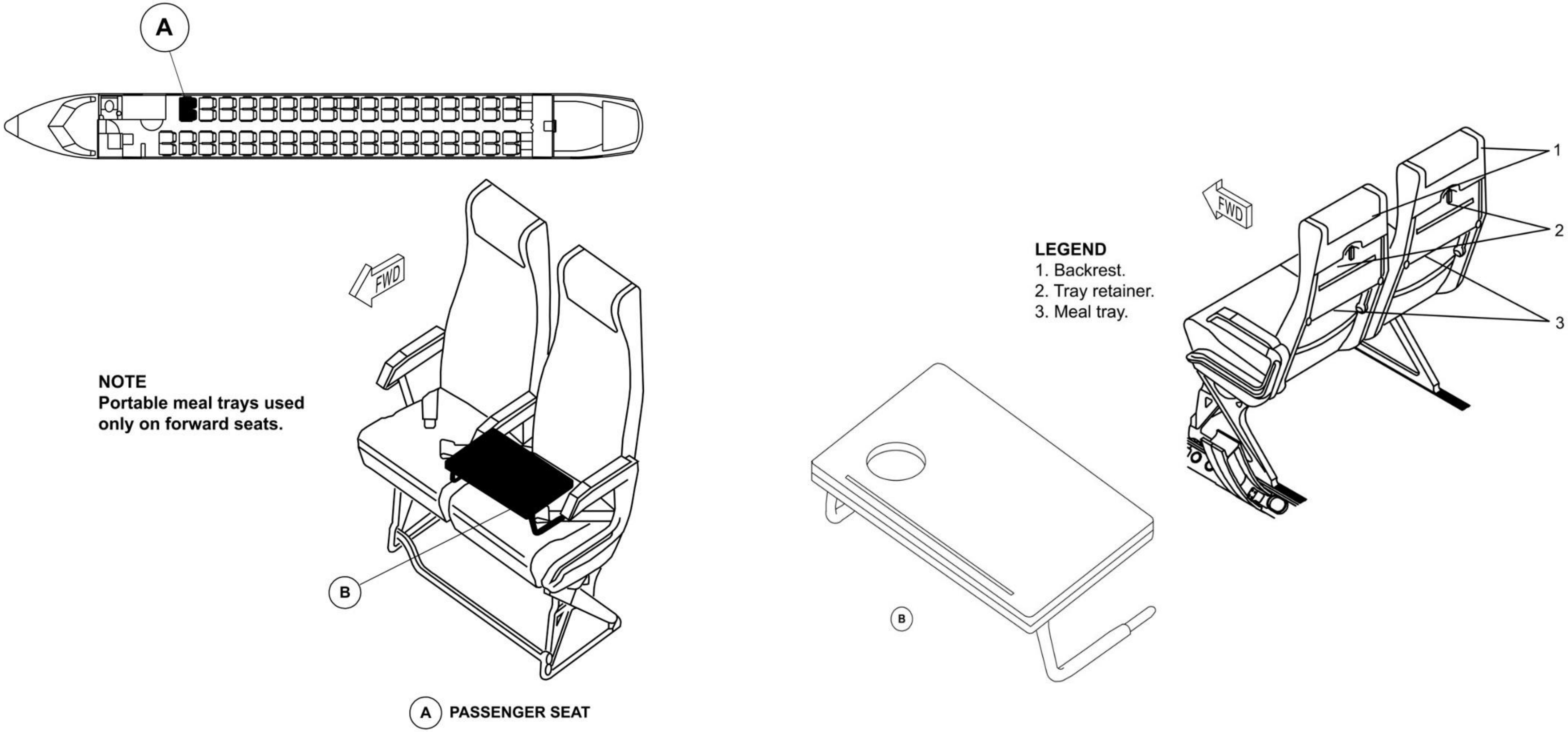


Figure 25-7 Portable Meal Tray

Flight Attendant Seats

Refer Figure 25-8 & Refer Figure 25-9

The forward flight attendant seat is aft facing and forms the aft wall of the wardrobe adjacent to the airstair door. The seat back is 19 in. (482.6 mm) wide and is part of the wardrobe bulkhead. It has a contoured energy absorbing back cushion and headrest which supports the arms, shoulders, head and spine. The seat portion has
a fold−down seat cushion, which is spring−loaded to stow automatically when the seat is not occupied. The seat cushions are secured to the seat pan by Velcro tape. The flight attendant faces rearward when sitting and has a direct view of the cabin area.

The seat is also equipped with a restraint harness consisting of shoulder restraints and a lap belt. The shoulder restraint is connected to an inertia reel fitted below the head cushion.

Stowage for a life jacket and flashlight is provided below the seat

An attendant's interphone headset and control panel is installed to the right of the head cushion.

The aft flight attendant seat is installed on the G3 galley bulkhead behind the two aft doors (aft passenger door and aft service door). When in use, the seat is located close to the centerline of the aircraft to give a direct view of the aft cabin area.

When stowed, the seat is latched to the left of the G3 galley. When the latch is released, the seat assembly rotates 180 degrees to latch to the galley countertop. Lowering the seat cushion engages a pin into a receptacle on the floor. The upper latch mechanism must be fully engaged in the countertop fitting before the seat pan can be lowered. A handle on the right side of the seat frame releases both latches, enabling the seat to return to its stowed position. The seat may not be deployed unless the aisle carts in the G3 galley are secured behind the seat.

On aircraft with ModSum 4−458825 OR 4−459126 OR SB 84−25−148 incorporated, there is one aft flight attendant seat which is foldable. The aft flight attendant seat is installed on the flat bulkhead. When the aft flight attendant seat is not in use, a spring in the bottom of the seat automatically retracts the seat bottom to the vertical position. The seat has a shoulder harness and lap belt which are joined by a metal to metal buckle. When released, the harness is retracted by inertia reels after the seat is stowed.

The aft flight attendant seat has stowage room for emergency equipment within the seat base (Refer SDS 25–60–00).

The seat has a shoulder harness and lap belt which are joined by a metal to metal buckle. When released, the harness is retracted by inertia reels after the seat is stowed.

An attendant's interphone headset and control panel is installed on the G3 galley bulkhead to the left of the seat.

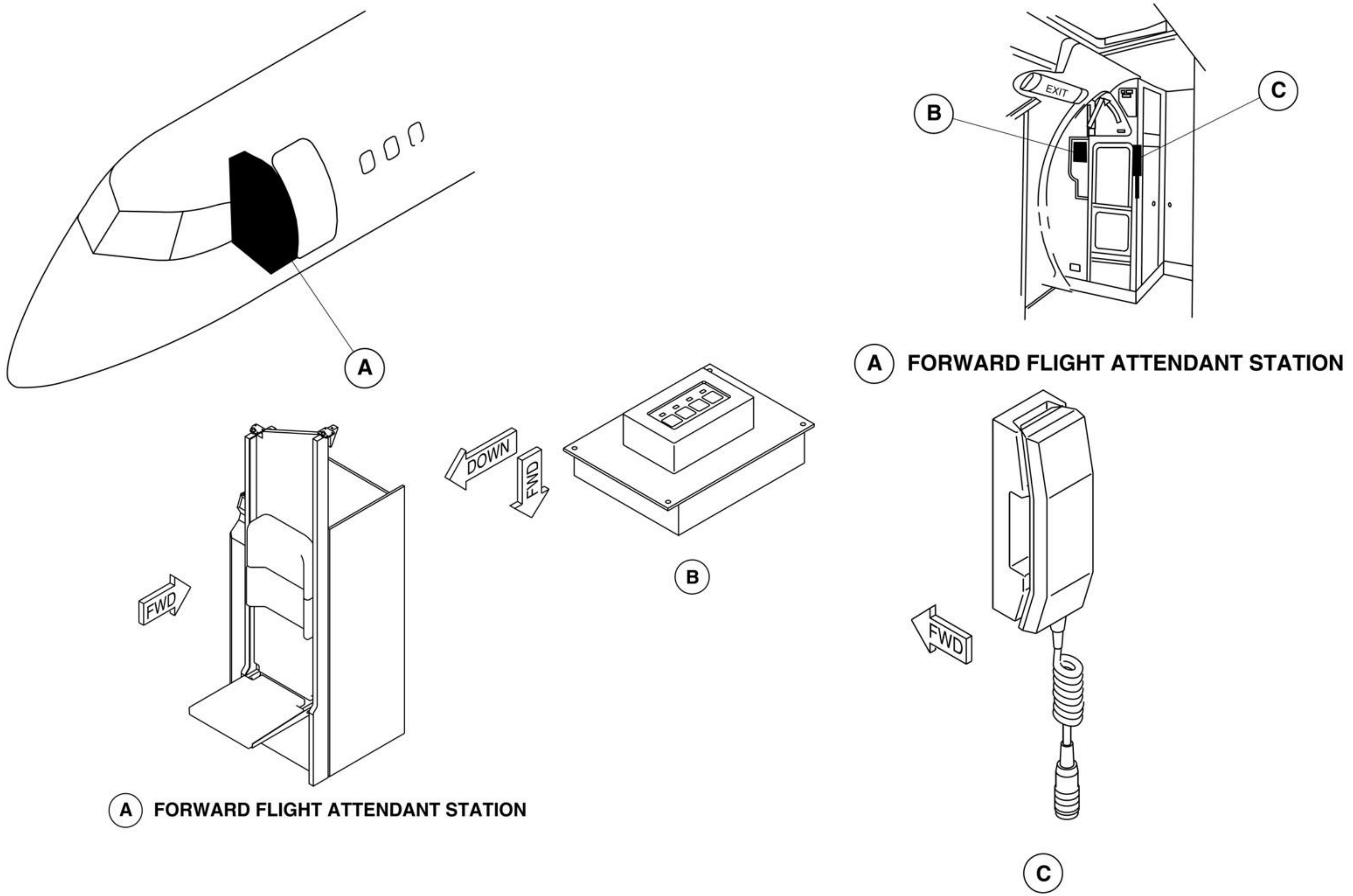


Figure 25-8 Forward Flight Attendant Seat

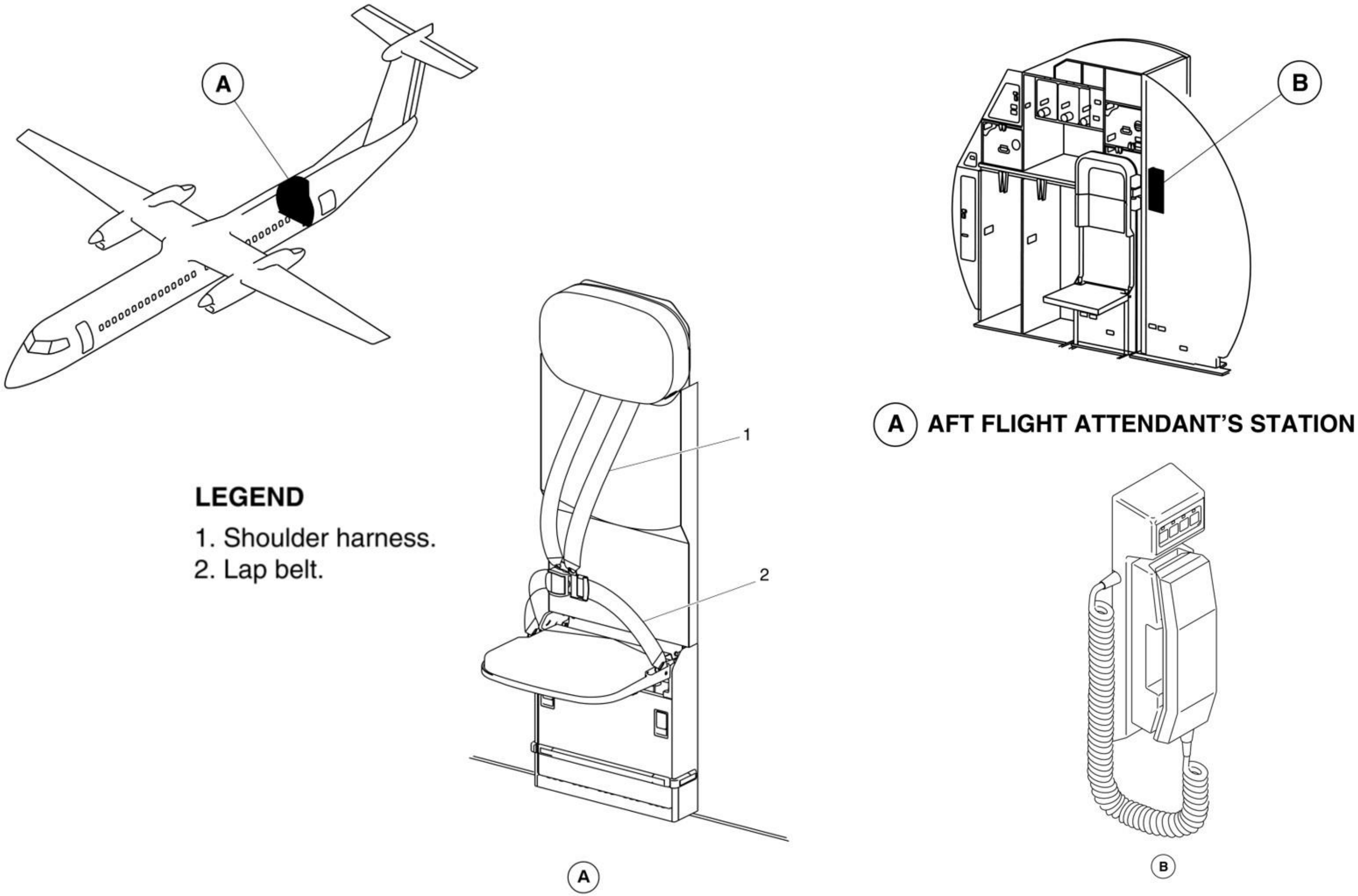


Figure 25-9 Aft Flight Attendant Seat

Overhead Storage

Refer Figure 25-10 & Refer Figure 25-11

The overhead bins are installed in the passenger compartment above the seats. They extend the full length of the passenger compartment on both sides of the aisle. They are used for carry−on luggage and portable emergency equipment storage.

The bins are made from a fibreglass composite material with a honeycomb core. The interior of the bin is painted while the exterior has a plastic laminate decorative finish.

The bins are attached to the aircraft structure with support brackets and bin hangers through anti−vibration mounts. Spacers are installed to keep an equal distance between each bin.

There are 12 individual assemblies (six on each side of the aircraft) which have two separate bin compartments. This gives a total of 24 bins (12 on each side of the aircraft). These bins are 47 in. (1.19 m) in length and have a maximum weight capacity of 60 lb (27 kg). There are two 15 in. (381 mm) long aft bins located on the left and right side, forward of the aft draft bulkhead, which have a maximum weight capacity of 19 lb (8.6 kg). There is one 38.5 in. (978 mm) long forward bin on the left side with a maximum weight capacity of 49 lb (22 kg). This gives a total of 27 individual bins.

Each bin has a self−latching door which opens upward and pivots at the front and rear of each bin compartment. The door is attached to the bin with gas spring actuators. The door is held in the closed position by the latch. When the latch is released, the actuators open the door at a controlled rate.

The door latch assembly has a spring loaded handle which hinges on a latch pin. A set screw is located on the inside and underneath the handle and can be adjusted to prevent the handle from rattling and vibrating. Lifting the handle firmly upwards gives access to the set screw.

Cove panel retention springs and fluorescent lights are located above the bins. The fluorescent lights are protected from accidental breakage by a transparent light guard. Air condition system headers with outlets and ceiling lights ballast are installed on the outboard side of the bins. An electrical wire harness with receptacles and connectors are installed on each bin. The wiring harness for the ceiling lights ballast and the PSU panels is connected to the outboard side of the bins by clips and tie−wraps.

On aircraft with ModSum 4−458919 OR 4−459413 OR 4−459622 incorporated, there are eight bin assemblies on the left side and on the right side of the aircraft. The first bin assembly from the airstair door is 39 in. (990.6 mm) long and has a maximum weight capacity of 49 lb (22.23 kg). The first right bin assembly is 69.25 in. (1758.95 mm) long. This bin assembly has two separate bin compartments. The smaller bin compartment has a maximum weight capacity of 28.22 lb (12.80 kg) and the other bin compartment has a maximum weight capacity of 60 lb (27.22 kg). The last two bin assemblies (one on the left side and the other on the right side of the aircraft) at the aft seats of the passenger compartment are 18.375 in. (466.73 mm) long and each bin has a maximum weight capacity of 21 lb (9.53 kg). The remaining twelve bin assemblies (six on the left side and six on the right side of the aircraft) are 95.75 in. (2432.05 mm) long. These twelve bin assemblies individually have two separate bin compartments and each bin compartment has a maximum weight capacity of 60 lb (27.22 kg).

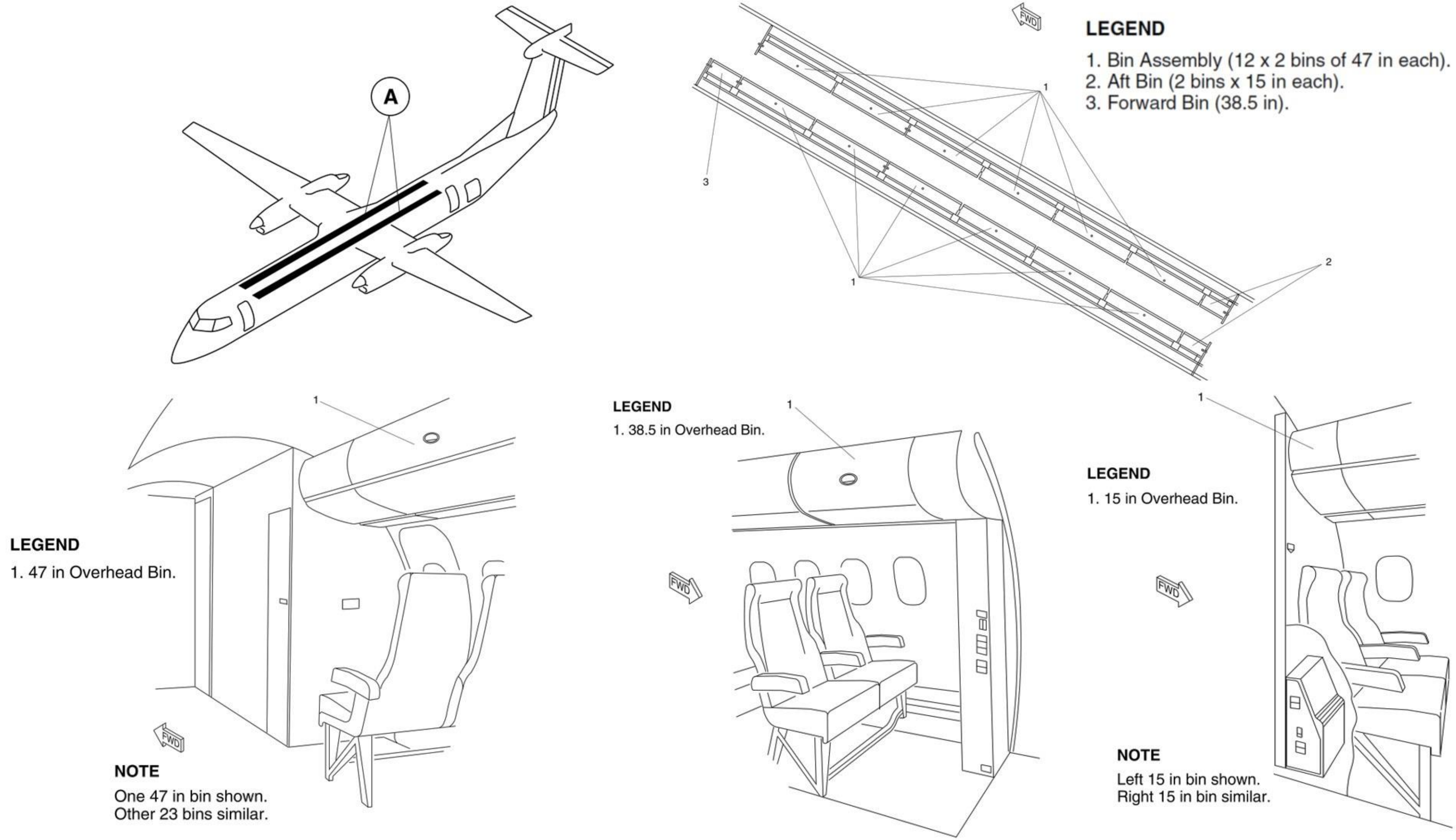
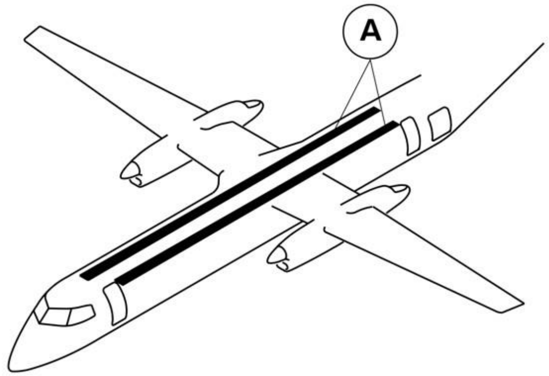
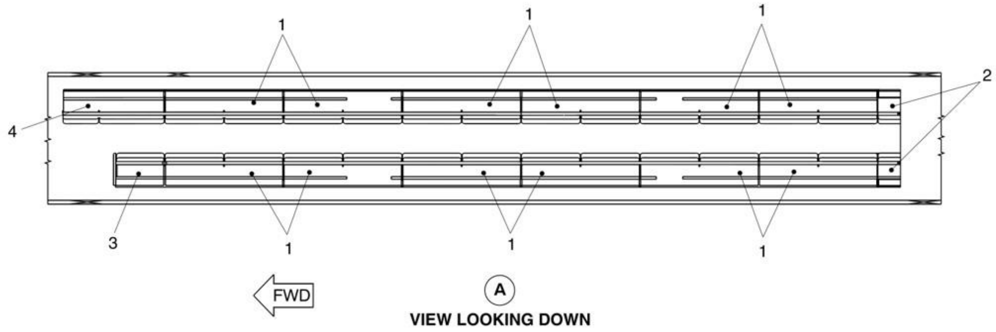


Figure 25-10 Overhead Storage Bin



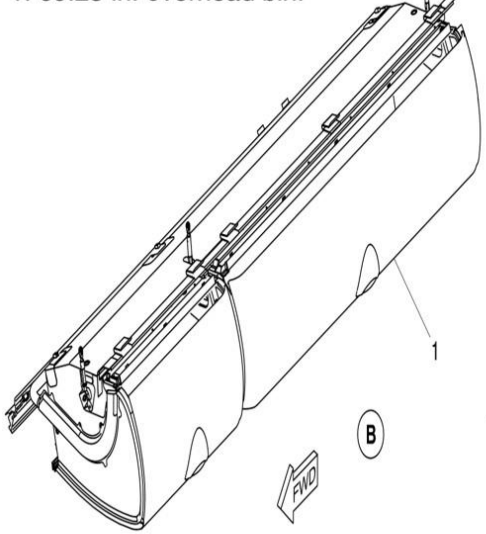
LEGEND

1. Bin assembly (6 X 2 bins of 95.75 in. each).
2. Aft bin (2 bins of 18.375 in. each).
3. Forward bin (39 in.).
4. Forward bin (69.25 in.).



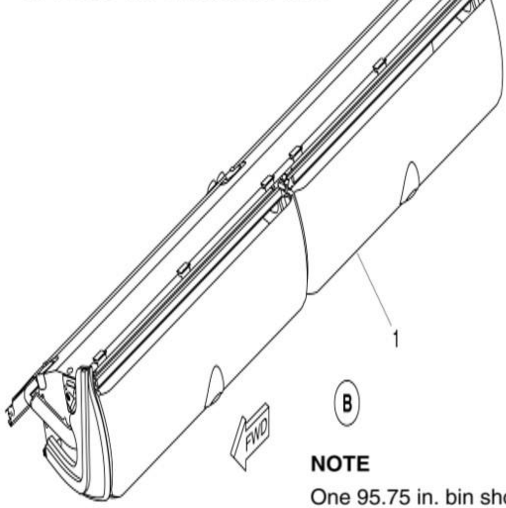
LEGEND

1. 69.25 in. overhead bin.



LEGEND

1. 95.75 in. overhead bin.

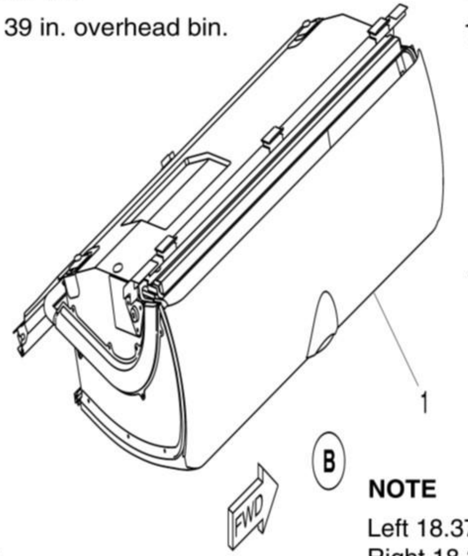


NOTE

One 95.75 in. bin shown.
Other 11 bins similar.

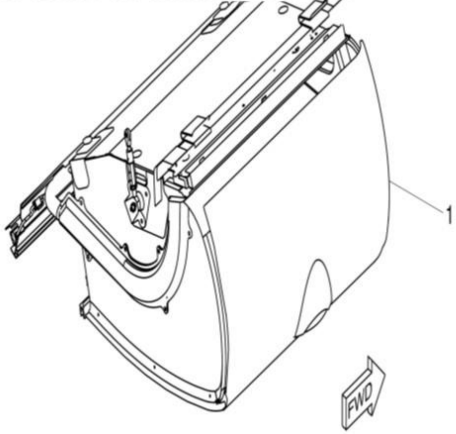
LEGEND

1. 39 in. overhead bin.



LEGEND

1. 18.375 in. overhead bin.



NOTE

Left 18.375 in. bin shown.
Right 18.375 in. bin opposite.

B

Figure 25-11 Overhead Storage Bin - ModSum 4–458919 OR 4–459413 OR 4–459622 incorporated

Wardrobes and Partitions

Refer Figure 25-12

The forward wardrobe is located on the forward left side of the passenger compartment between the flight attendant seat and the avionics equipment rack.

The wardrobe is used for garment hanger bags and briefcase size bags. The space above the wardrobe is for avionics equipment.

The wardrobe base is attached to the floor with screws and to the ceiling structure with fittings. The wardrobe has a self-contained shelf unit with a hanger bar, honeycomb composite panels with quick disconnect latches, a floor pan and a door.

The door pivots at the top and bottom on two Teflon bushing and pin assemblies and opens into the aisle.

Panels within the closet compartment can easily be removed to give access to electrical equipment and utility lights. The wardrobe overhead light is controlled by a rocker switch in the flight attendants panel and powered from the 28 Vdc left secondary bus through a circuit breaker.

The wardrobe enclosure is formed by the upper shelf unit, the floor pan and quick release sidewall panels. The floor pan is attached to the floor structure and the quick release sidewall panels connect the upper shelf to the floor pan.

Removal of the sidewall panels gives access to the electrical equipment and the utility lights located in the sidewall forward of the airstair door.

The wardrobe side panels are made of a honeycomb core and have a composite laminate surface. The flight attendant seat back forms part of the aft, inboard wardrobe compartment.

On aircraft with Modsum 4−459647 incorporated, the forward wardrobe has the provision for the installation of:

- An Emergency Medical Kit (EMK) with velcro strapping system below the lower shelf
- A standard first aid kit on the forward aft facing wall
- Two fire extinguishers on the forward aft facing wall beside the first aid kit
- One megaphone in the upper forward corner on the inboard face of the wardrobe door
- Two Protective Breathing Equipment (PBE) units on the forward aft facing wall
- One Automatic Electronic Defibrillator (AED) including its mounting bracket in the forward section of the forward wardrobe door.

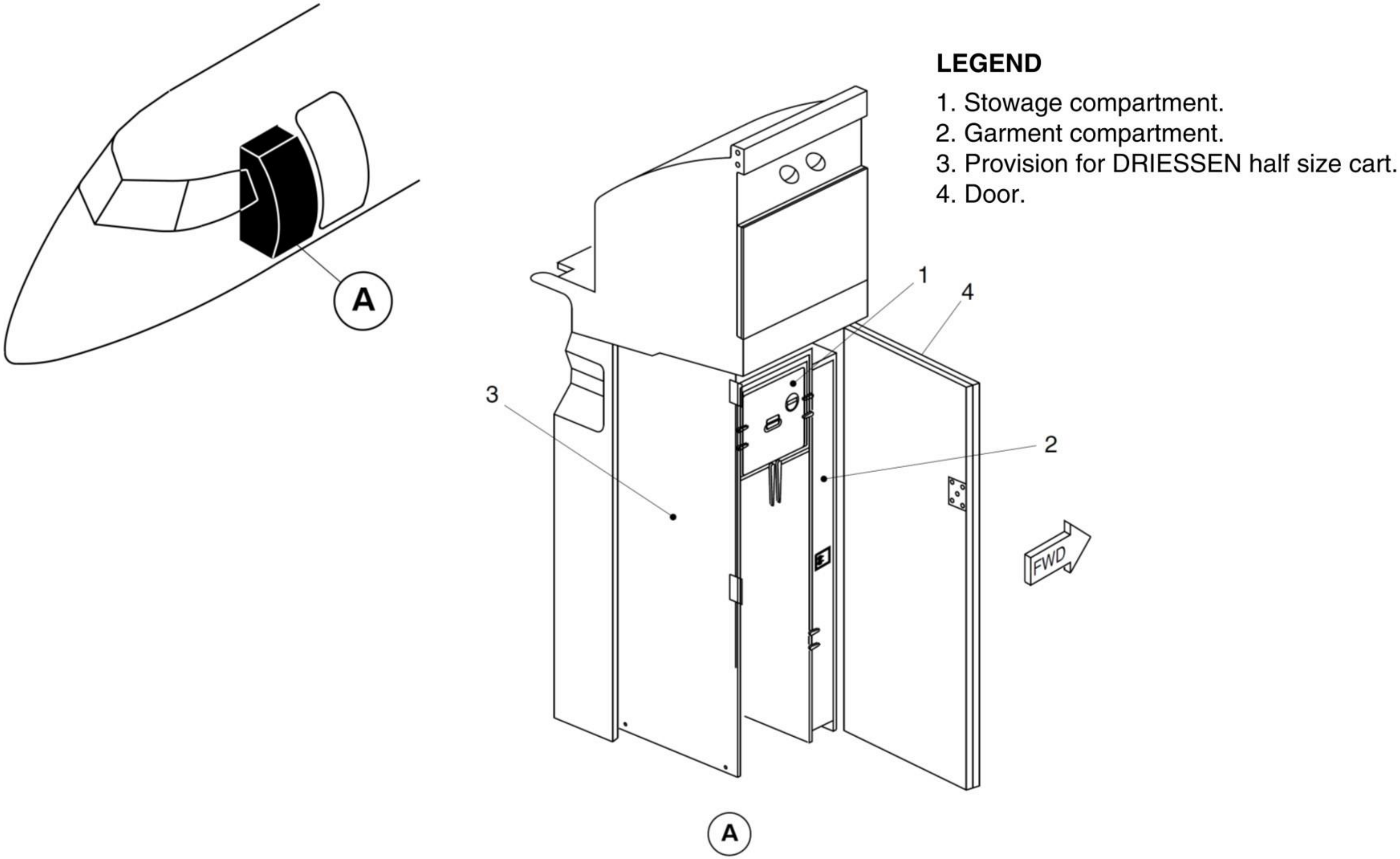


Figure 25-12 Wardrobe with Insert

Forward Draft Bulkhead

Refer Figure 25-13

The forward draft bulkhead forms a partition between the forward passenger door and the adjacent passenger seats. It shields the passengers from cold drafts when the forward passenger door is open. It also supplies a forward stowage area for the passenger compartment emergency equipment.

The bulkhead is made from a fibreglass/honeycomb material covered with a decorative plastic laminate. Its edges are trimmed with aluminum and a kick strip is attached at the floor line.

The bulkhead has an edge molding on the aft side to make a cover for the forward passenger door support mechanism.

The bottom of the bulkhead is installed with two pins which engage with vibration isolators in the floor panels.

The top of the bulkhead is attached to the forward left hand overhead stowage bin with two screws. A document holder is installed on the aft side of the bulkhead. The forward draft bulkhead contains pamphlet pocket and kick plate.

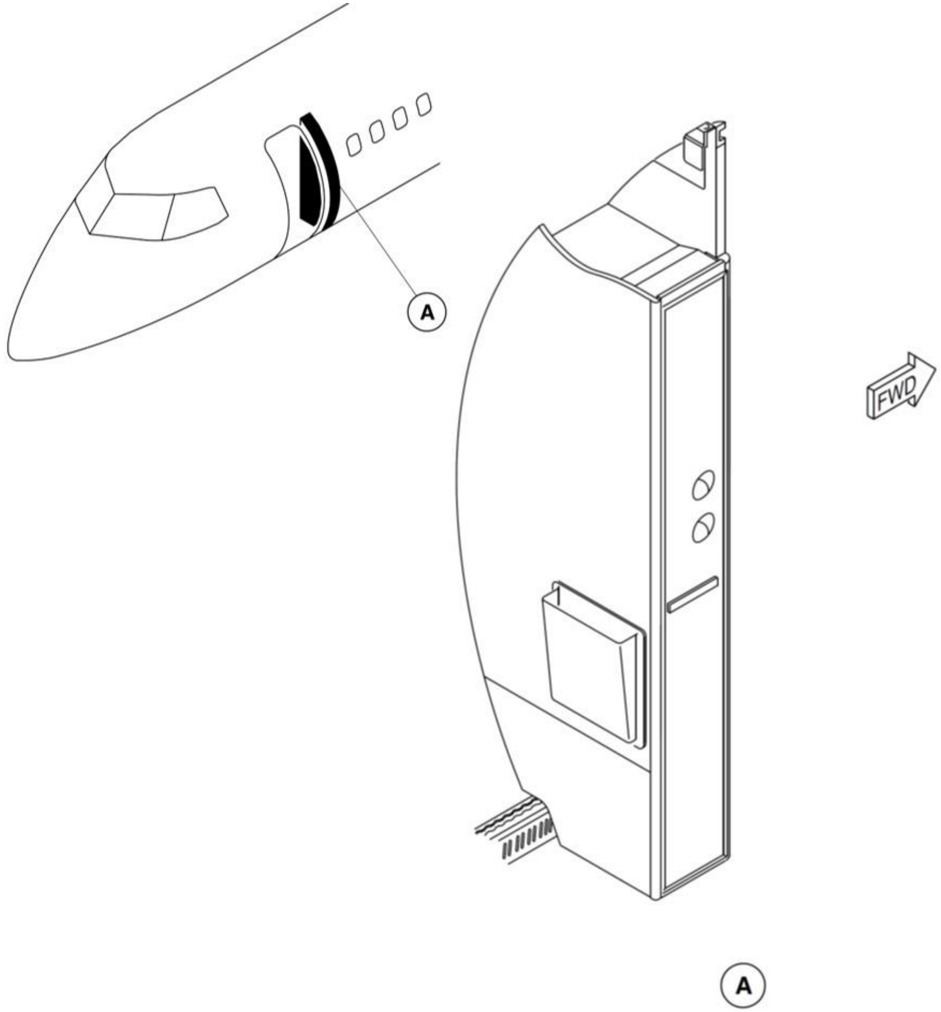


Figure 25-13 Forward Draft Bulkhead - Location

Aft Draft Bulkhead

Refer Figure 25-14

The two aft draft bulkheads are located forward of the aft service door and the aft passenger door and immediately behind the rear passenger seats. They make a partition between the G3 galley area and the adjacent passenger seats.

The bulkheads are made from a fiberglass composite material and extend from the floor to ceiling outboard of the aisle

They shield the passengers from cold drafts when either of the two doors are open. Each of the aft draft bulkheads contains an aft opening emergency equipment drawer. Also, it contains a F/A interphone handset.

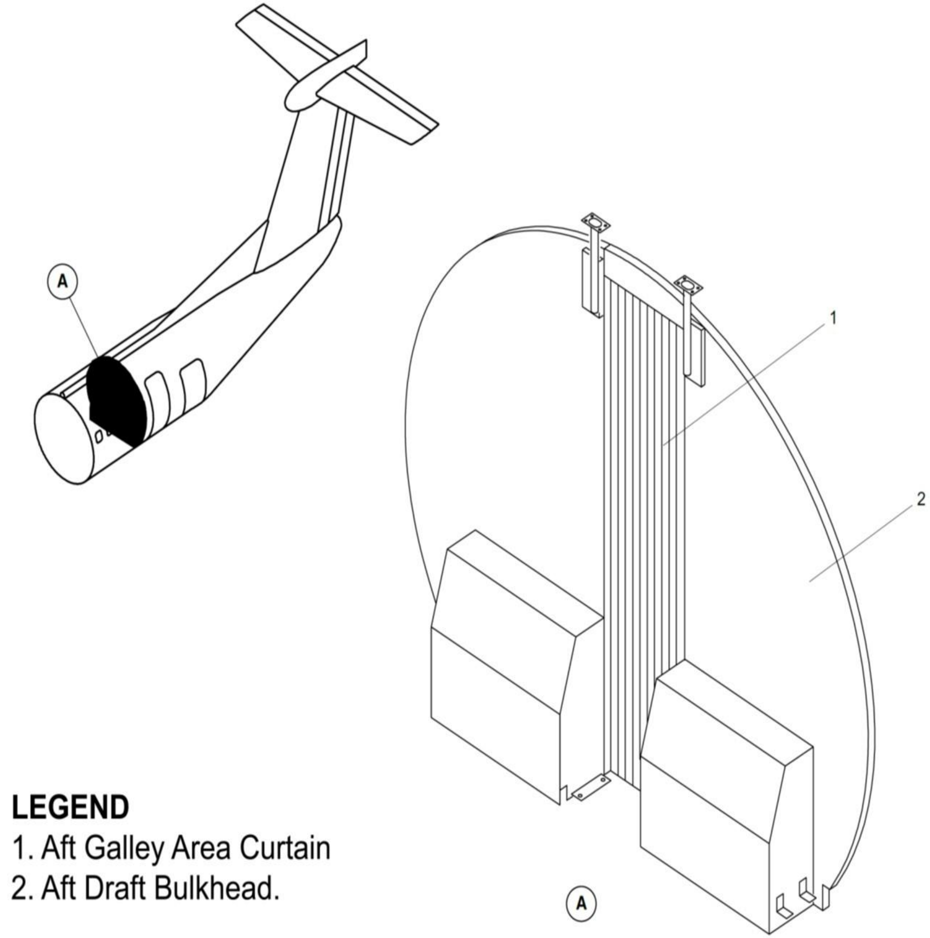


Figure 25-14 Aft Draft Bulkhead - Location

Floor Panels and Coverings

Refer Figure 25-15

There are two types of floor panels installed in the passenger compartment. Type 1 or Type 2 panels may be installed, depending on the cabin interior noise and vibration levels, the required stress resistance and the location.

COMP 27 Type 1

Floor panels used under the seats are made of unidirectional carbon/phenolic faces and aramid honeycomb core. These panels withstand stress of 37.5 lb/ft² (1.8 kN/m²).

COMP 27 Type 2

Floor panels used in the center aisle are sandwich construction with unidirectional carbon/phenolic faces and aramid honeycomb core. These panels withstand stress of 75 lb/ft² (3.6 kN/m²).

COMP 28 Type 1

Floor panels used in the cabin entry way area are sandwich construction with unidirectional S–glass/epoxy faces and aramid honeycomb core. These panels withstand stress of 75 lb/ft² (3.6 kN/m²).

COMP 28 Type 2

Floor panels used in the galley and lavatory areas are sandwich construction with unidirectional S–glass/epoxy faces and aramid honeycomb core. These panels absorb moisture with negligible mechanical property degradation and withstand stress of 125 lb/ft² (6.0 kN/m²)

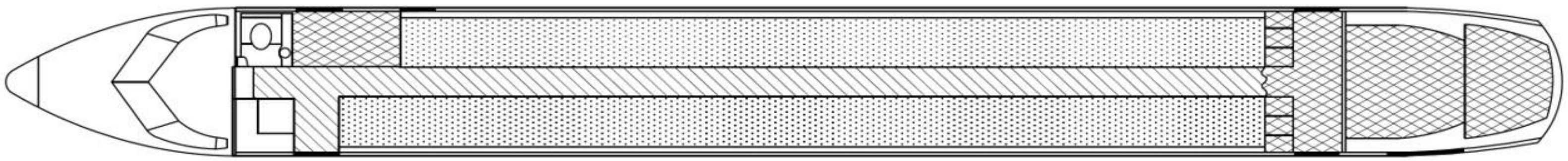
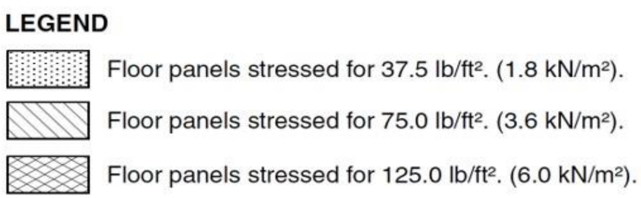


Figure 25-15 Floor Panels



No−Slip Floor Covering

Refer Figure 25-16

A vinyl slip resistant mat is used in the heavily travelled areas of passenger compartment. The no−slip covering is installed on the steps of the airstairs, on the threshold of the airstair door and in the aisle leading to the airstair door. It is also installed in the lavatory and galley areas. The vinyl mats are sealed with a tape and caulked to prevent moisture from seeping down onto the floor panels.

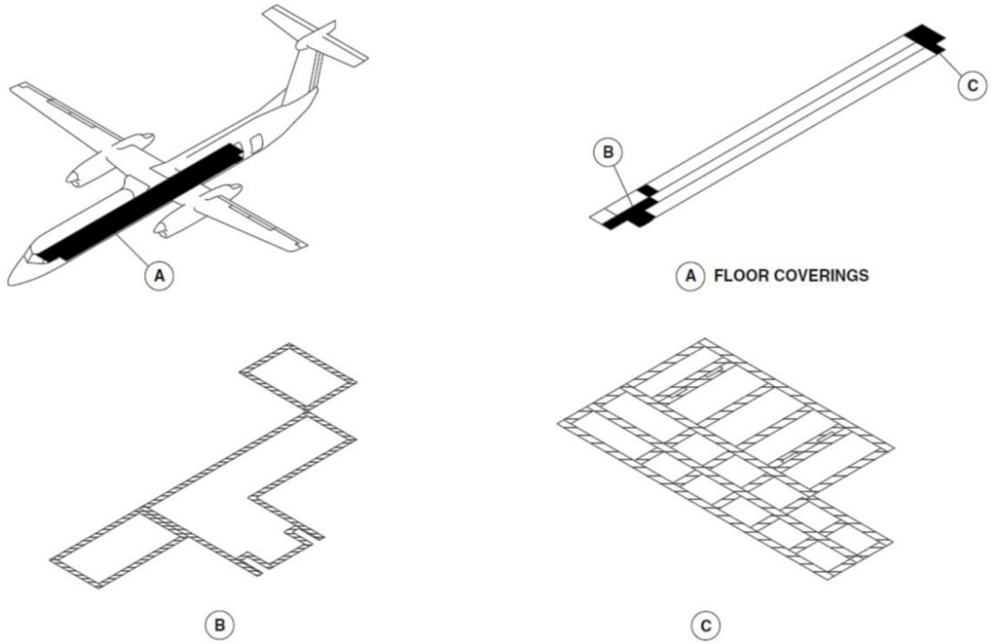


Figure 25-16 No-slip Floor Covering – Location

Carpets

Refer Figure 25-17

The passenger compartment carpets are installed in the aisle and under the seats. The carpets are slip resistant and have a anti−static filament in the weave. The color of the carpet matches the color of the vinyl mats.

Seat Track Covers

A urethane plastic cover protects the inboard and outboard seat tracks. The seat track covers are installed on the sections of the track between the seat legs and prevent dirt and foreign objects from plugging up the seat tracks.

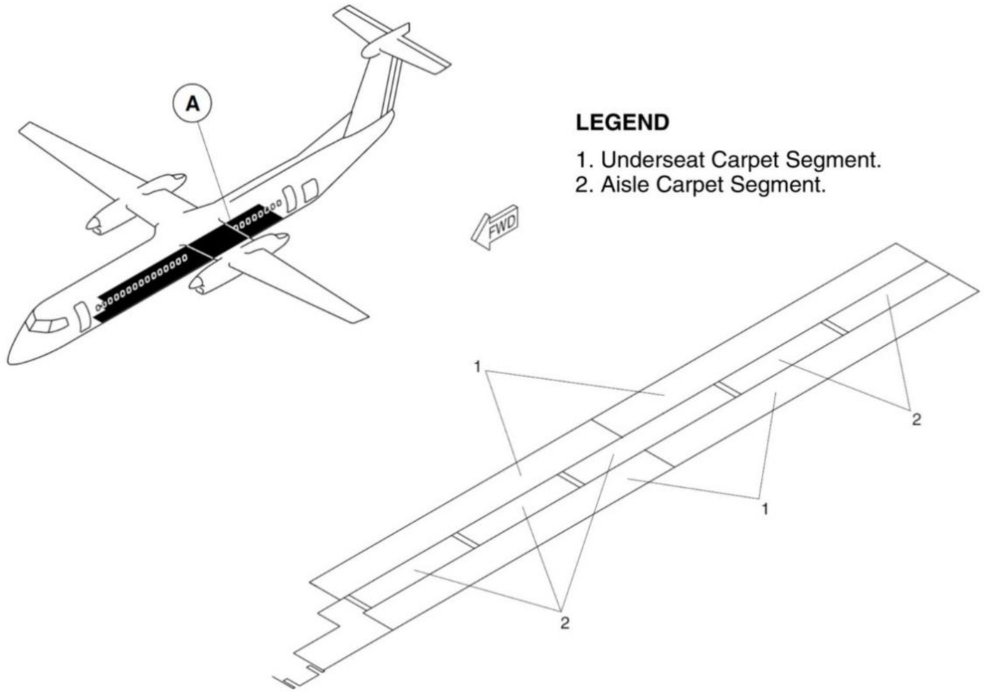


Figure 25-17 Carpets – Location

Sidewall Panels

The passenger compartment sidewall panels are made from two layers of fiberglass reinforced plastic with a decorative plastic laminate on the cabin side. Where required, a window reveal with a mar resistant window lens is installed. The lower portion of the sidewall panels has a durable fabric to protect it from passengers'
feet.

On NextGen aircraft, the sidewall panels are installed to the structure in the hooks of the dado panel rail at the bottom and sidewall panel rail on the top. They are held in position with or without the screws and washers or captive screws, and dual locks. They are easily removed based on their location without the need to remove the overhead storage bins.

There is a protective window reveal panel installed between the sidewall panels opening and the aircraft structure. The window reveal is removable without removing the sidewall panel. Infill strips are installed between each of the sidewall panels.
Dado Panels (25–26–06)

On NextGen aircraft, the dado panels are made of the composite material with a decorative plastic laminate cover on the cabin side. The panels are installed above the outboard seat track rails and prevent the aircraft structure from the damage.

The top of the panels are engaged behind the lower edges of the sidewall panels and the bottom of the panels are attached to the aircraft structure with fastening tape. The grills in the dado panels are connected to the Environmental Control System (ECS) inlet duct to supply the passenger compartment with conditioned air.

Ceiling Panels

Refer Figure 25-18

On NextGen aircraft, the ceiling is covered with the ceiling panels and transition panels. The emergency flood lights are installed on the ceiling panels. The ceiling panels and the transition panels are of the same shape as the top of the aircraft structure. They are made of two layers of glass reinforced plastic with a decorative cover. These ceiling panels and transition panels are installed with the latches, spring clips, captive screws and normal screws with or without washers.

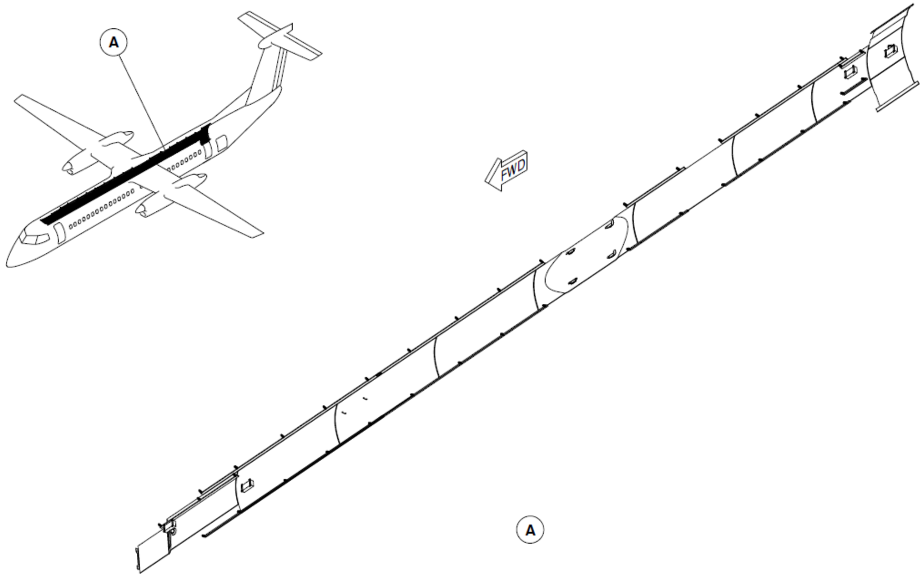


Figure 25-18 Ceiling Panels - Location

Passenger Service Units

Refer Figure 25-19

The Passenger Service Units (PSUs) are made of polycarbonate black lexan and are attached to the structure by a butt joint on the outboard side and by rubber tubing on the inboard side.

The units have passenger service panels that contain:

- Two adjustable reading lamps and operating switches
- Two adjustable air outlets (gaspers)
- Flight attendant call switch and light indicator
- NO SMOKING sign
- FASTEN SEAT BELT sign
- Passenger address loudspeakers

The PSUs are movable in one inch (24.5 mm) increments to allow for variable seat pitches.

Stowage Box Assembly

On aircraft with ModSum 4−459213 incorporated, there are two stowage box assemblies installed at the floor level one on the left side and one on the right side. The stowage box assemblies are located immediately forward of the aft cargo compartment and behind the rear passenger seats. Each stowage box assembly consists of a pair of back−to−back doghouse stowage boxes.

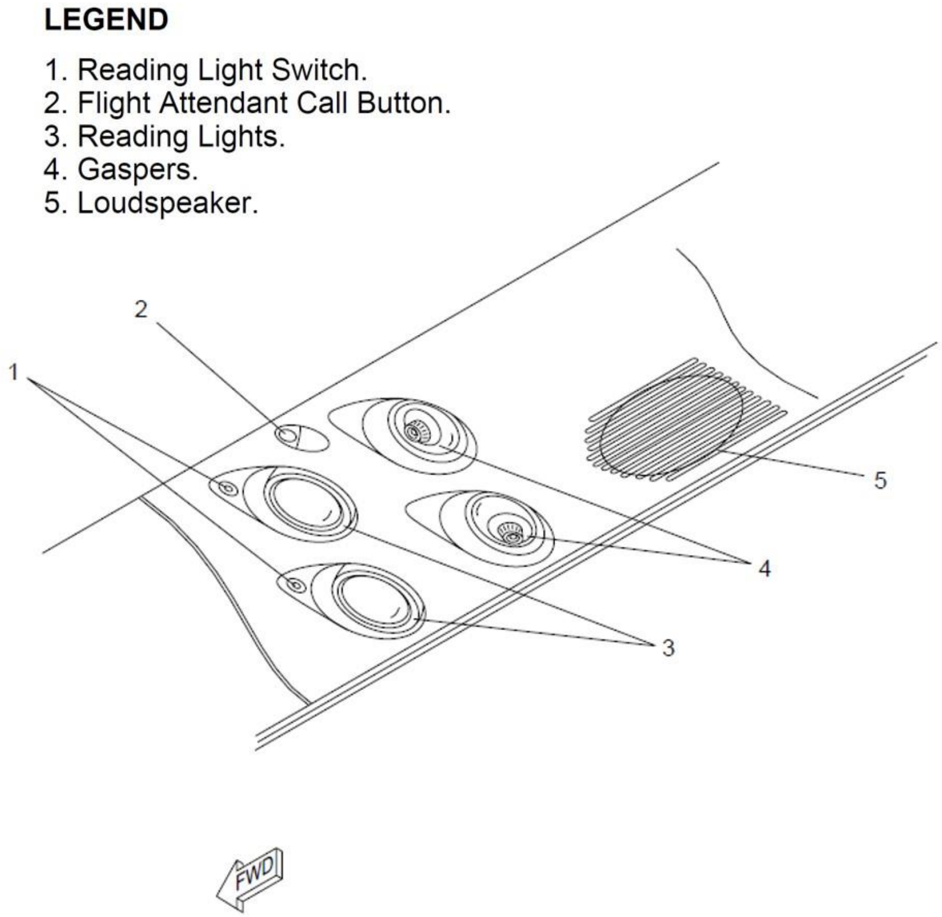


Figure 25-19 Passenger Service Units - Location

Galleys

Introduction

The galleys on board the aircraft supply a place to store and prepare food and beverages for the passengers and flight crew.

On aircraft with ModSum 4−459363 incorporated, the G1 galley is removed to increase the volume of the aft baggage compartment to 828 ft^3 (23.45 m^3).

On aircraft with ModSum 4−459364 incorporated, the G2 galley is removed to increase the volume of the aft baggage compartment to 828 ft^3 (23.45 m^3).

On aircraft with SB84−25−181 incorporated, the G1 & G2 galleys are removed and the G3 galley is installed to increase the passenger seating capacity from 74 to 78.

On aircraft with SB84−25−175 incorporated, the G6 galley is removed to increase the volume of FWD baggage compartment from 51 ft^3 (1.44 m^3) to 91 ft^3 (2.58 m^3).

General Description

The galleys allow for the storage of food and beverages (hot and cold) and for the removal and storage of waste material.

The galleys also supply adequate space and the required equipment for the preparation and serving of the food and beverages.

The galleys are supplied with AC and DC electrical power for the ovens and water heaters.

Galleys - Component Description

Aft Galley (G3)

Refer Figure 25-20

The G3 galley is located behind the aft service door. It contains three dc powered hot jugs, a removable waste container, miscellaneous stowage, three ovens and four Atlas standard half carts. The carts are retained in the galley by 1/4 turn latches and have a pull−out work shelf. The aft flight attendant seat is a part of the G3 galley.

On aircraft with ModSums 4−458077 OR 4−458032 OR 4−458890 incorporated, the G3 galley is located behind the aft service door. It contains two AC powered coffee makers, a removable waste container, three Atlas standard service units, miscellaneous stowage compartments, four Atlas standard half carts and a pull−out work table. Each half size cart has seven pull−out drawers with a flip−top tray and all the half size carts are retained in the galley by the 1/4 turn latches. The two miscellaneous stowage compartments are lockable. The aft flight attendant seat is the part of the G3 galley.

On aircraft with ModSums 4−458175 OR 4−457921 incorporated, the G3 galley is located behind the aft service door. It contains five liquid container hot jugs powered by 28 Vdc, a removable waste container, two Atlas standard service units, a pull−out work table, four Atlas standard half carts and miscellaneous stowage compartments.

Each half size cart has seven drawers with a flip−top tray and all the half size carts are retained in the galley by the 1/4 turn latches. The hot jugs and standard units are located in the galley compartments. The galley compartments have the doors which are lockable by 1/4 turn latches. The aft flight attendant seat is the part of the G3 galley.

On aircraft with ModSums 4−458600 incorporated, the G3 galley is located behind the aft service door. It contains three hot jugs, two stowage units, one oven, one removable waste container, four miscellaneous stowage compartments, four half size trolleys, and a pull−out work table. A gasper pipe is connected at the bottom of the waste container for air circulation in the galley. The aft flight attendant seat is the part of the G3 galley.

On aircraft with ModSum 4−459614 incorporated, the G3 galley is located behind the aft service door. It contains three hot jugs and one oven. There is a provision for two standard service units and four half sized trolleys. The aft flight attendant seat is the part of the G3 galley.

On aircraft with ModSum 4−458949 incorporated, the G3 galley is located behind the aft service door. It contains one AC powered coffee maker and one AC powered espresso maker.

There is also provision for four Atlas standard half trolleys, three Atlas standard service units and a pull−out work table. It also contains miscellaneous stowage compartments and one waste compartment. Each half size cart has seven pull−out drawers with a flip−top tray and all the half size carts are retained in the galley by the 1/4 turn latches. The aft flight attendant seat is a part of G3 galley.

On aircraft with ModSum 4−459302 OR 4−459315 OR SB84−25−181 incorporated, the G3 galley is located behind the aft service door. It contains two AC powered coffee makers. There is also provision for four Atlas standard half trolleys, three Atlas standard service units and a pull−out work table. It also contains miscellaneous stowage compartments and one waste compartment.

Each half size cart has seven pull−out drawers with a flip−top tray and all the half size carts are retained in the galley by the 1/4 turn latches. The two miscellaneous stowage compartments are lockable. A gasper pipe is connected at the bottom of the waste container for air circulation in the galley. The aft flight attendant seat is a part ofG3 galley.

On aircraft with ModSum 4−459448 incorporated, the G3 galley is located behind the aft service door. It contains an oven, one removable waste container, six miscellaneous stowage compartments and a pull−out work table. There is also provision for four Atlas standard half trolley carts and three Atlas standard stowage units. Each half size cart has seven pull−out drawers with a flip−top tray and all the half size carts are retained in the galley by the 1/4 turn latches. The six miscellaneous stowage compartments are lockable. A gasper pipe is connected at the bottom of the waste container for air circulation in the galley. The aft flight attendant seat is a part of G3 galley.

On aircraft with ModSum 4−459588 incorporated, the G3 galley is located behind the aft service door. It contains three hot jugs powered by 28 Vdc, a removable waste container, one oven, two Atlas standard stowage units, a pull−out work table, four Atlas standard half size trolleys with four−wheel brake systems and miscellaneous stowage compartments. Each half size cart has seven drawers with a flip−top tray and all the half size carts are retained in the galley by the 1/4 turn latches. The galley compartments have the doors which are lockable by 1/4 turn latches. A gasper pipe is connected at the bottom of the waste container for air circulation in the galley. The aft flight attendant seat is a part of G3 galley.

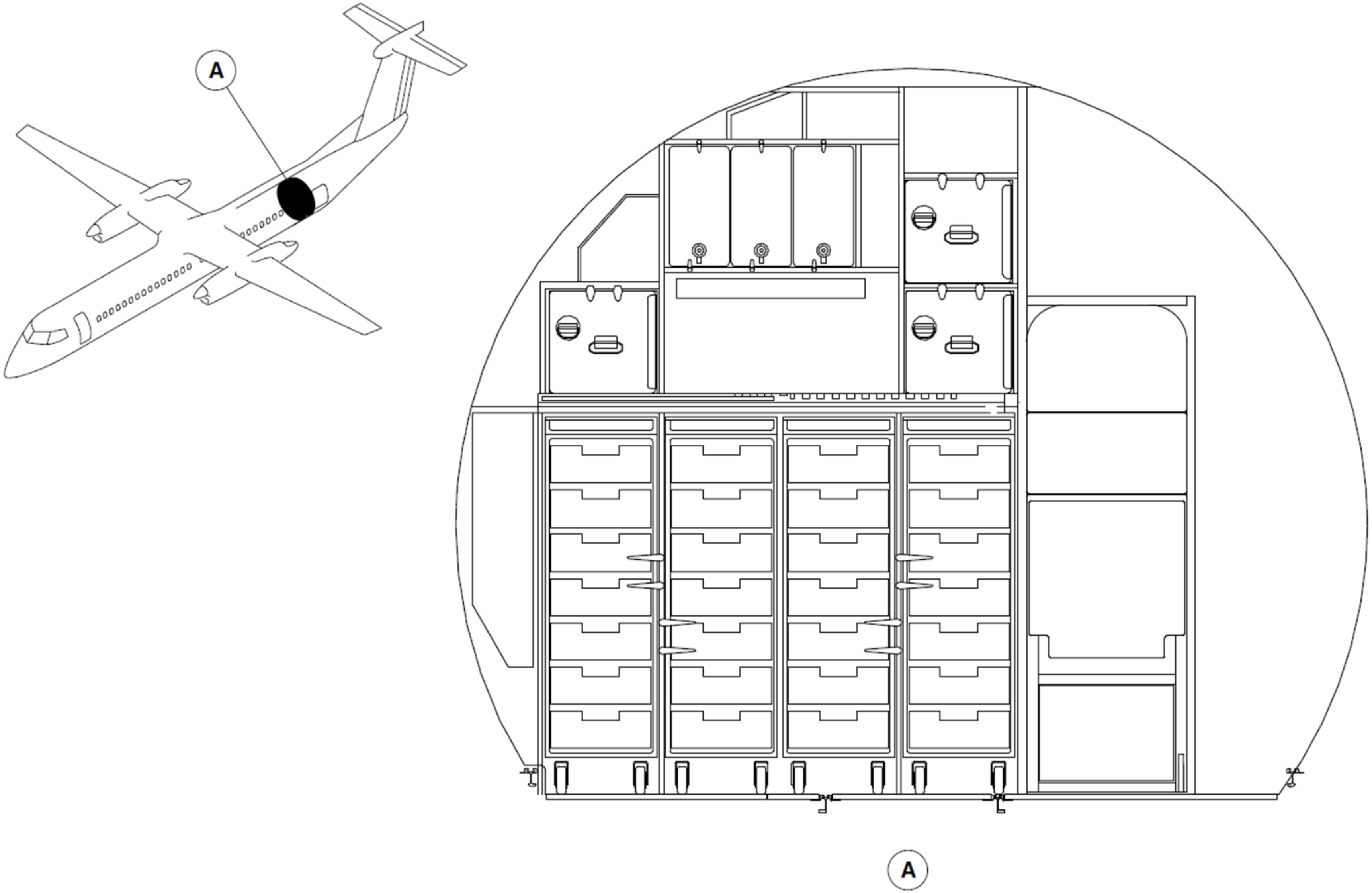


Figure 25-20 Aft Galley(G3) - Location

Lavatories

Introduction

The lavatory compartment supplies the crew and passengers with washroom facilities.

General Description

Refer Figure 25-21

The lavatory compartment is installed on the right forward side of the passenger compartment, aft of the flight compartment. The lavatory compartment can also be optionally installed at the rear of the passenger compartment. The lavatory compartment contains the items that follow:

- Externally serviced toilet unit
- Wall mirror
- Ashtray (Optional)
- Two towel dispensers
- Wet wipe dispenser
- Toilet paper dispenser
- Waste disposal
- Smoke detector
- Fire Extinguisher
- Baby change table (optional).

The lavatory compartment has an entrance door with an external handle, an internal handle and dual action lock. If necessary, the door can be unlocked from outside the compartment. An opaque OCCUPIED sign is installed on the door exterior and a LAVATORY OCCUPIED sign is installed in the passenger compartment. When the lavatory compartment door is locked, the OCCUPIED sign on the door and the LAVATORY OCCUPIED sign in the passenger compartment come on.

Any announcements made over the Passenger Address (PA) system will be heard from the PA speaker installed in the ceiling. A lavatory compartment RETURN TO SEAT sign and an attendant call button are installed on the upper amenity. The lavatory compartment RETURN TO SEAT sign comes on with the passenger compartment fasten seat belt signs. The lavatory call button can be used if assistance is necessary. The air conditioning system supplies conditioned air to the lavatory compartment through an gasper outlet in the upper amenity unit (Refer SDS 21–22–00).

A metal waste disposal container, with a door assembly spring loaded to the closed position, is designed to contain fires. A fire extinguisher automatically operates when the temperature in the waste disposal container reaches a set point. If there is smoke from a fire or other source in the lavatory compartment, the smoke detector will cause an audible tone to sound, and the repeater lights in the passenger compartment ceiling flash (Refer SDS 26–24–00).

The lavatory lighting is supplied from an overhead light panel and by a light unit. The overhead light has two 28 Vdc lamps, which come on when power is applied to the lavatory. The lavatory light unit is installed on the aft wall and it is adjacent to the compartment door and the mirror.

On aircraft without Mod Sums 4−457787 AND 4−126450 incorporated, the lavatory light unit has a ballast inverter and two fluorescent lights. The fluorescent lights come on when the latch on the lavatory door is moved to the OCCUPIED position (Refer SDS 33–20–00).

On aircraft with Mod Sums 4−457787 OR 4−126450 incorporated, the lavatory light unit has a ballast inverter and two LED lights. The LED lights come on when the latch on the lavatory door is moved to the OCCUPIED position (Refer SDS 33–20–00). An optional warm water wash system may be installed with a sink and faucet..

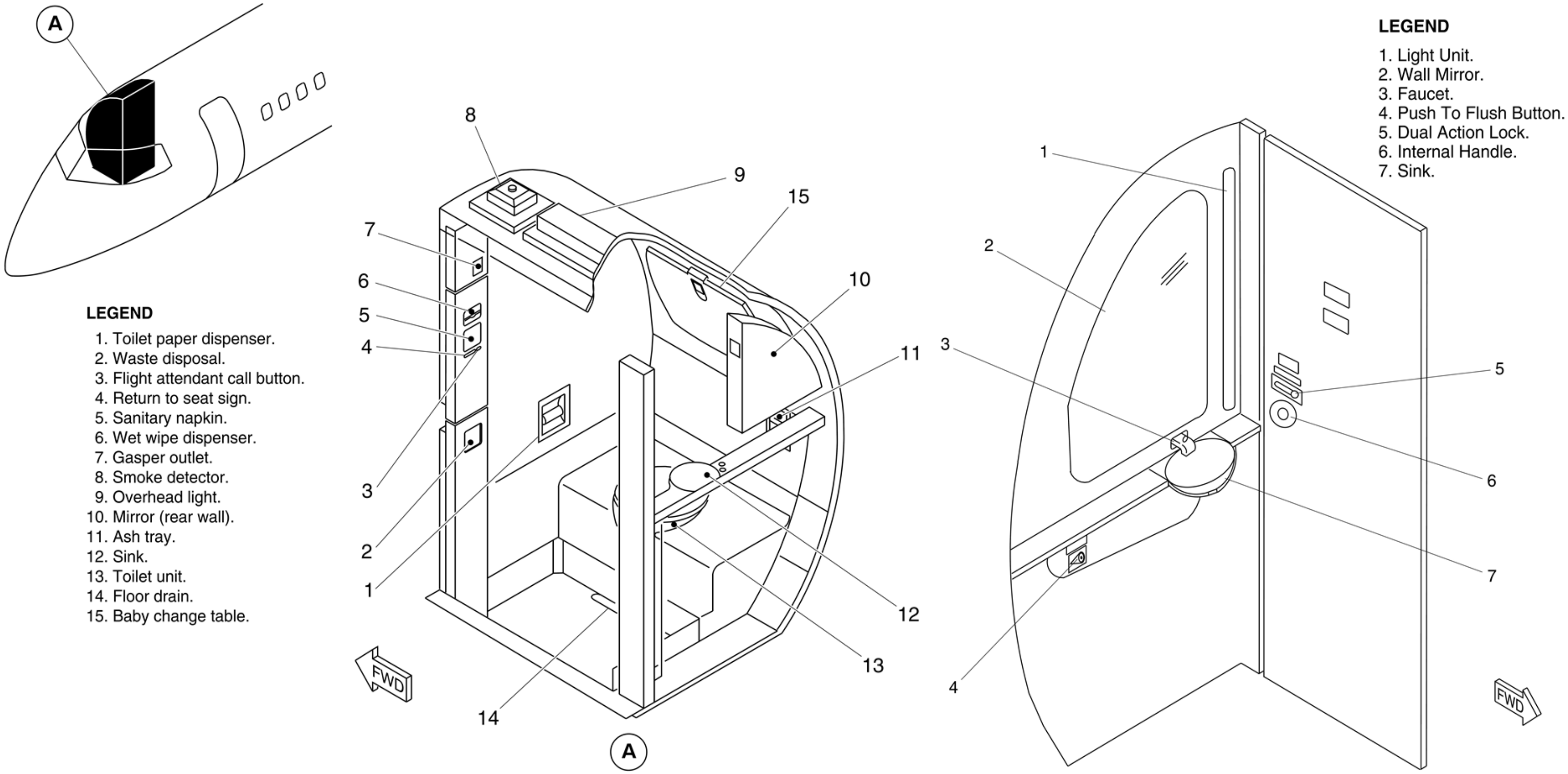


Figure 25-21 Lavatory Compartment

Lavatories - Component Description

Lavatory Compartment Structure

The lavatory structure is made of No max−Kevlar. All other lavatory components are attached to this structure. The structure is attached to the passenger compartment seat rails and to the fuselage by six bolts. The lavatory smoke detector is attached to the lavatory ceiling.

Lavatory Entry Door

Refer Figure 25-22

The lavatory door gives access to the lavatory and has a kick strip on both sides. The dual action lock turns on an OCCUPIED sign adjacent to the lock and a remote LAVATORY OCCUPIED sign in the forward passenger compartment. The door is held in the closed position by a latch, which is unlatched with the door handle. Grilles on both sides of the door let air move between the lavatory and the passenger compartments.

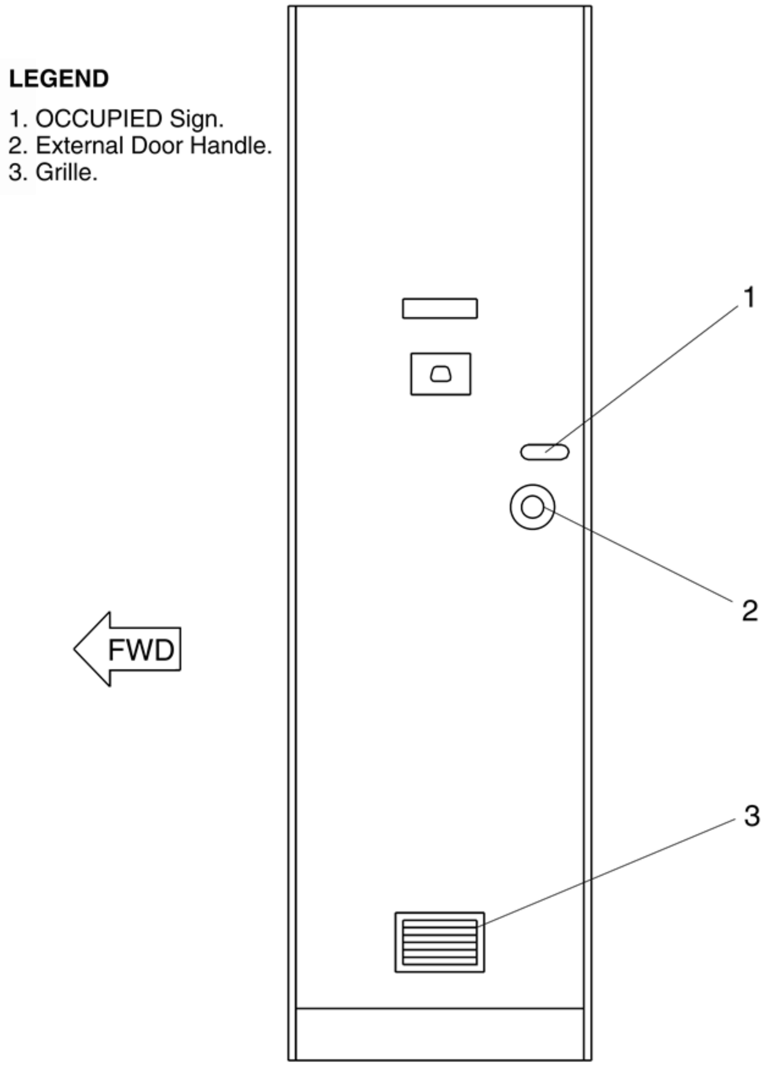


Figure 25-22 Lavatory Entry Door

Toilet Shroud

Refer Figure 25-23

The toilet shroud covers the toilet tank and is sealed to the lavatory compartment structure during assembly.

Upper Amenity Assembly

The upper amenity assembly contains a gasper to supply conditioned air, dispensers for wet wipe towels and napkins, a RETURN TO SEAT sign and a call button. On aircraft with SB84−25−98 or ModSum 4−458569 incorporated, the upper amenity assembly also contains a drop down oxygen assembly.

Lower Amenity Assembly

The lower amenity assembly has doors that enclose a towel box and a waste disposal container. The lavatory waste disposal fire extinguisher is installed between the towel box and waste container. The extinguisher discharge tube passes through a shelf just below the fire extinguisher bottle, and is directed towards the waste container. Any space between the tube and the shelf is sealed with silicon compound.

The main panel of the assembly is hinged and held closed with catches so that it can be opened for servicing. This panel also has silicon seals on the bottom and lower vertical edges for waste disposal fire containment. The waste disposal door is spring loaded to the closed position and has flanges which are held against the main panel when the door is in the closed position.

Avionics Access Panel

A part of the forward lavatory compartment wall has an avionics access panel for the rack, immediately aft of the copilot's seat. This panel has a support handle which can be used by lavatory compartment occupants. A toilet paper dispenser is also installed in a recess in this panel.

Lavatory Floor Pan

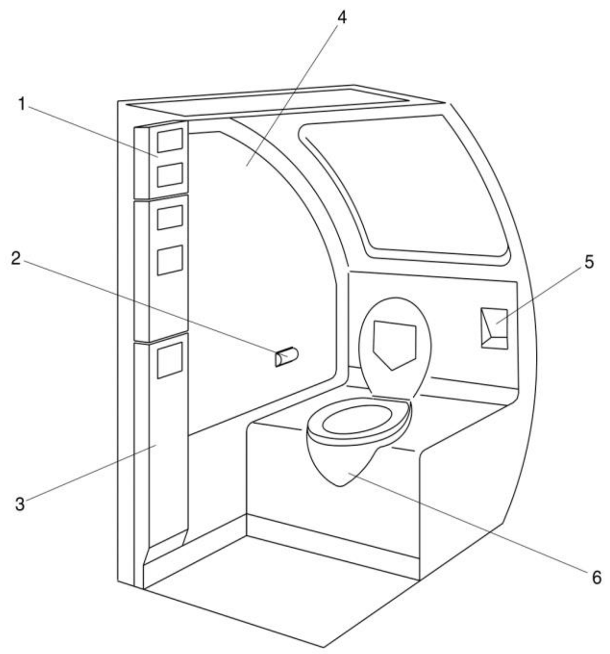
The lavatory floor pan is attached to the toilet assembly above the floor. Any crevices are filled with silicon sealant.

Service Access Door

A service door is installed below the sink, or counter top in the "Dry Lav" configuration. In the "Dry Lav" configuration only the aft face has two containers attached to it which can be used for servicing purposes.

Assist Handle

On aircraft with ModSum 4−459329 OR 4−459396 OR 4−459630 OR 4−459727 OR 4−459749 incorporated, an assist handle is mounted on the lavatory (aft facing) close out panel.



LEGEND
1. Upper Amenity Assembly.
2. Toilet Paper Dispenser.
3. Lower Amenity Assembly.
4. Avionics Access Panel.
5. Ash Tray.
6. Toilet Shroud.

Figure 25-23 Lavatory Interior

Baggage Compartments

Introduction

The forward and aft baggage compartments have sufficient space for the safe storage of the passengers' baggage and cargo.

General Description

The forward and aft baggage compartments are Class C compartments. They have the properties that follow:

- smoke detectors, one in the forward baggage compartment and two in the aft baggage compartment (Refer SDS 26–12–00)
- a built–in fire extinguishing system which is controllable by the crew (Refer SDS 26–22–00)
- sufficient sealing to ensure minimum leakage of smoke and fire extinguishing agent into the passenger compartment. On aircraft with the ModSum 4–458912 incorporated, forward baggage compartment is removed to accommodate additional passenger seats. On aircraft with ModSum 4–459141 incorporated, the aft baggage compartment 411 ft³ (11.63 m³) is removed to increase the volume of the aft cargo compartment to 828 ft³ (23.45 m³).

The aft compartment has dedicated air supply. It has an inflow valve and an outflow valve, which are automatically (or manually) closed when fire or smoke is detected (Refer SDS 21–21–00 and SDS 26–12–00).

Baggage loading for both compartments is accomplished through the related external door. The forward compartment also has an internal door which allows access to the passenger compartment

Baggage Compartment - Component Description

Forward Baggage Compartment

Refer Figure 25-24

The forward baggage compartment is located on the right side of the aircraft in front of the forward emergency exit. For the baseline interior configuration, the baggage compartment is located between the compartment rear bulkhead (Stn. 70) and the aft wall of the lavatory. This configuration gives the forward baggage compartment a volume of approximately 91 ft³ (2.58 m³). If the lavatory is not installed, the compartment extends forward to the flight compartment bulkhead (Stn.−33) for a total volume of 135 ft³ (3.82 m³). If the aircraft is equipped with a G6 galley (Refer SDS 25–30–00), the compartment rear bulkhead is moved forward to Stn. 39 to accommodate the galley. This configuration gives the forward baggage compartment a volume of 51ft³ (1.44 m³).

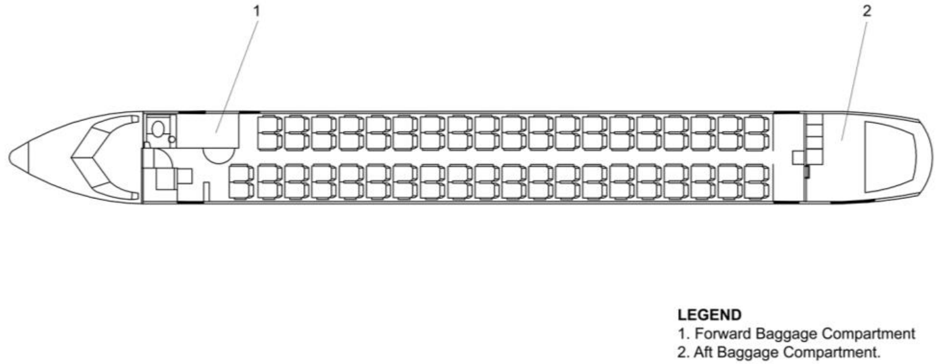
On the aircraft with SB84−25−175 incorporated, the 91 ft³ (2.58 m³) forward baggage compartment is installed on the right side of the aircraft in front of the forward emergency exit. This baggage compartment is converted from 51 ft³ (1.44 m³) baggage compartment and G6 galley.

On aircraft with ModSum 4−459101 incorporated, the 91 ft³ (2.58 m³) forward baggage compartment has a life raft stowage unit installed at the aft side. The access to the life raft stowage unit is through the internal door of the forward baggage compartment. The life raft stowage unit has shelves for the stowage of four life rafts.

The passengers' baggage is loaded into the forward baggage compartment through the exterior forward baggage door. The exterior door opens inward and upward and then translates outward and forward, to its locked open position (Refer SDS 52–32–00). The exterior door has an unlocked proximity sensor and a door open proximity sensor. The two proximity sensors send signals to the Proximity Sensor Electronic Unit (PSEU). Door position information is displayed on the doors page on the Multi Function Display (MFD) (Refer SDS 52–70–00).

On aircraft with the ModSum 4−458147 incorporated, additionally the 91 ft³ forward baggage compartment has two cart bumper assemblies, a shelf on top and provision for the five half carts and a stowage container for alpha vane covers. A smoke detector is installed in the ceiling area of the compartment and is connected to the Fire Protection Panel (FPP) in the flight compartment (Refer SDS 26–12–00). The smoke detector is protected by a metal grille.

Fire extinguishing for the forward baggage compartment is supplied by a High Rate Discharge (HRD) extinguisher bottle installed in the compartment forward roof area. The forward baggage compartment also shares a Low Rate Discharge (LRD) extinguisher bottle with the aft baggage compartment (Refer SDS 26–22–00). The forward baggage compartment has a lamp which is recessed for protection and to prevent possible fire from close contact with the baggage.



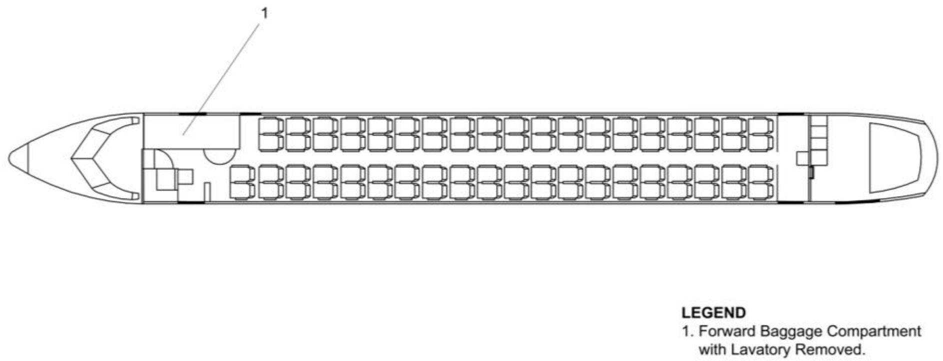


Figure 25-24 Forward Baggage Compartment - Location

Refer Figure 25-25

Forward Baggage Compartment Panels

The forward and aft bulkheads for the baggage compartment are made from regular aramid fibre panels. At floor level, the panels are attached to the inboard and outboard seat tracks by fitting pins. At the top, the panels are attached to a bearing by spigots. The bearing is installed in an adjustable housing which is attached to the fuselage structure.

The compartment sidewall panels are made from moulded semi−transparent glass cloth. The panels are attached to the fuselage structure by quick release fasteners.

The sidewall panels are attached to the adjacent floor panels by an angled piece of metal trim. The trim is also held in place by quick release fasteners. In the baseline configuration 91ft^3 (2.58 m^3), the forward and aft bulkhead panels and the sidewall panels can restrain up to 910 lb. (413.6 kg) of loose baggage. In the 135 ft^3 (3.82 m^3) configuration, the load limit is 1350 lb (613.6 kg). In the 51 ft^3 (1.44 m^3) configuration, the load limit is 510 lb (231.8 kg).

The floor panels are a sandwich construction with unidirectional S−glass/epoxy surfaces and an aramid honeycomb core. They can withstand a stress of 75 lb/ft^2 (3.6 kN/m^2).

Forward Baggage Compartment Liners

The compartment liner is made of fiberglass/phenolic resin laminate. The liners are attached to the floor angles and structural rails and brackets with screw fasteners. Sound proofing and insulation material are secured by clips to the fuselage structure throughout the compartment.

The sealing of the compartment consists of liners, floor panels, baggage bulkhead and fiberglass/silicone foam seals.

Forward Baggage Compartment Door

Access to the forward baggage compartment from the passenger compartment is through a hinged, internal door. A key operated lock is installed in the door.

The door must be closed during taxi, take−off, flight and landing. To ensure that the door is closed, there is an INTERNAL BAGG DOOR caution light on the caution and warning panel. When the door is closed it contacts a microswitch located on the compartment panel.

If the microswitch is not contacted, the INTERNAL BAGG DOOR light comes on. If the microswitch fails, the INTERNAL BAGG DOOR light automatically comes on. However, the door can be locked and manually checked for MMEL dispatch relief.

On aircraft with ModSum 4−309222 OR 4−309227 OR SB84−52−48 incorporated, the caution light indication INTERNAL BAGG DOOR on the caution and warning panel is changed to INTERNAL DOORS. This light monitors the open condition of the forward baggage compartment internal door and the unlatched condition of the fortified flight compartment door.

Also, an advisory dual lights indication BAGG DOOR/CKPT DOOR is located on the internal doors control panel to show the open condition of the forward baggage door and the unlatched condition of flight compartment door. The internal doors control panel is located on the overhead console panel.

The forward baggage door latch can only be operated and locked from the passenger compartment side of the door. Installed above the interior door is a decompression panel. The panels designed to detach into the baggage compartment, if the pressure differential between passenger and baggage compartments becomes too great. The panel will be caught by a containment cage if the panel is detached.

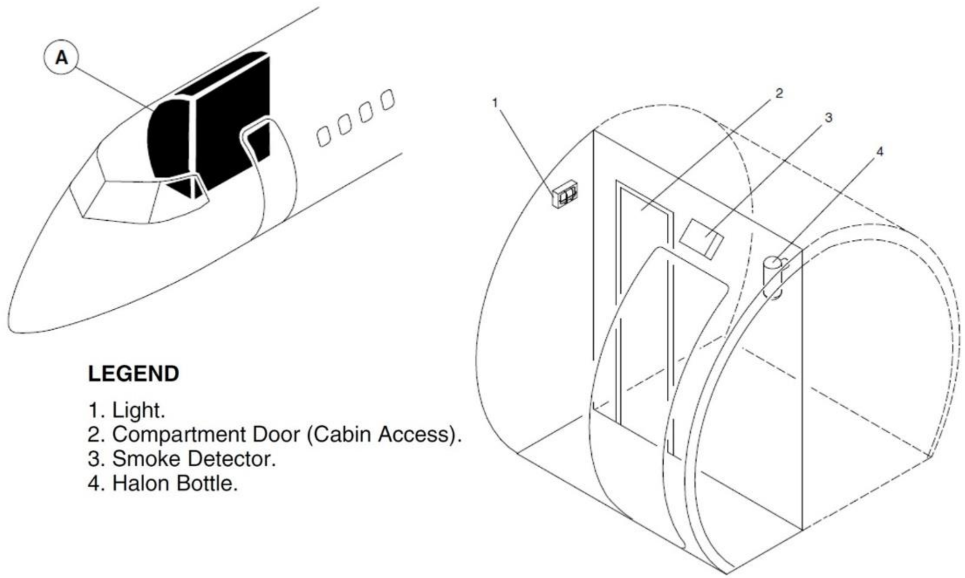


Figure 25-25 Forward Baggage Compartment

Aft Baggage Compartment

Refer Figure 25-26

The aft baggage compartment is located at the rear of the pressurized part of the fuselage immediately forward of the aft pressure dome. The forward bulkhead of the compartment is located at Stn. 701 and the aft (canted) bulkhead is located at Stn. 775. The total baggage compartment volume is 411 ft^3 (11.64 m^3). If the G3 galley is installed (Refer SDS 25–30–00), the total volume is reduced to 365 ft^3 (10.33 m^3). The floor panels are a sandwich construction with unidirectional S−glass/epoxy surfaces and an aramid honeycomb core. The floor panels from the forward bulkhead to the step have a load capacity of 125 lb/ft^2 (6.0 kN/m^2). The panels from the step to the aft (canted) bulkhead have a load capacity of 75 lb/ft^2 (3.6 kN/m^2).

The aft bulkhead of the passenger compartment serves as a divider between the passenger compartment and the baggage compartment. It has a service access panel and decompression blow out grilles, located on the left side of the aft baggage bulkhead. This bulkhead is designed to restrain up to 3,500 lb (1587.6 kg) of loose baggage.

The aft baggage compartment is provided with two lamps, a smoke detector and a built−in fire extinguishing system. The lamps are recessed and the smoke detectors, as well as the lamps, are protected from damage by metal grilles.

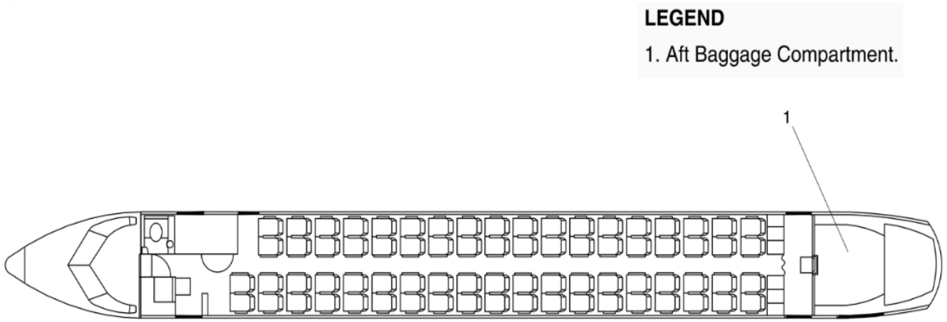


Figure 25-26 Aft Baggage Compartment - Location

Canted Bulkhead

Refer Figure 25-27

The canted bulkhead (rear wall of the aft baggage compartment) is located at Stn. 775. It is made from three detachable panels which are attached to the fuselage. The panels are made from regular aramid fibre.

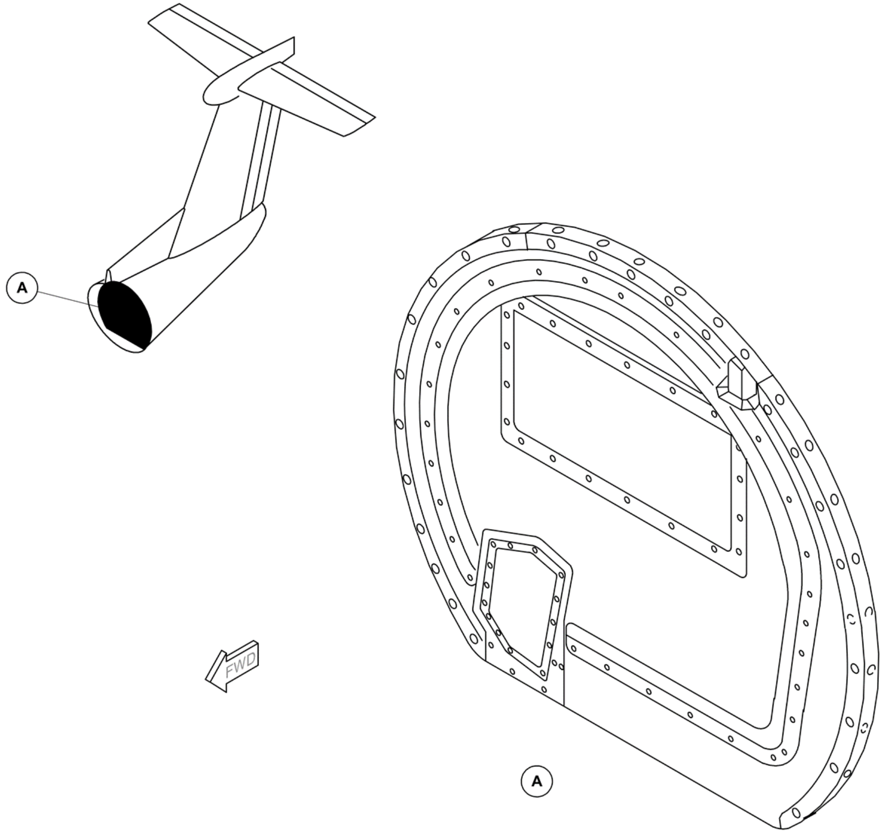


Figure 25-27 Canted Bulkhead - Location

G3 Galley Bulkhead

Refer Figure 25-28

The G3 galley bulkhead (forward wall of the aft baggage compartment) is located at Stn. 701. The bulkhead is made from regular aramid fibre panels. At floor level, the panels are attached to the inboard and outboard seat tracks by fitting pins. At the top, the panels are attached to a bearing by spigots. The bearing is installed in an adjustable housing which is attached to the fuselage structure.

G3 Galley Bulkhead Blowout Panel

A blowout (decompression) panel is installed in the G3 galley bulkhead to the left of the flight attendant's seat. The panel is designed to detach into the baggage compartment, if the pressure differential between passenger and baggage compartment becomes too great. The panel will be caught by a containment cage if the panel is detached.

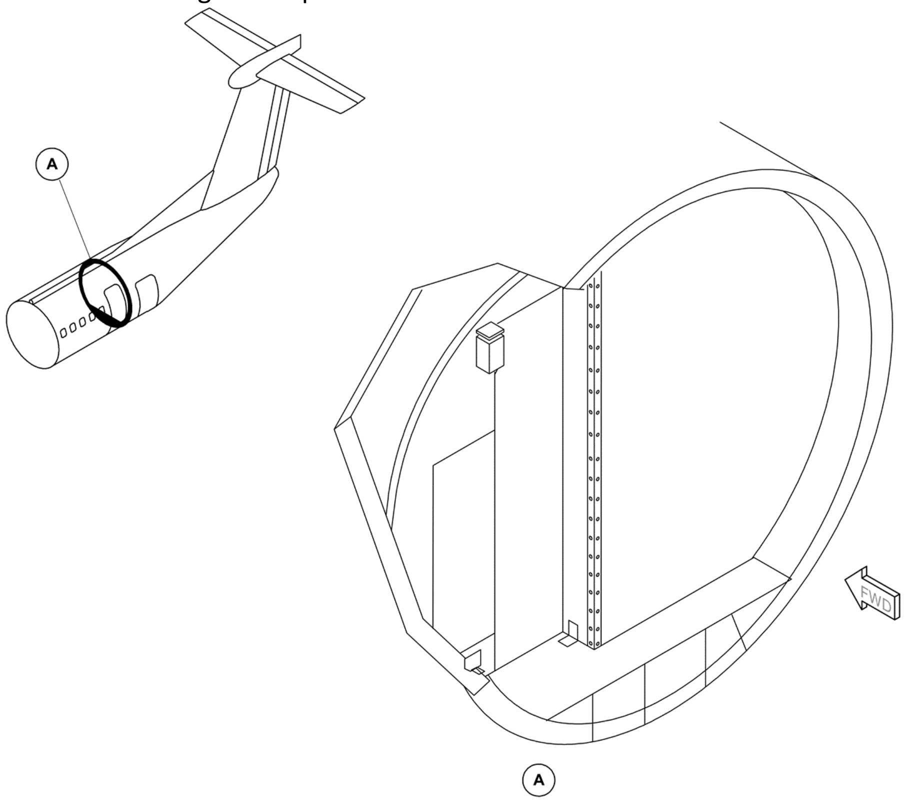


Figure 25-28 G3 Galley Bulkhead - Location

Aft Baggage Compartment Door Liners and Surrounds

Refer Figure 25-29

The compartment liner consists of fiberglass/phenolic resin laminate. The edges of the panels at the floor line are metal reinforced to prevent damage to the panels from the baggage. An easily removed covering panel protects the aft pressure bulkhead. A large label indicating the loading limit and load distribution is attached to the left side of the aft face of the passenger cabin/cargo compartment bulkhead.

Aft Baggage Compartment Liners

On aircraft without ModSum 4−459223 incorporated, the Baggage compartment is configured to 411 ft³ (11.64 m³) (baggage compartment forward bulkhead installed at station X701.00). The compartment liners are installed from the station X703.00.

The compartment liner is made of fiberglass/phenolic resin laminate to protect the fuselage structure. The liners are attached to the structural cleat, liner rail attachments, bulkhead angle and dado angle attachments with screw fasteners.

The edges of the panels at the floor line are metal reinforced to prevent damage to the panels from the baggage. An easily removed covering panel protects the aft pressure bulkhead.

On aircraft with ModSum 4−459223 and ModSum 4−459235 incorporated, the Baggage compartment is configured to 827.70 ft³ (23.44 m³) (baggage compartment forward bulkhead installed at station X586.00). The compartment liners are installed from the station X588.40. The compartment liner is made of fiberglass/phenolic resin laminate to protect the fuselage structure.

The liners are attached to the structural cleat, liner rail attachments, bulkhead angle and dado angle attachments with screw fasteners. The edges of the panels at the floor line are metal reinforced to prevent damage to the panels from the baggage. An easily removed covering panel protects the aft pressure bulkhead.

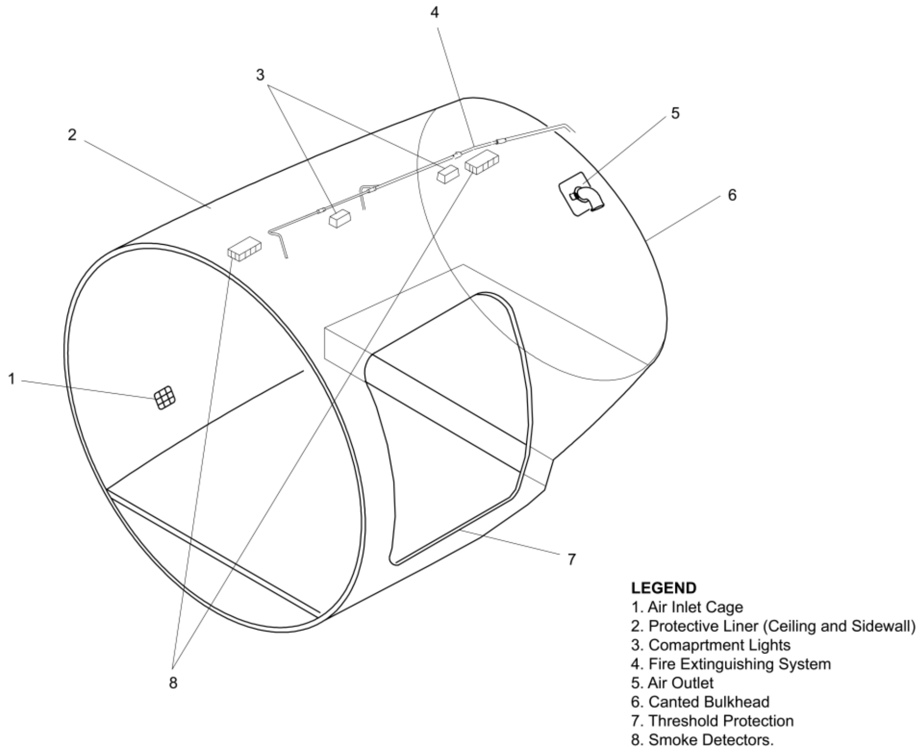


Figure 25-29 Aft Baggage Compartment

Baggage Restraint

[Refer Figure 25-30]

The baggage restraint system has tie−down rings, spring−loaded tubes an upper and lower net and a threshold protector. It is designed to restrain the baggage in flight and to protect the threshold during baggage loading and unloading. The nets cover the area of the aft baggage compartment door. The lower net is attached by clips to the threshold protector at the bottom and to a spring−loaded drawbolt tube at the top. The upper net is attached to the airframe structure at the top and to a short tube at the bottom. When the baggage compartment has been loaded, Ty−clip fasteners attach the short tube to the drawbolt tube which is attached to the airframe structure.

The threshold protector is raised up and hangs from the drawbolt. The threshold protector is a heavy gauge metal plate which is attached longitudinally to the baggage compartment floor by a hinged bracket. Protective strips and bumper pads are attached to the plate. An aluminum ninety degree extrusion is attached to the edge of the plate opposite the bracket. The drawbolt tube and lower net are suspended from this extrusion during the baggage loading process.

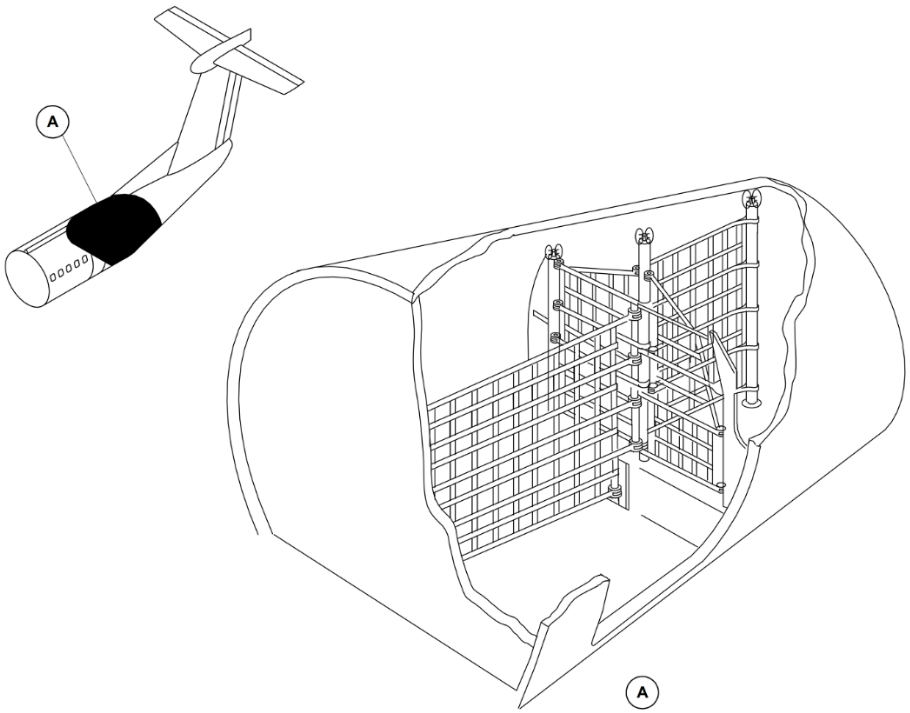


Figure 25-30 Aft Baggage Restraint

Cargo Tie–Down Fittings

Refer Figure 25-31

Sixteen 2,000 lb (907.2 kg) baggage tie–downs are supplied, ten of them located above the cargo floor level, four on the lower floor (behind the aft bulkhead) and two on the step floor.

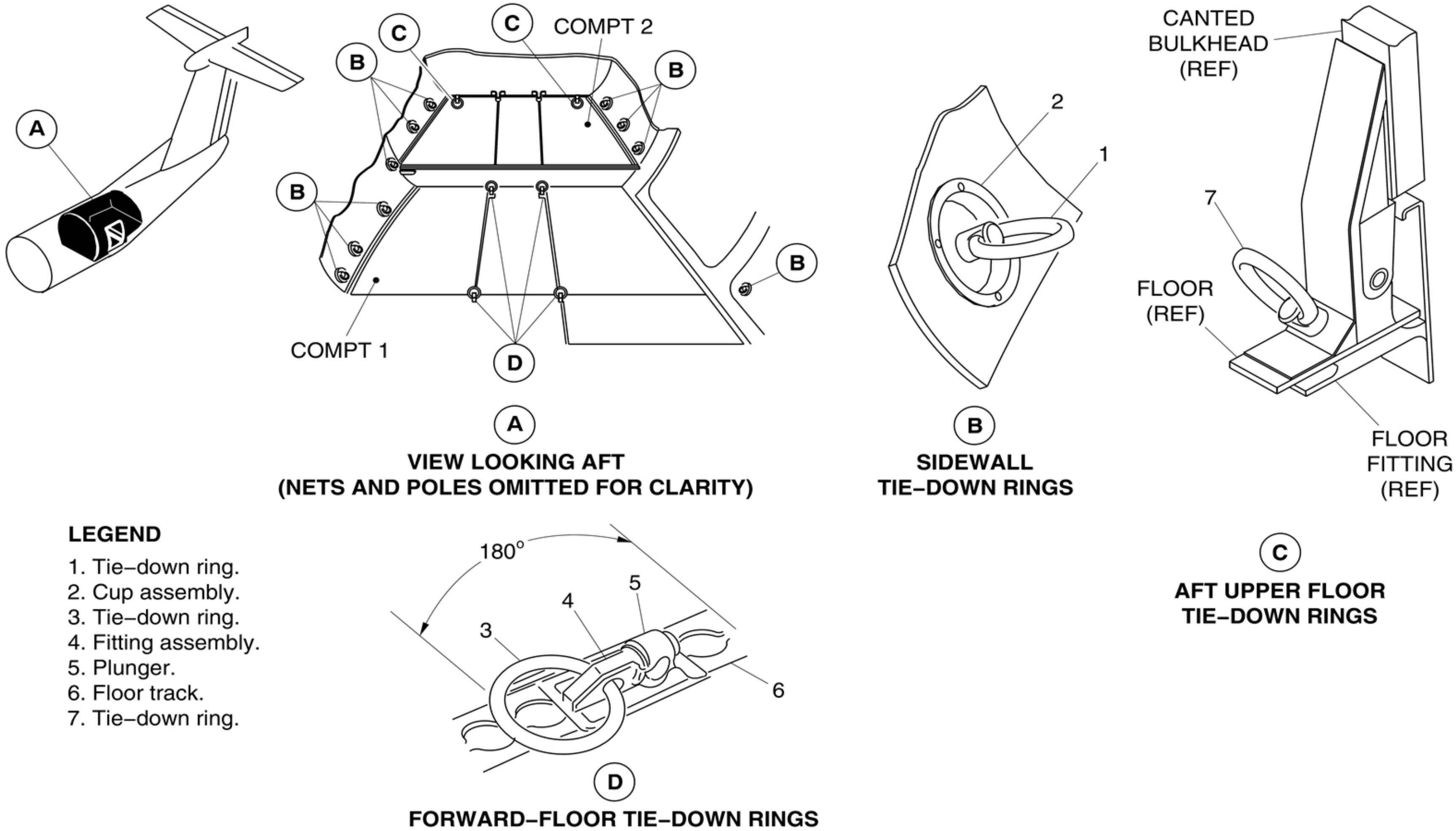


Figure 25-31 Cargo Tie–Down Fittings

Emergency Equipment

Introduction

The emergency equipment located in the flight compartment and the passenger compartment is used by the flight crew during emergency situations.

General Description

Refer Figure 25-32

The emergency equipment located in the flight compartment includes the items that follow:

- Axe, Fire
- Flashlights
- Vest, Life
- Rope, Escape
- Hood, Smoke
- Megaphone
- Crew Headset

The emergency equipment in the passenger compartment includes the items that follow:

- Dam, Ditching–Airstair Door
- Flashlights
- Vests, Life
- Kit, First Aid
- Megaphone

LEGEND

1. Co-pilot's Oxygen Mask
2. Observer's Oxygen Mask
3. Observer's Life Vest Stowage
4. Pilot's Oxygen Mask
5. Flashlights
6. Fire Axe
7. Fire Extinguisher
8. PBE.

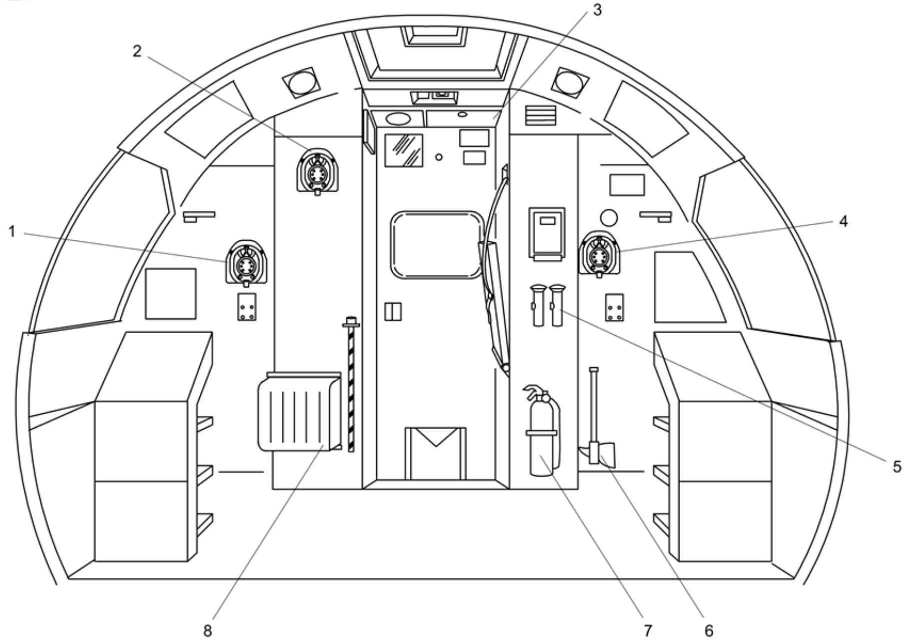


Figure 25-32 Flight Compartment Emergency Equipment – Aft Bulkhead

Flight Compartment Emergency Equipment

Refer Figure 25-33

Flight compartment emergency equipment is generally installed on the forward face of the flight compartment bulkhead. Any equipment not visible behind the pilots' seats is clearly marked, at eye level, as to its location. All emergency equipment is located within reach of atleast one crew member. The equipment is placarded and is protected from inadvertent damage.

Vest, Life

The life vests are used as an aid to flotation for the flight crew in the event of evacuation into water. The life vests are inflated by compressed gas that is connected to a toggle cord activated inflation valve on the jacket. A manual inflation valve and tube are installed, for use in the event the jacket has to be inflated by mouth.

There are two life vests stowed in individual boxes, behind the left and right ceiling panels above the pilot's seat. There is a third life vest stowed on the rear flight compartment bulkhead above the door. The vests are held in the stowage boxes by quick release Velcro fastener.

LEGEND

1. Pilot Life Vest Stowage.
2. Emergency Escape Rope Storage.
3. Main Landing Gear Alternate Release Door.
4. Copilot Life Vest Stowage.
5. Landing Gear Alternate Extension Door.

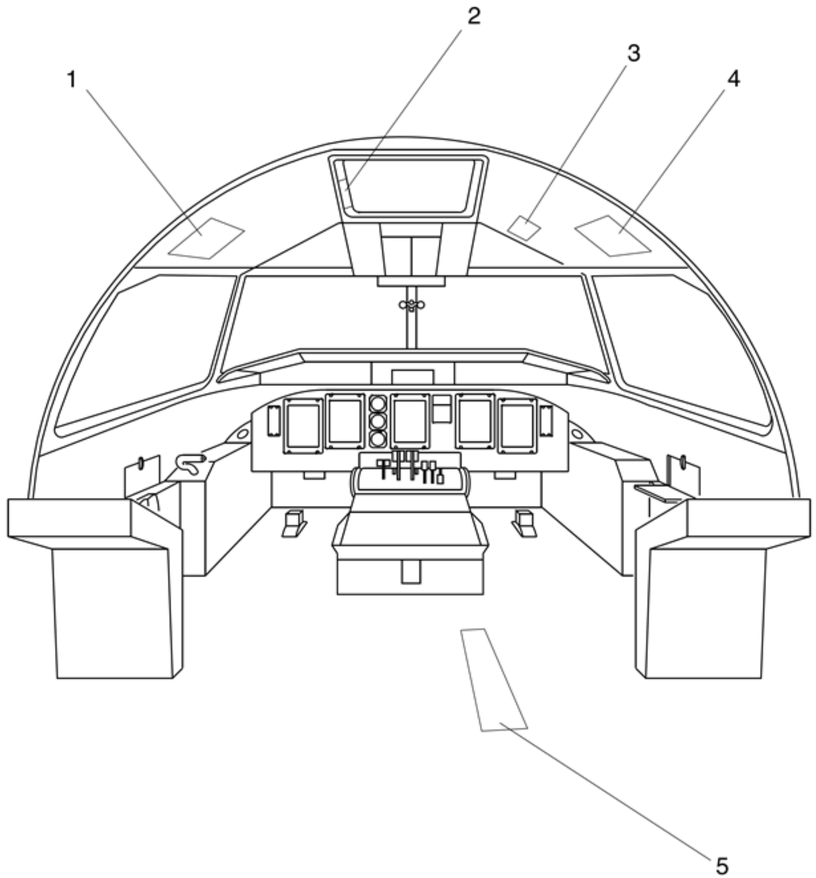


Figure 25-33 Flight Compartment Emergency Equipment – Forward

Axe, Fire

Refer Figure 25-34

The single bladed fire axe is installed on the flight compartment bulkhead behind the pilot's seat. It is secured to the bulkhead by a Velcro strap.

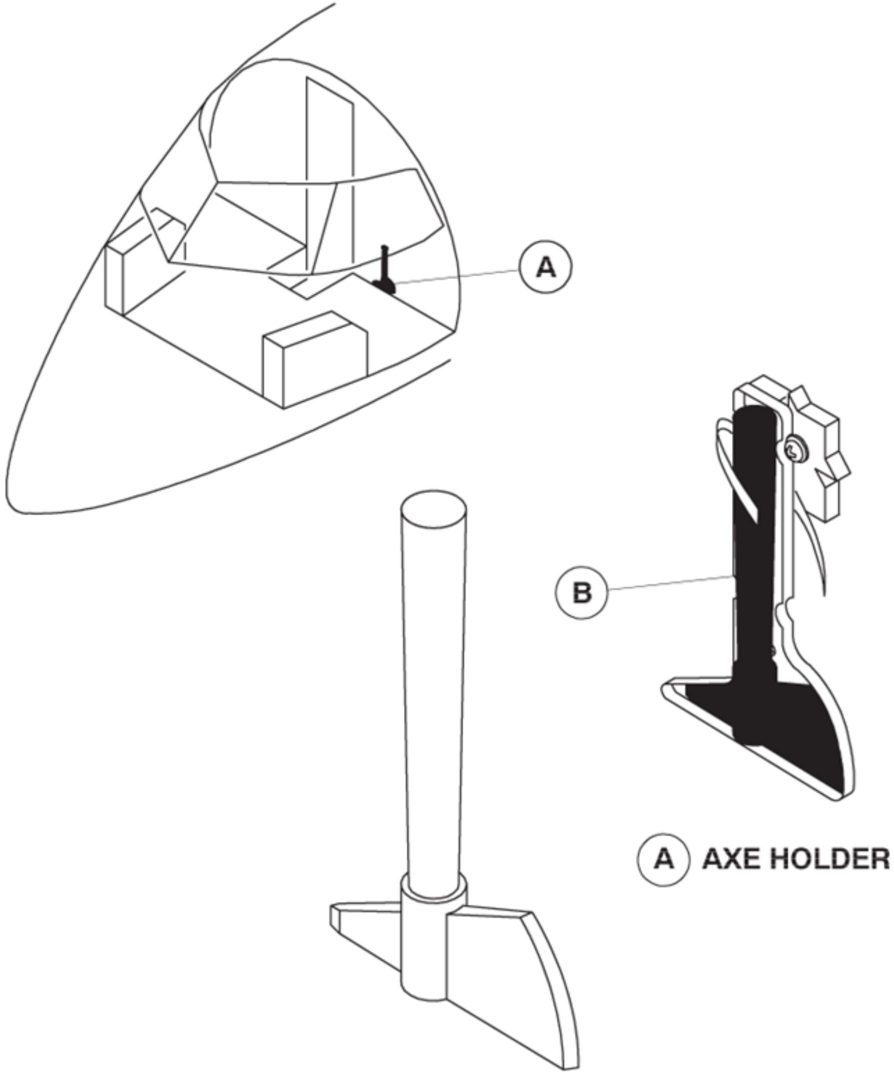


Figure 25-34 Fire Axe – Location

Flashlights

Refer Figure 25-35

There are two flashlights installed on the aft flight compartment bulkhead behind the pilot's seat. They are powered by two size D batteries and secured in their housing by a Velcro strap.

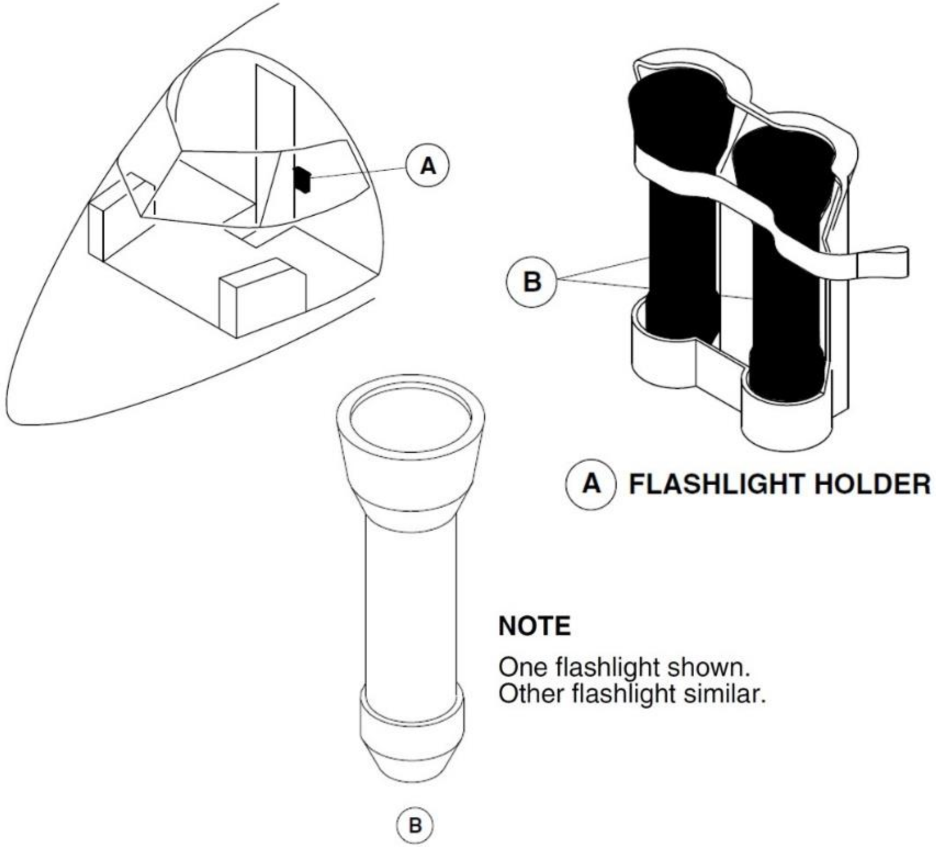


Figure 25-35 Flash Lights – Location

Refer Figure 25-36

On aircraft with Modsum 4−458593 or 4−458729 or 4−459025 incorporated, two flashlights are installed on the left hand bulkhead of the flight compartment behind the pilot's seat. The flashlights are secured in a retention bracket. They are powered by 'Model 213' battery pack. Activation of the flashlight is automatic when it is removed from the retention bracket and deactivated when re−installed.

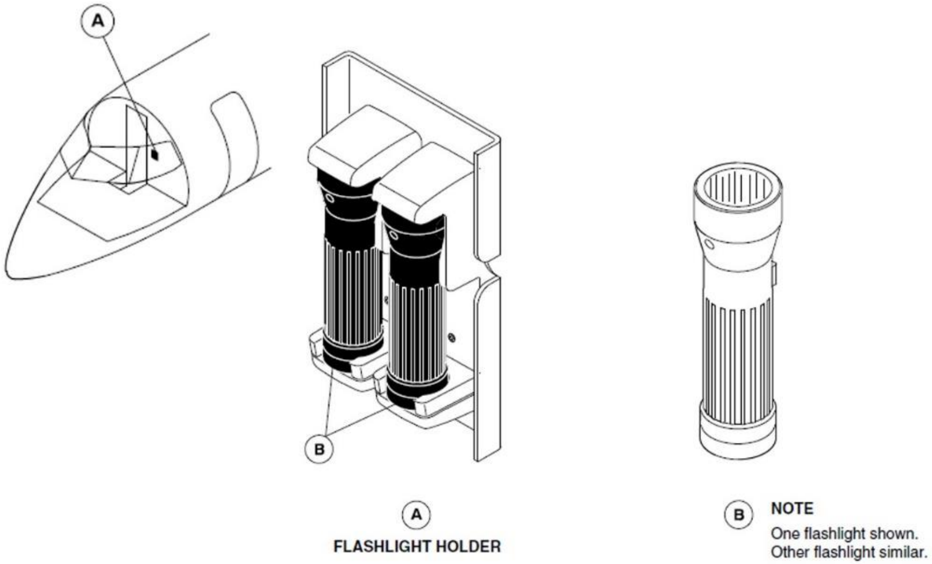


Figure 25-36 Flash Lights – Location Modsum 4−458593 or 4−458729 or 4−459025 incorporated

Rope, Escape

Refer Figure 25-3

The escape rope is used by the flight crew after evacuating the flight compartment through the emergency escape hatch. It is 10 feet (3.05 meter) long web strap with 8 knots throughout its length. It is connected to a shoulder bolt installed in the roof structure.

The rope is coiled to fit in a box located at the left side of the escape hatch. The cover of the box is secured by Velcro and labelled EMERGENCY ESCAPE ROPE STOWAGE.

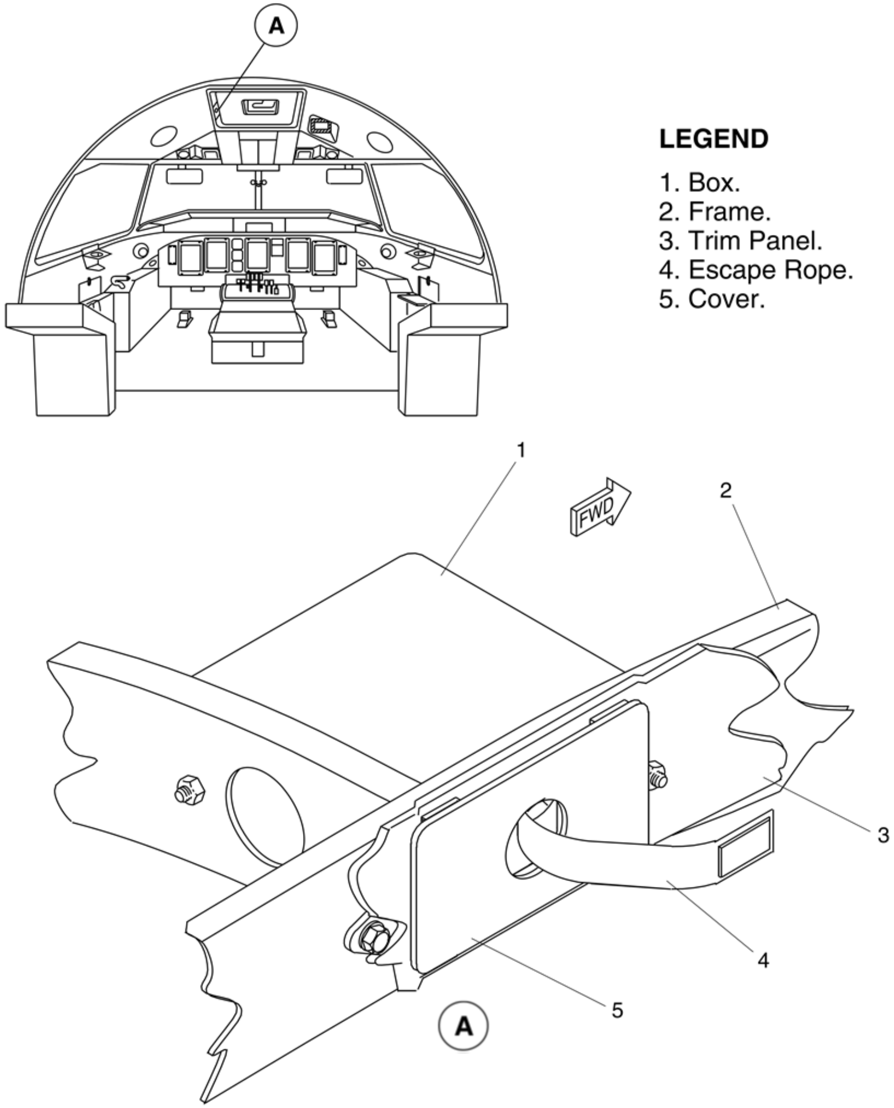


Figure 25-37 Escape Rope – Location

Hood, Smoke

Refer Figure 25-38

The smoke hood is a self-contained, portable, disposable breathing device installed in a quick release stowage on the aft flight compartment bulkhead. The system will supply oxygen for approximately 15 minutes.

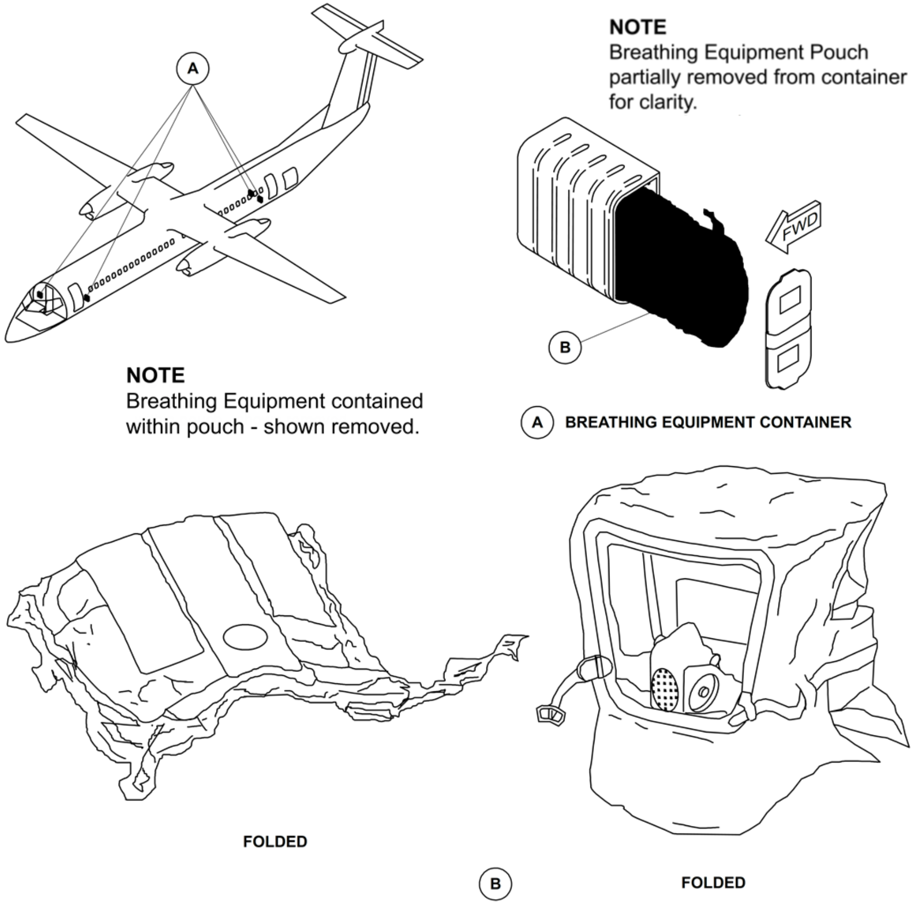


Figure 25-38 Smoke Hood – Location

Megaphone

On aircraft with SB 84−25−189 incorporated, one megaphone is installed on the left hand side bulkhead of the flight compartment behind the pilot's seat.

Crew Headset

Refer Figure 25-39

On aircraft with Modsum 4−459933 incorporated, one crew headset is installed on the upper aft bulkhead of the flight compartment behind the pilot's seat and two headsets in the roof structure.

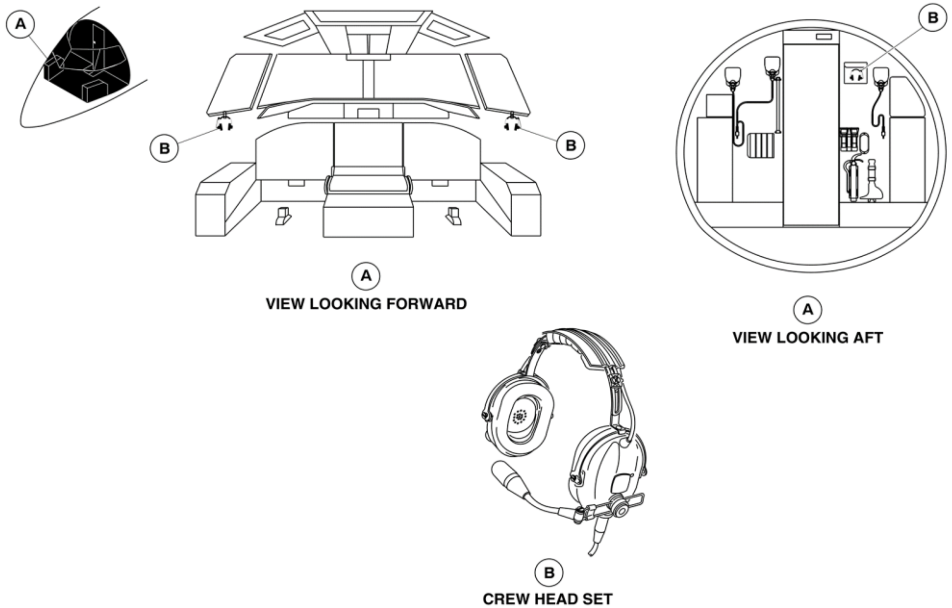


Figure 25-39 Crew Headset – Location

Passenger Compartment Emergency Equipment

Refer Figure 25-40

The emergency equipment is installed in the emergency equipment stowage compartments in the passenger cabin. They are used when an emergency situation arises during flight.

The locations of the five compartments are:

- One compartment in the forward draft bulkhead, aft of the airstair door
- Two compartments on the forward face of the aft draft
- Bulkhead, on either side of the aisle
- One compartment under the forward flight attendant seat
- One compartment under the aft flight attendant seat

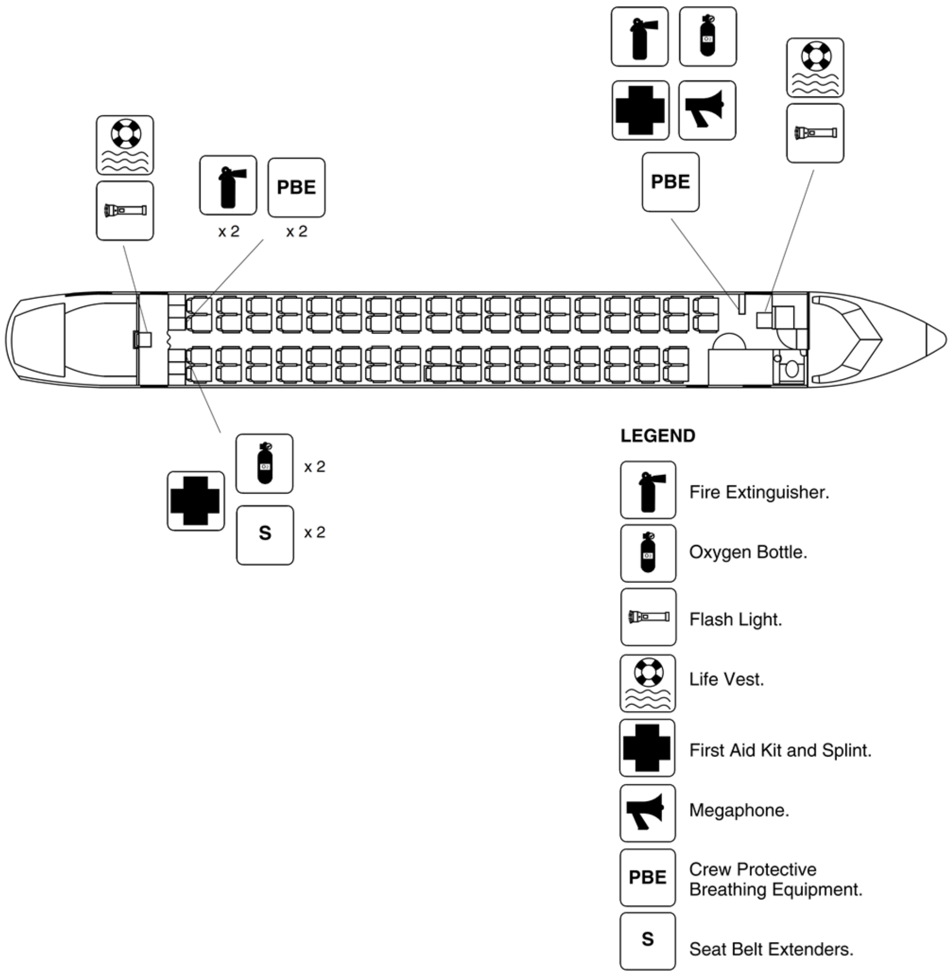


Figure 25-40 Passenger Compartment Emergency Equipment – General Layout

Refer Figure 25-41

The emergency equipment stowage compartment in the forward draft bulkhead has:

- One fire extinguisher
- One oxygen bottle with two Puritan Bennet masks
- One PBE unit
- One first aid kit
- One megaphone.

First Aid Kit

There are two first aid kits located in the passenger compartment. One kit is located in the forward emergency equipment compartment, and the other kit is located in the right rear emergency equipment compartment.

The first aid kits are sealed to ensure their completeness and do not contain unsealed consumable items.

Megaphone

The battery operated megaphone is installed in the forward draft bulkhead. The megaphone allows the flight attendant to communicate instructions to the passengers during emergencies when excessive noise may be present.

In optional configurations the megaphone may be installed in forward left overhead bin or in the forward wardrobe.

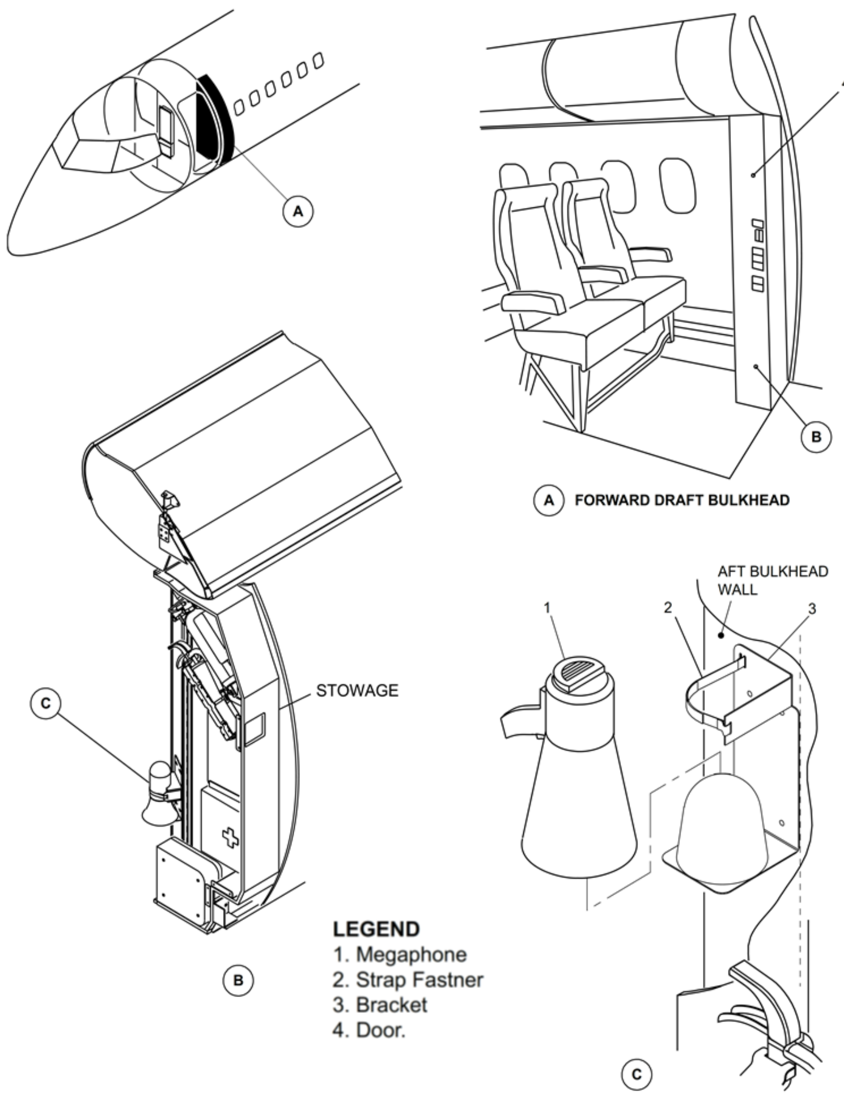


Figure 25-41 Emergency Equipment – Forward

Draft Bulkhead

Refer Figure 25-42

The emergency equipment stowage compartment on the left side of the aft draft bulkhead has two Flag fire extinguishers and two PBE units.

The right side compartment has two oxygen bottles with Puritan Bennet masks and one first aid kit.

Each flight attendant station has a life vest and a flashlight stored under the fold−up seat and protected by a quick release panel. All five emergency equipment stowage compartments are clearly placarded as to their contents.

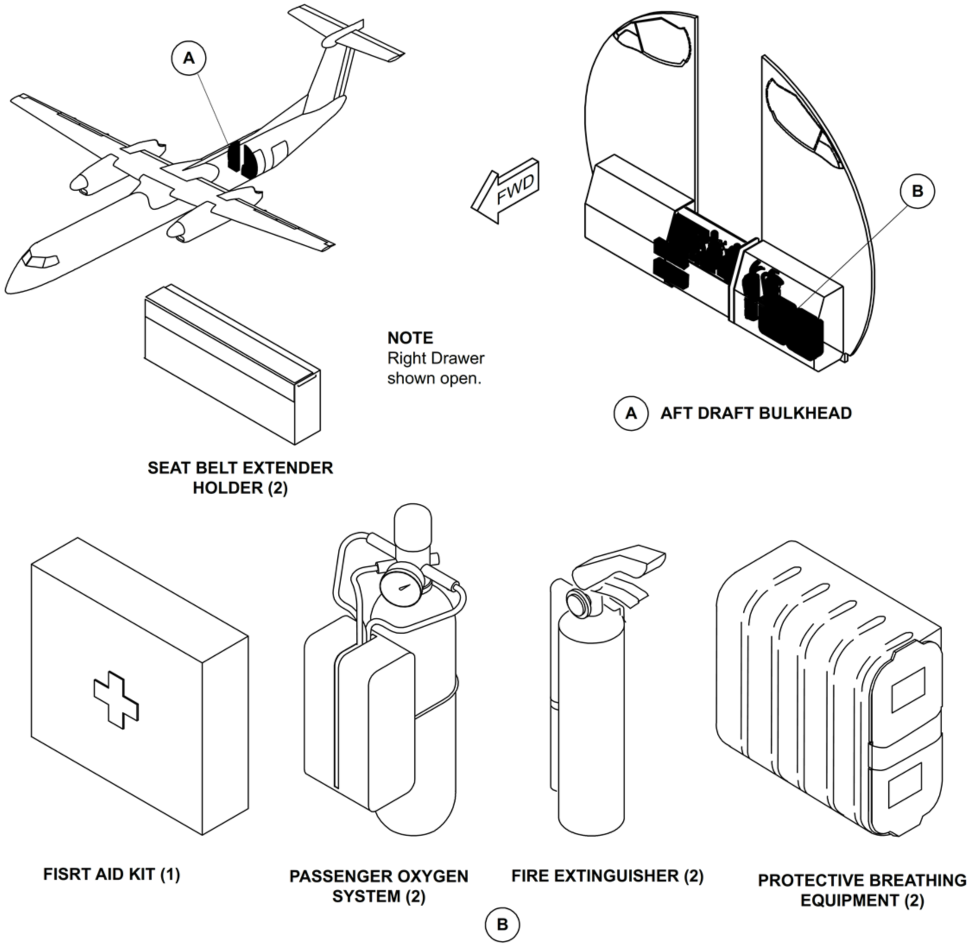


Figure 25-42 Emergency Equipment – Aft Draft Bulkhead

Flashlights

Refer Figure 25-43 & Refer Figure 25-44

There are two flashlights stowed in compartments located under the flight attendants' seats at the front and rear of the passenger compartment.

They are secured in the stowage by velcro straps and powered by two size D batteries.

Life Vests

There are two life vests, stowed in compartments located under the flight attendants' seats, at the front and rear of the passenger compartment. The vests are identical to those located in the flight compartment. Passenger life jackets are available only as a customer option.

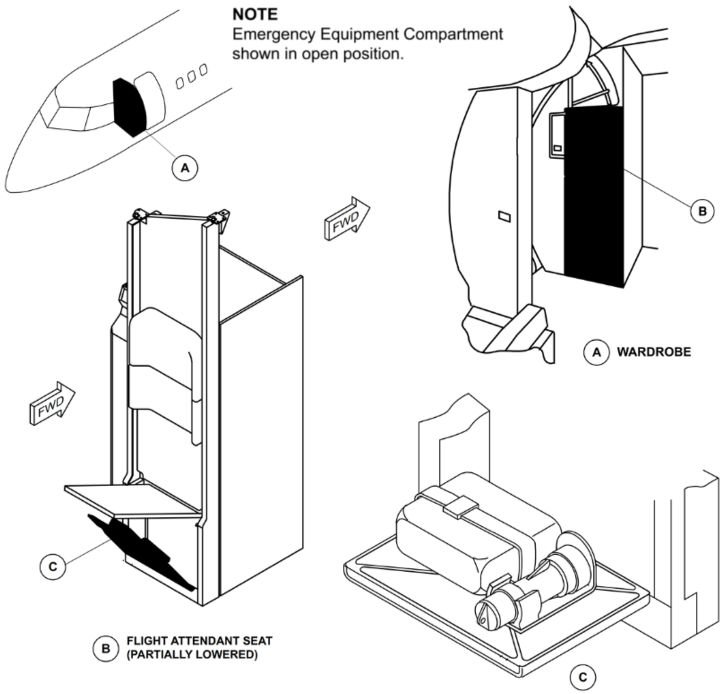


Figure 25-43 Emergency Equipment – Forward Flight attendant Seat

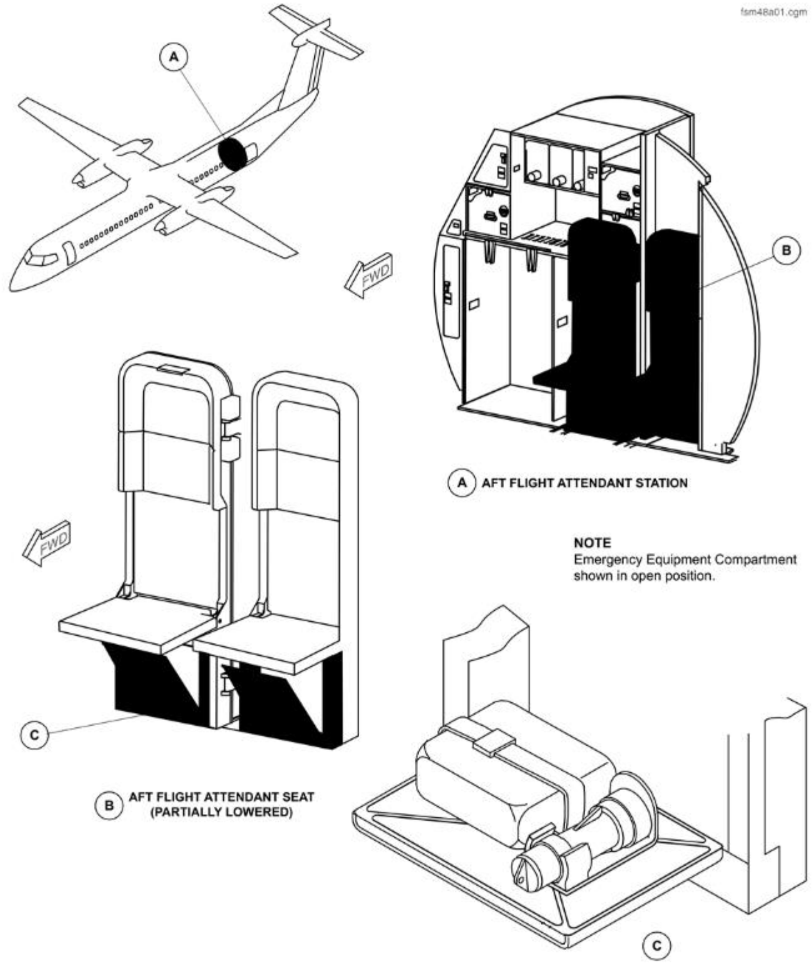


Figure 25-44 Emergency Equipment – Aft Flight attendant Seat

Dam, Ditching–Airstair Door

Refer Figure 25-45

The ditching dam prevents water from flowing into the cabin, when the airstair door is opened after an aircraft ditching. With the aircraft floating left wing down, the airstair door exit threshold will be several inches underwater. In this case, the ditching dam will raise the freeboard.

The ditching dam is a composite panel with rollers installed on each corner. The dam is located above the forward airstair door and is latched in place. The dam, when not in use, forms the ceiling panel above the door and is retained in tracks forward and aft of the entryway.

The rollers installed on the dam fit in the tracks along each side of the door. When required, the dam is released from its locking clips and pulled down the tracks to rest on the floor, where it is latched in place. This creates a 16 in. (406 mm) high barrier 13 in. (330 mm) inboard of the airstair door threshold.

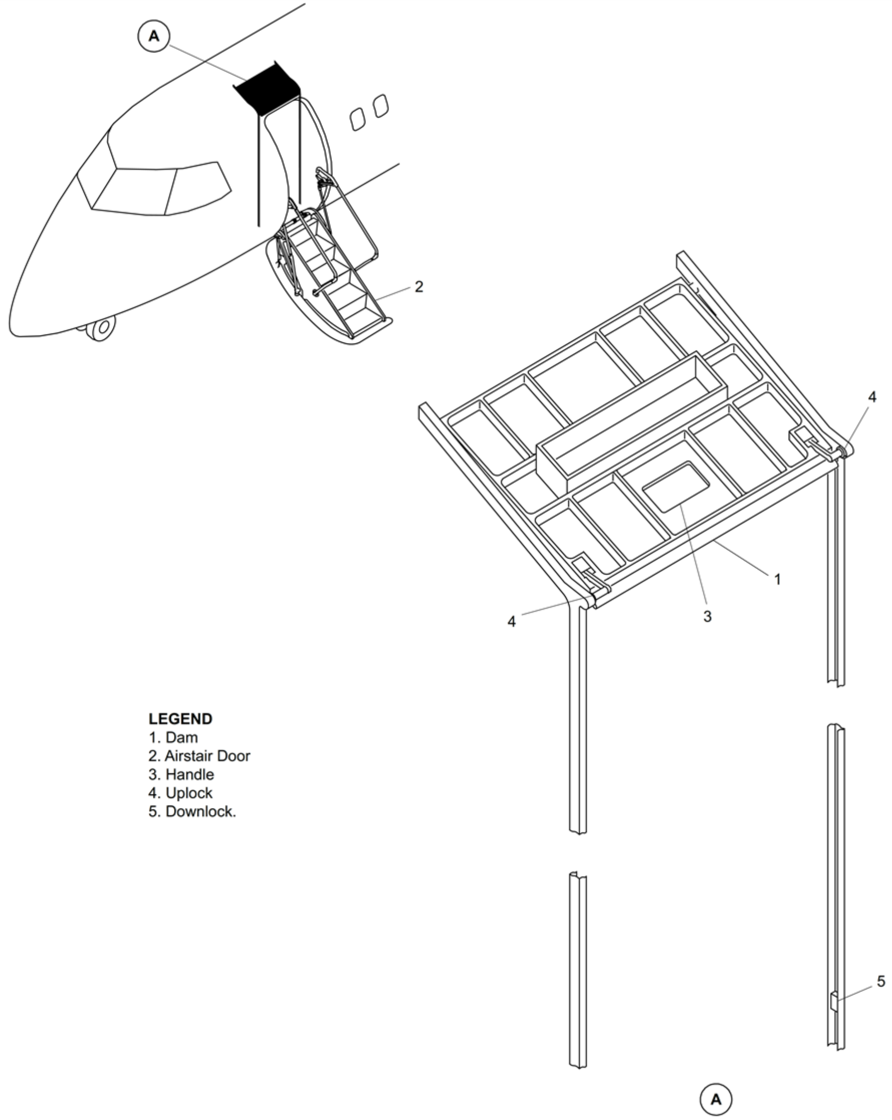


Figure 25-45 Air Stair Door Ditching Dam

Dam, Ditching–Type 1 Emergency Exit Door

Refer Figure 25-46

On aircraft with Modsum 4Q458968 incorporated, the ditching dam is a composite panel installed in the bottom part of the type I emergency exit door. The emergency exit door is located at the forward RH side of the cabin.

During normal flight mode, the ditching dam handle is in horizontal position and the ditching dam is attached to the type I emergency exit door using four pin brackets which are installed on the ditching dam structure. The guide pins on the door rest inside the pin brackets and the lock pins are retracted. For ditching dam evacuation, the handle on the ditching dam rotated 60° clockwise to engage the lock pins on either side of the door and guide pins are retracted. This action will lock the ditching dam with the door surround structure and release it from the type I emergency exit door.

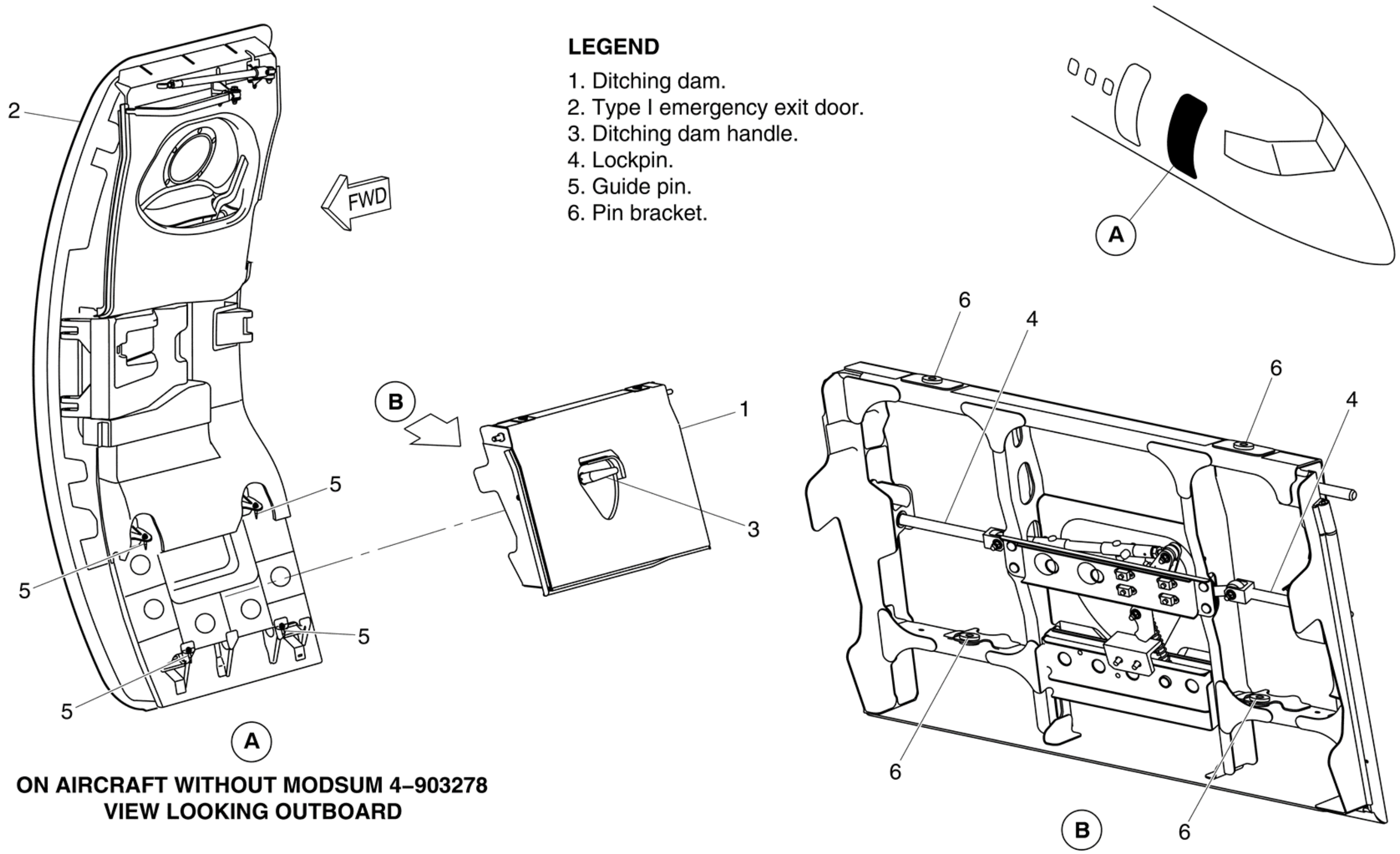


Figure 25-46 Type 1 Emergency Exit Door Ditching Dam

Refer Figure 25-47

On aircraft with Modsum 4−903278 incorporated, a handle cover is installed on the ditching dam to cover the handle. A ball stud clip and a ball stud are installed to engage th handle cover over the ditching dam. The ball stud clip is installed on the handle cover and the ball stud is installed on the ditching dam structure.

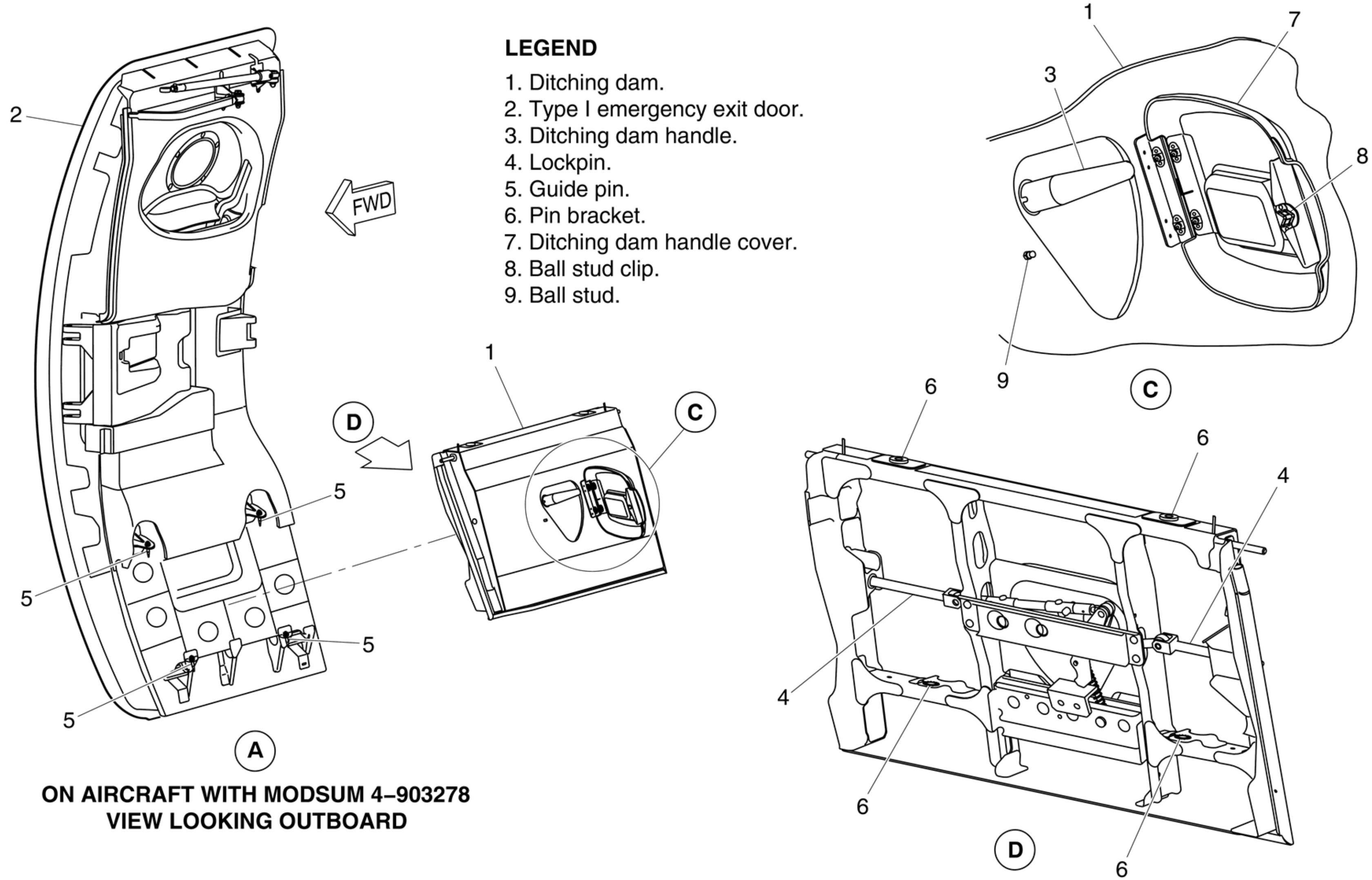


Figure 25-47 Type 1 Emergency Exit Door Ditching Dam Modsum 4−903278 incorporated

Emergency Locator Transmitter (ELT)

Introduction

The Emergency Locator Transmitter (ELT) automatically transmits a modulated radio signal to aid in locating an aircraft in distress. The fixed ELT is installed in the dorsal fin of the aircraft.

On aircraft with ModSums 4−458047 OR 4−458134 OR 4−458174 OR 4−457986 OR 4−458228 OR 4−458082 OR SB84−25−83 OR 4−458788 incorporated, an additional portable ELT (KANNAD 406 AS) is supplied.

The portable ELT (KANNAD 406 AS) is installed in the most forward RH overhead bin in addition to the fixed ELT installed in the dorsal fin of the aircraft.

On aircraft with ModSum 4−458008 incorporated, an additional portable ELT (ELTA ADT 406 S) is supplied. The portable ELT (ELTA ADT 406 S) is installed in the forward wardrobe in addition to the fixed ELT installed in the dorsal fin of the aircraft.

The portable ELT is a standalone system intended for activation by the survivors after a crash. The portable ELT can be removed by the survivor and activated manually to help the Search and Rescue (SAR) operation.

The description and operation of the portable ELT (KANNAD 406 AS) is given under the title Portable Emergency Locator Transmitter (Kannad 406 AS).

The description and operation of the portable ELT (ELTA ADT 406 S) is given under the title Portable Emergency Locator Transmitter (ELTA ADT 406 S).

General Description

Refer Figure 25-48

The fixed ELT system provides the aircraft with an independent and automatically latched continuous distress signal transmission. An inertial force, similar to those experienced in a crash, activates the transmitter. The ELT system also supplies the flight crew with the functions that follow:

- Visual annunciation of operation
- System reset capability when operating due to an automatically−latched initiation
- Manual system operation.

The ELT system has the components that follow:

- ELT
- Antenna and Cable Assembly
- Switch, Remote Control

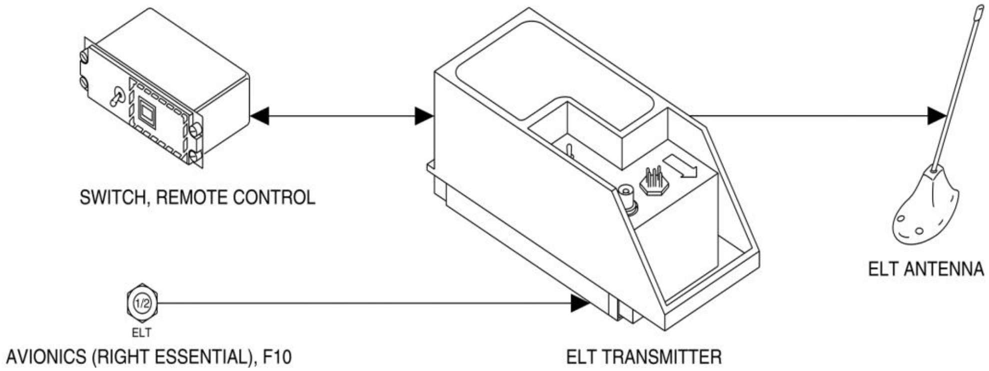


Figure 25-48 ELT – Block Diagram

Detailed Description

Refer Figure 25-49

An inertia switch within the ELT automatically activates the system when subjected to excessive longitudinal inertia forces. The ELT transmits at the assigned international civil and military emergency frequencies of 121.5 MHz and 243.0 MHz.

The transmission modulates with a distinctive downswept audio tone. The ELT power supply is independent of the aircraft electrical system. It uses an internal 7.5 Vdc Alkaline and Manganese battery pack.

The Remote Control Panel (when connected) is also energized by this battery pack.

When activated, the ELT may be switched off by momentarily placing the remote switch in the RESET position. A remote reset is possible only if a 28 Vdc signal is supplied through a resistor to the ELT.

The ELT system normally operates in an inactive state in order to maximize reliability. It does not supply specific failure indications or annunciation. Periodic operational tests gives the operational status of the ELT system.

For normal operation, the three position remote switch, in the flight compartment and the master switch located on the ELT are set to AUTO.

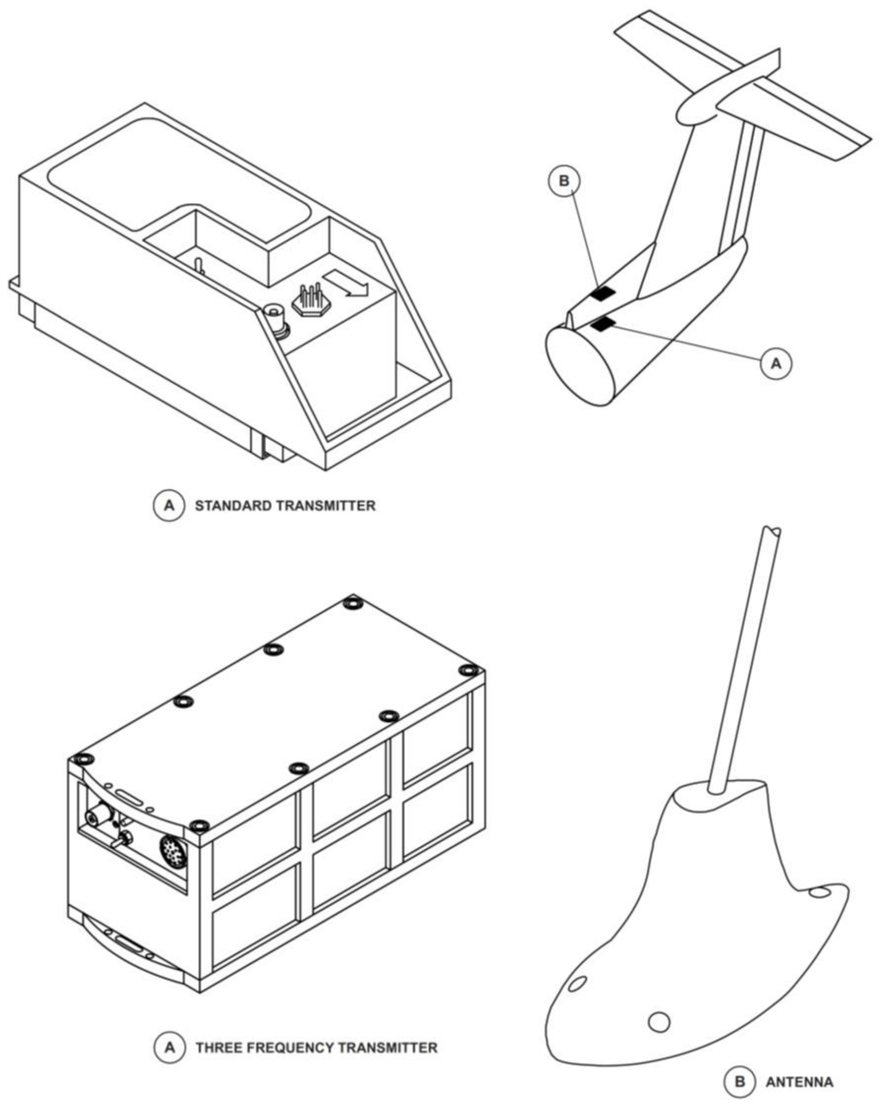


Figure 25-49 ELT – Locations

ELT - Component Description

ELT

Refer Figure 25-50

The ELT is a self−contained, battery−powered unit. When activated either automatically or manually, the ELT transmits an amplitude modulated down−sweeping audio tone on the international civil and military frequencies of 121.5 MHz and 243.0MHz respectively. The carrier frequency is pulse−modulated with a down−sweeping audio tone. The audio tone sweeps within 1600 to 300 Hz, at a 3 Hz rate.

An inertia switch is connected to the transmitter printed circuit board. It is activated by longitudinal inertia forces between 5 and 7 G (gravity) acceleration. The output pulse of the inertia switch triggers a Silicon Controlled Rectifier (SCR) transmitter activation latch. In turn, the SCR enables the battery pack to power the transmitter.

Once activated the transmitter remains operative until OFF bias to the transmitter activation latch is applied. Momentarily selecting RESET on the remote switch biases the transmitter to off.

The ELT connects directly to the Antenna through the coaxial antenna cable assembly. The battery pack that powers the transmitter is attached to the base of the ELT. It is easily accessed by removing the ELT base plate and hen disconnecting the battery connector. A label attached to the side of the ELT shows the expiry date of the battery pack currently installed in the ELT.

The ELT is housed within a high−impact, fire retardant waterproof orange coloured case. The case is designed to withstanding forced landing and crash environmental conditions in an operable condition. The ELT attaches to a mounting bracket.

The mounting bracket secures to the airframe directly below the ELT Antenna in the dorsal fin at station X+874.5, Y0 and Z+200.1. Access to the ELT is gained through removable panels on the top, and the LH and RH sides of the dorsal fin. The Access Panel Locations are: 321 CT, 321 BR 321 BL. The ELT is installed with the "Direction of Flight" arrow on the ELT pointing forward.

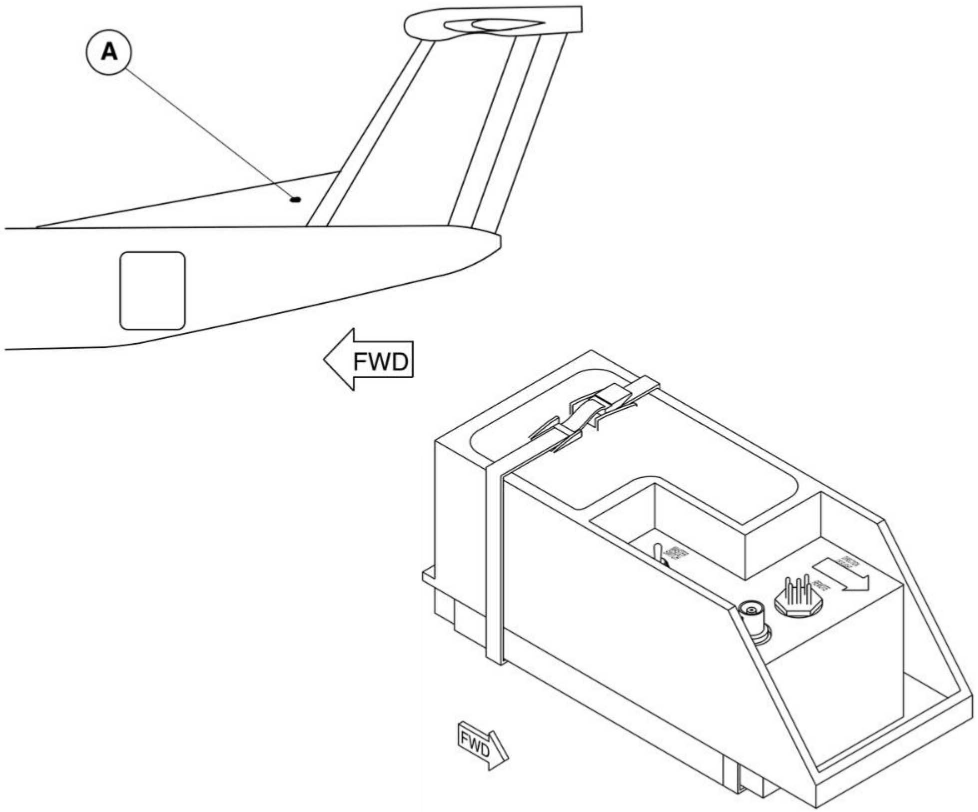


Figure 25-50 ELT

Antenna

Refer Figure 25-51

The antenna operates at 121.5 and 243.0 MHz frequencies. It radiates in an omni–directional azimuth pattern. The ELT Antenna is a rigid rod antenna weighing 6 ounces.

The ELT antenna fits to the airframe immediately below the outer skin on the dorsal fin. It is forward of the vertical stabilizer at station X+874.7, Y0 and Z+210.8. The antenna protrudes through the aircraft skin and is sealed by a grommet. The antenna connects directly to the ELT through the coaxial antenna cable assembly.

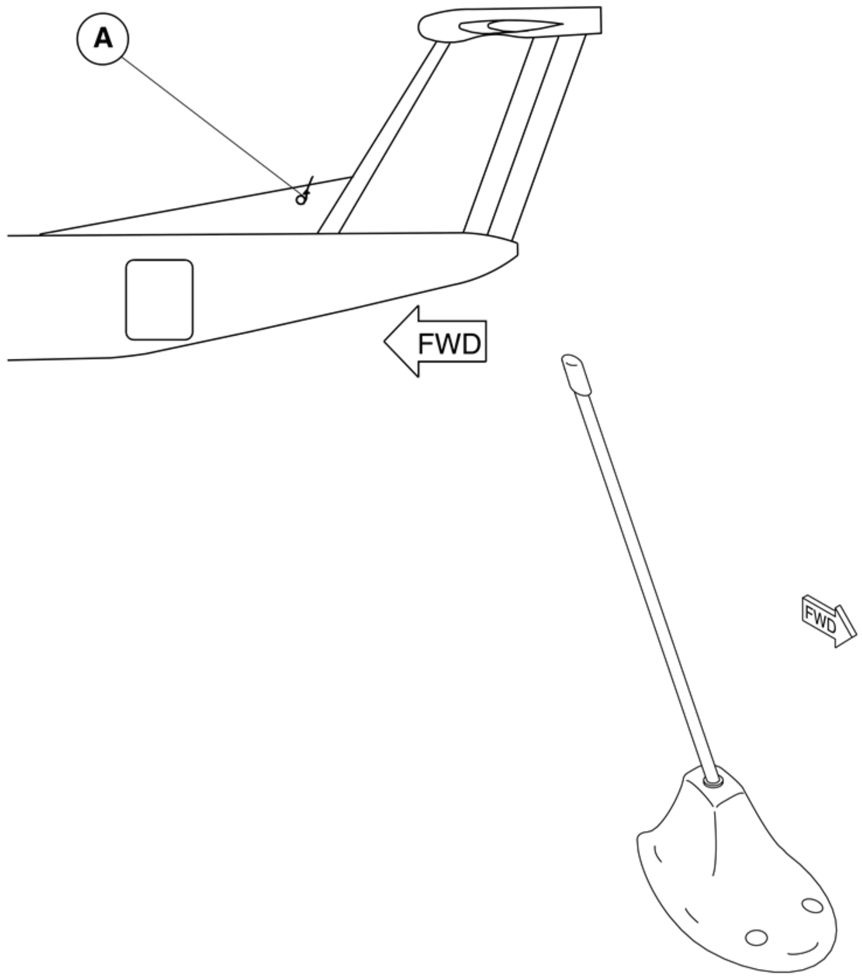


Figure 25-51 Antenna – Location

Switch, Remote Control

Refer Figure 25-52

The remote switch is a three position rocker switch that provides remote selection of all the Emergency Locator Transmitter (ELT) functions.

A red warning light in the switch indicates when the Emergency Locator Transmitter (ELT) is operating.

The remote switch is part of a combined Flight Data Recorder (FDR) and Emergency Locator Transmitter (ELT) panel assembly. The panel is located on the overhead console in the flight compartment.

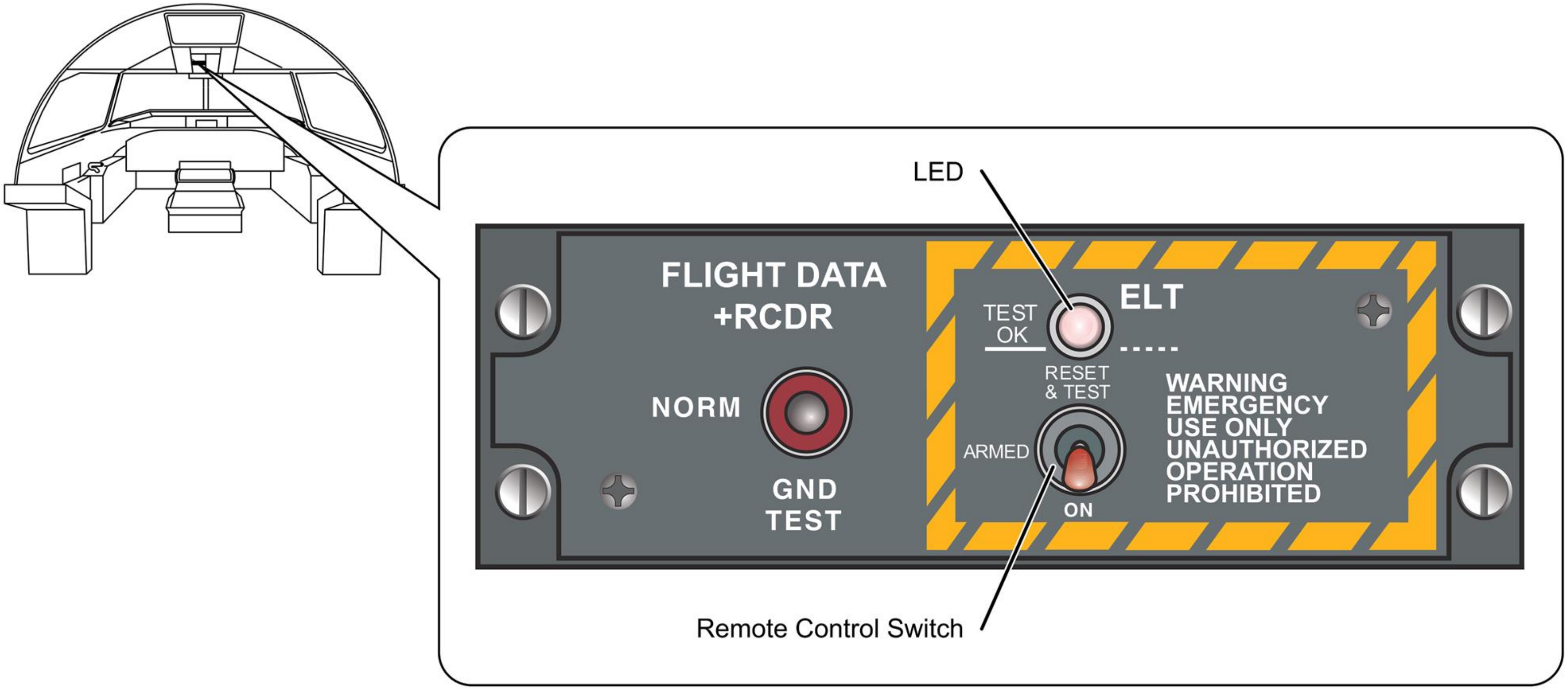


Figure 25-52 Remote Control Switch – Location

Portable Emergency Locator Transmitter (KANNAD 406 AS)

Refer Figure 25-53

The portable ELT is attached with a Velcro strap tightly to the mounting bracket installed in the most forward face in the RH overhead bin. The ELT is installed with the front panel facing towards the outboard. The portable ELT is installed with a floating collar to enable the ELT to float if used in water. This floating collar can be removed if the ELT is attached to a life raft or buoyant part. The portable ELT is not installed with a G−Switch (shock detector).

The portable ELT system has the components that follow:

- Emergency Locator Transmitter with Antenna
- Mounting Bracket with Locking Pin
- Switch, Indicator and Buzzer
- Programming Dongle.

The portable ELT is designed to transmit on three frequencies 121.5, 243.0 and 406.0 MHz. The two frequencies 121.5 MHz and 243.0 MHz are international civil and military emergency frequencies for homing. The transmission on frequencies 121.5 MHz and 243.0 MHz are modulated with a distinctive downsweeping audio tone. The ELT transmission on 406.0 MHz frequency is used by the COSPAS−SARSAT satellites to pinpoint the location and to determine identity of the aircraft in distress.

The portable ELT can transmit 121.5/243.0 MHz frequencies for 48 hours minimum at −20°C with new batteries. The portable ELT can transmit 406.0 MHz distress signal for 24 hours minimum at −20°C. The transmission on 406.0 MHz is automatically stopped after 24 hours to keep the battery power for 121.5/243.0 MHz homing frequency transmissions as long as possible to help the rescue operations.

When the ELT is active, the transmitter operates continuously on
121.5 and 243.0 MHz. At 121.5/243.0 MHz, the portable ELT transmits 100 to 400 mW power on each frequency. The 121.5 and 243.0 MHz transmission frequencies are modulated with down sweeping audio frequency from 1420 Hz to 490 Hz with a repetition rate of 4 Hz. During the first 24 hours of operation, the 406.0 MHz signal is transmitted for 440 short messages (ms) every 50 seconds to the COSPAS−SARSAT satellites. The output power on 406.0 MHz is approximately 5 W (approximately 50 times more powerful than the VHF signal).

The portable ELT has a flexible antenna. The antenna is attached to the front face of the portable ELT unit with a DNC or TNC connector. When it is inactive, the antenna is slid through the hole provided in the floating collar. To activate the portable ELT, keep the antenna in the vertical position. The ELT power supply is independent of the aircraft electrical system. It uses an internal Lithium Manganese battery pack. The ELT battery expiry date is fixed at 6 years from the date of manufacture. If no activation occurs during battery life time, the battery must be replaced every 6 years. The portable ELT is not connected to other aircraft components or wiring. An ARM−OFF−ON switch on the front face of the ELT controls the operation of the ELT. The locking pin is installed across the front face through the two holes provided on the portable ELT cover plate to prevent the inadvertent operation of the switch

The ELT front panel has the controls that follow:

- Position toggle switch ARM/OFF/ON
- TNC connector for the antenna.
- Visual indicator (red)
- DIN 12 connector for dongle or programming equipment Connection

The portable ELT unit switch has the positions that follow:

- ARM
- OFF
- ON

The switch position ARM is used for the self−test. The OFF/ON switch positions are used to de−activate/activate the portable ELT unit.

To operate the portable ELT, put the antenna in vertical position and put the switch to ON position. The portable ELT operates as follows:

- The ELT starts with the self−test sequence.
- The 121.5/243.0 MHz transmission starts immediately after the self−test.
- The 406.0 MHz transmission starts after 50 seconds.
- During transmission, buzzer operates and visual indicator (red) flashes periodically.

Once activated the portable ELT can be de−activated manually by selecting the switch to OFF (center) position. The indicator in the front panel of the portable

ELT gives the indication on the working mode of the ELT as follows:

- After the self−test, long flash indicates that the ELT is operational and has no error.
- After the self−test, the series of short flashes indicate that the self test has failed.
- During the operation of the portable ELT, periodic flashes of the indicator indicate that the ELT is transmitting at 121.5/243.0 MHz frequencies. One long flash (1 in every 50 seconds) indicates 406.0 MHz transmission by the ELT.

The buzzer gives audio information on the portable ELT working mode as follows:

- The buzzer produces continuous tone during the self−test.
- The buzzer produces 2 beeps per second during 121.5/243.0 MHz transmission and remains silent when ELT transmits 406.0 MHz signal.

The programming dongle is programmed with hexadecimal operator code and is attached to the ELT. If the ELT is removed from the aircraft and replaced, the programming dongle shall be transferred to the new ELT and a maintenance dongle shall be installed on the removed ELT.

Training Information Points

Direct manual activation of the Emergency Locator Transmitter (ELT) is possible. The operator must disconnect the remote control cable from the ELT and turn the master switch ELT from AUTO to the ON position. Testing times are allocated in Canada. The ELT may be tested within 5 minutes of the hour. Notify Air Traffic Control (ATC) of intentions. The duration of the test should not exceed 3 down sweeping audio tones or 1 second.

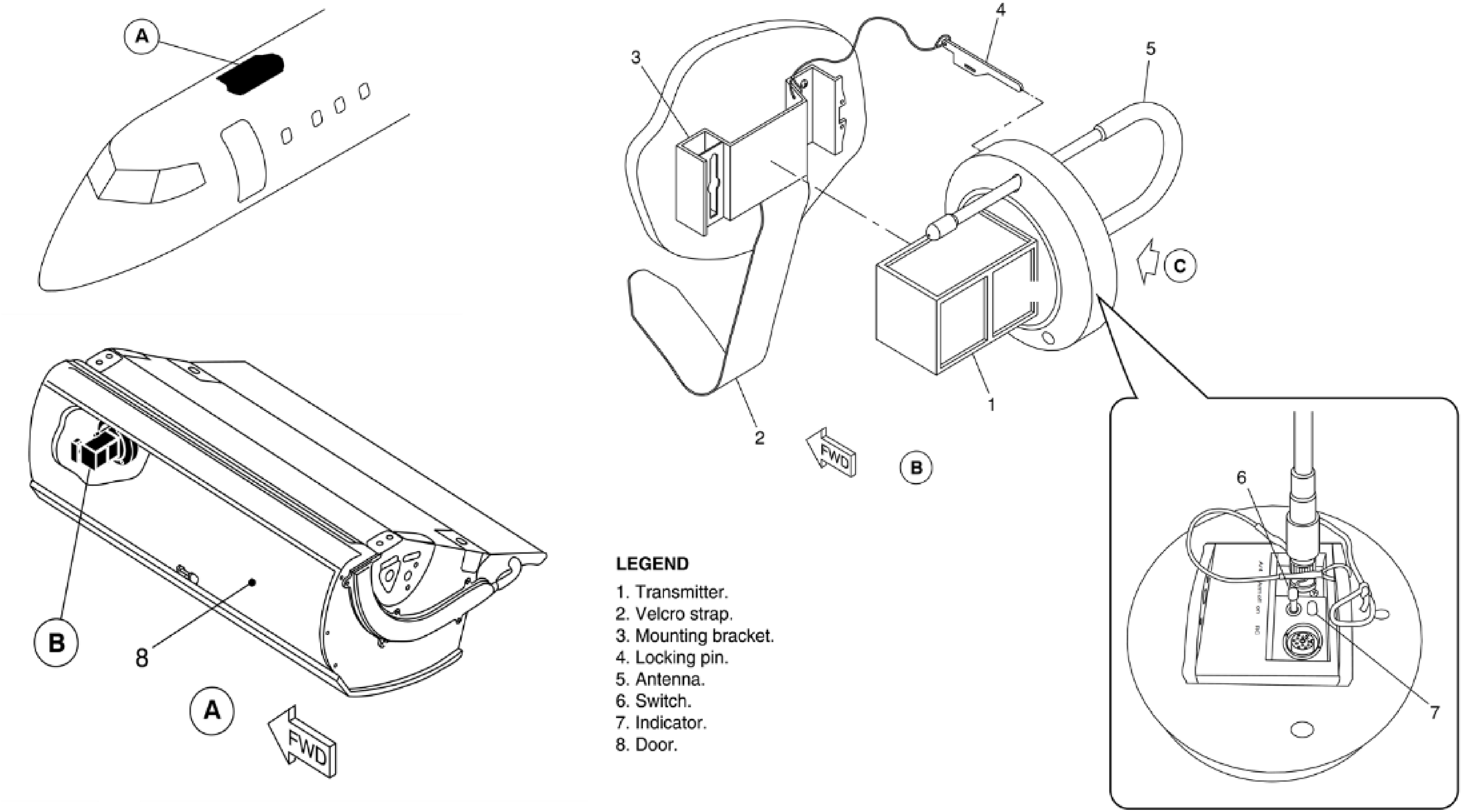


Figure 25-53 Potable ELT – Location

Controls and Indications

The remote switch is used to select these ELT operating modes:

- Remote AUTO
- Remote ON
- Remote RESET

Remote AUTO Mode: The ELT operates in the automatic mode when the master and the remote switches are selected to the AUTO position. The ELT transmits when the inertia switch activates. The red ELT monitor light located in the remote switch illuminates.

Remote ON Mode: The ELT can be manually activated in the event of an emergency by selecting the remote switch to the ON position. The ON selection overrides the automatic inertia switch. The monitor light located within the remote switch illuminates.

The Remote ON Mode selection provides a means to test the ELT during a preflight test. For test purposes, the ELT may be momentarily activated by placing the remote switch to the ON position. Returning the remote switch to the AUTO position terminates ELT operation. The monitor light goes out.

Remote RESET Mode: The remote RESET mode lets a reset of an inadvertent ELT activation. Momentarily placing the remote switch in the RESET position and then to the AUTO position re-arms the ELT and the ELT monitor light goes out.

Direct manual activation of the ELT is possible. By turning the master switch ELT from AUTO to the ON position. Testing times are allocated.

System Operation

ELT Modes of Operation

1. The ELT has two modes of operation:
   - Inactive mode – ELT armed
   - Active mode – ELT transmitting the emergency signals.

Automatic (normal) Operation

1. The ELT remote switch on the ELT control panel in the flight compartment is normally set to the ARM position. In this position the ELT is armed and ready for the active mode.

2. The ELT automatically goes into the active mode when the inertia G switch in the transmitter unit is closed due to an impact.

NOTE: When the ELT transmits the emergency signals, the red LED comes on and flashes continuously, and the buzzer gives out a distinct audible sound.

3. The ELT transmits the signals on the emergency frequencies of 121.5 MHz, 243.0 MHz and 406 MHz to help the search and rescue personnel to locate the aircraft.

4. The third signal (406 MHz) is transmitted every 50 seconds to a satellite. This signal is encoded with a digital message.

5. The ELT stops transmission of the emergency signals when it is reset.

6. After the ELT is reset with the remote switch, the red LED stops flashing, stays illuminated for approximately one second and then goes out.

7. When the ELT system senses a problem, the LED flashes a coded signal after the one second pulse (Refer the Operational Test of the ELT System).

Manual Operation

1. Manual operation of the ELT system is usually done for the test purposes or to activate the ELT in an emergency.

2. The ELT can be manually operated with the remote switch in the flight compartment or with the transmitter unit ON/ OFF switch in the tail section. The ELT must be connected with the coaxial cables to the antenna for operation

3. The ELT operates when the remote switch in the flight compartment is put to the ON position. When the ELT operates, it transmits the signals on the emergency frequencies of 121.5 MHz, and 243.0 MHz.

4. After 50 seconds of normal ELT operation, the third signal (406 MHz) is transmitted to the satellite, which is considered to be a valid distress signal.

5. To stop the ELT from transmission of the emergency signals, the ELT must be reset. Refer Reset the ELT.

6. The standard emergency frequency of 121.5 MHz and 243.0 MHz can be heard on the VHF and UHF communication systems that are in range and tuned to the emergency frequencies. The third signal (406 MHz) is encoded with a digital message and is sent to a satellite.

Reset the ELT

1. For manual reset, switch to position off on the ELT

2. To reset using the Remote Control Panel, the switch has to be in position "ARM" on the ELT than press the "RESET & TEST" on the remote control

Insulation

Introduction

Insulation is supplied for the flight compartment and cabin for the purposes of acoustic attenuation and thermal energy control.

Each blanket is constructed to fit as closely to structure as possible and require minimal, if any on−floor modifications. Each blanket is custom fit to a particular location and structural geometry.

General Description

Refer Figure 25-54 (Insulation – Fuselage Cross Section)

The insulation is installed in the areas that follow:

- Flight Compartment Insulation
- Passenger Cabin Insulation
- Baggage Compartment Insulation

The insulation package consists of primary blankets which fit in the wall cavities and top cover blankets to cover frames.

The primary blankets are composed of 1.5” of 0.6 lb. fiberglass and 0.5” polyimide foam. The top covers are made of 0.5” polyimide foam. The secondary blankets are made from the solimide foam (AC550, 0.55 pcf).

On aircraft with ModSum 4−457894 incorporated, primary blankets are composed of 1.0” of 0.6 lb. fiberglass and 0.5” polyimide foam.

The top covers are made of 0.5” polyimide foam. The secondary blankets are made from the fiber glass (Microlite AA, 0.6 pcf).

Each insulation bag has a covering of Orcon AN49R vapor barrier film or equivalent. They are heat sealed and constructed with drains that preclude buildup of moisture in the bags and which direct moisture outboard and down the bilges. The drain holes are located on the bottom bevel of each bag.

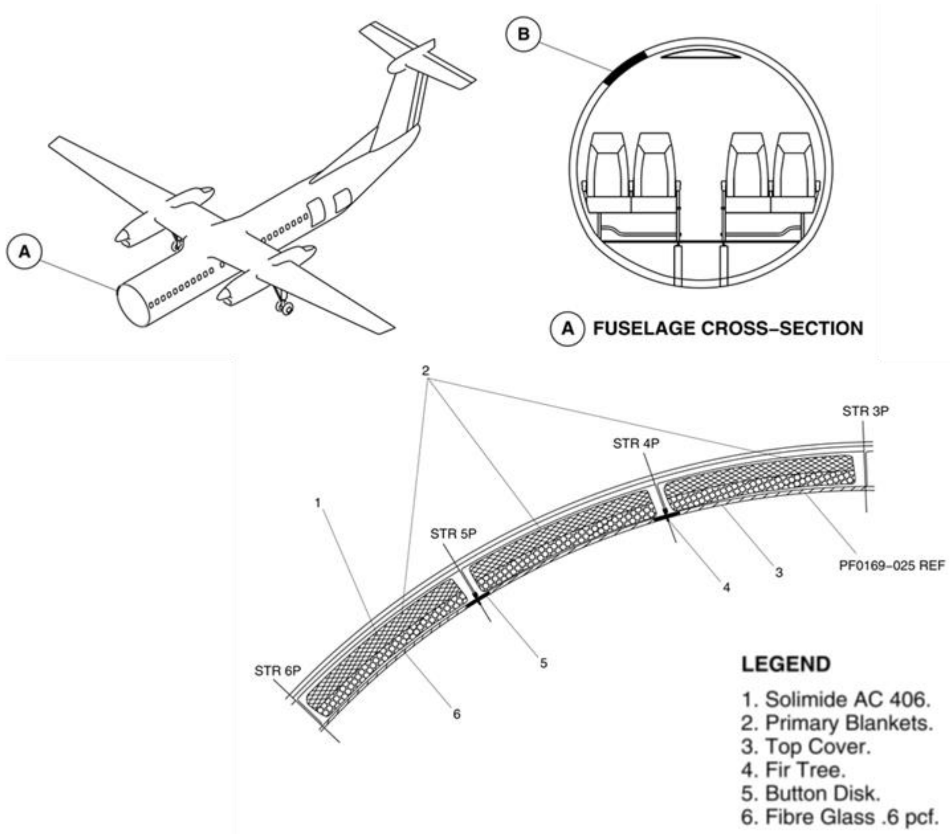


Figure 25-54 Insulation – Fuselage Cross Section

All blankets are closely tailored to the structure so that noise channels within the wall cavity are eliminated. Each blanket is identified with the precise location by frame station and stringer so that installation time is enhanced. This will also permit for easy removal for zonal inspection requirements on the aircraft systems and structure. The primary blankets are pressed−fit between frame stations and stringers. The top covers are installed with fir trees through sealed penetrations and secured with plastic tree clips.

Refer Figure 25-55, Refer Figure 25-56 & Refer Figure 25-57

For areas around doors and in door structure, individual cavities are filled with 3” fiberglass blankets and an overframe blanket provided to cover any bare metal structure.

All cut−outs and channels to accommodate structure and internal hardware are heat sealed with Orcon AN49R film or approved tape.

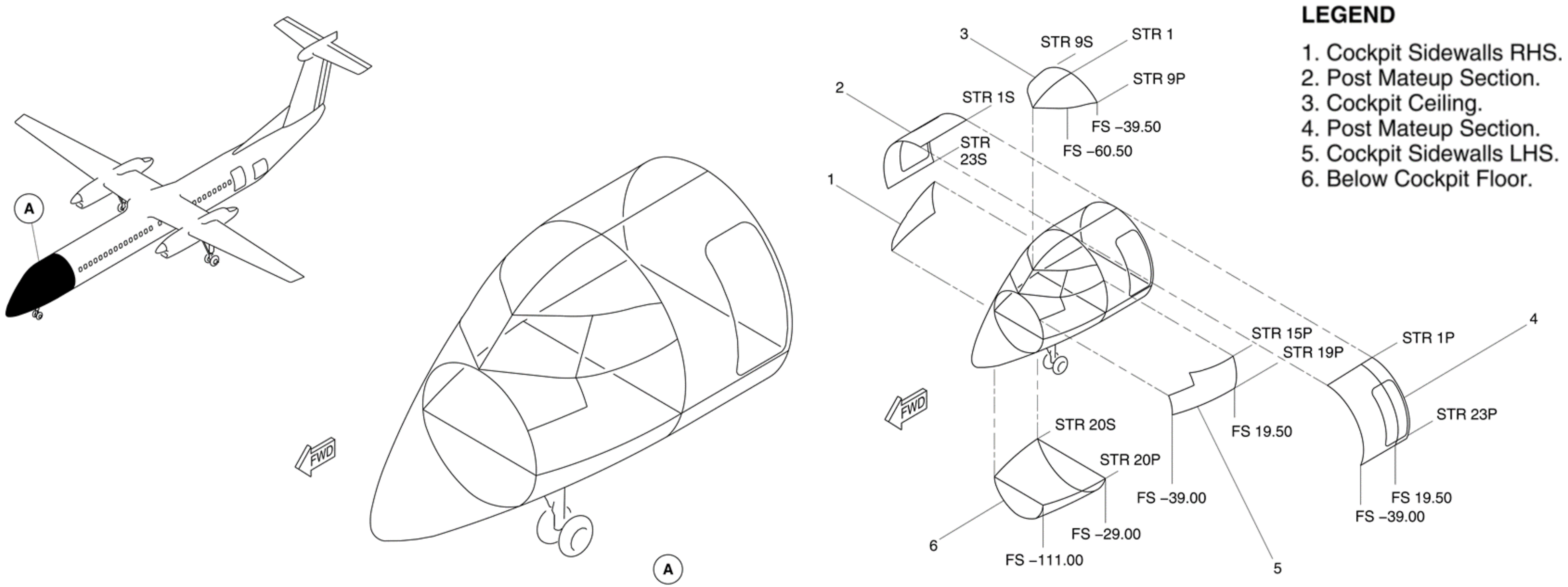


Figure 25-55 Flight Compartment Insulation

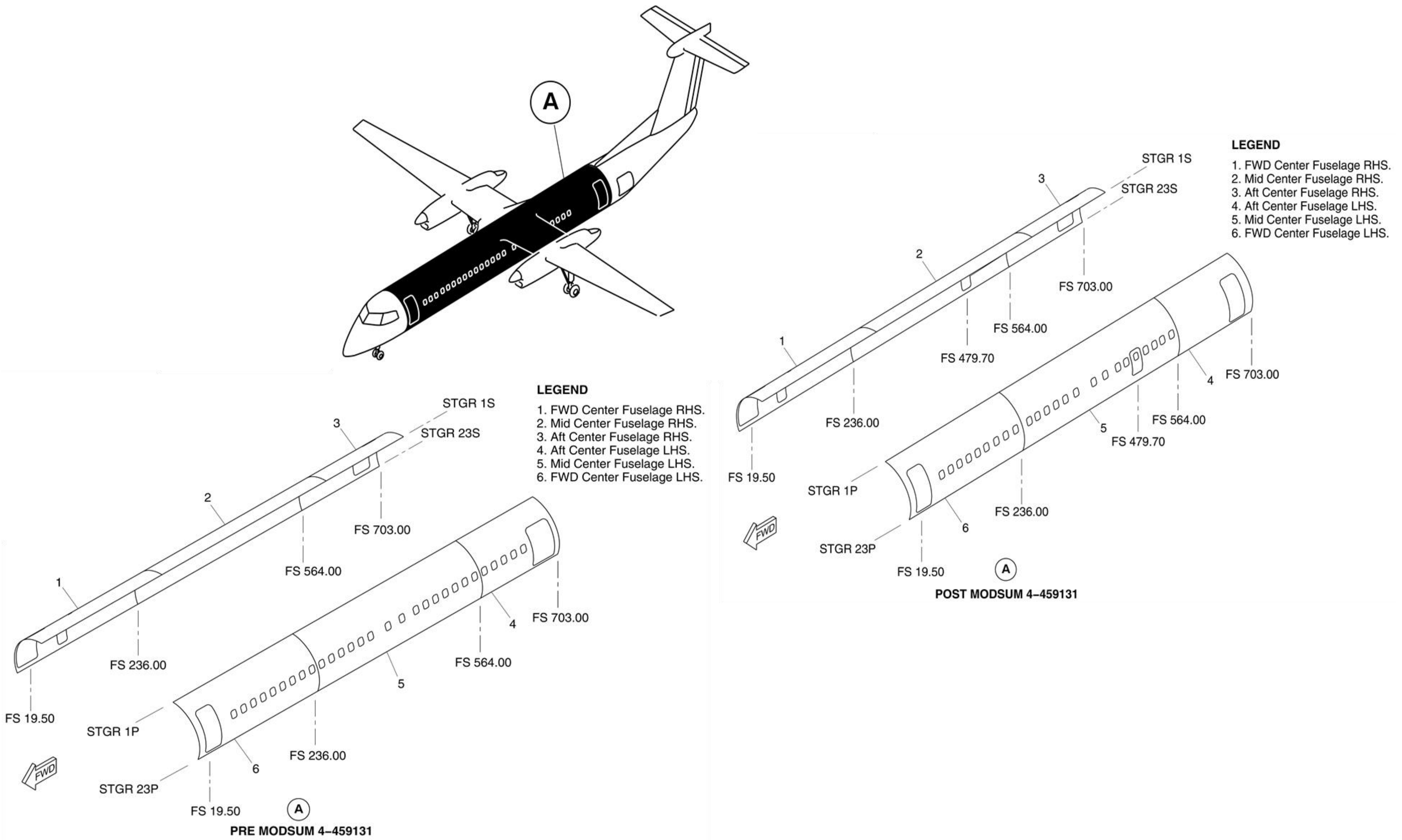


Figure 25-56 Main Fuselage Insulation

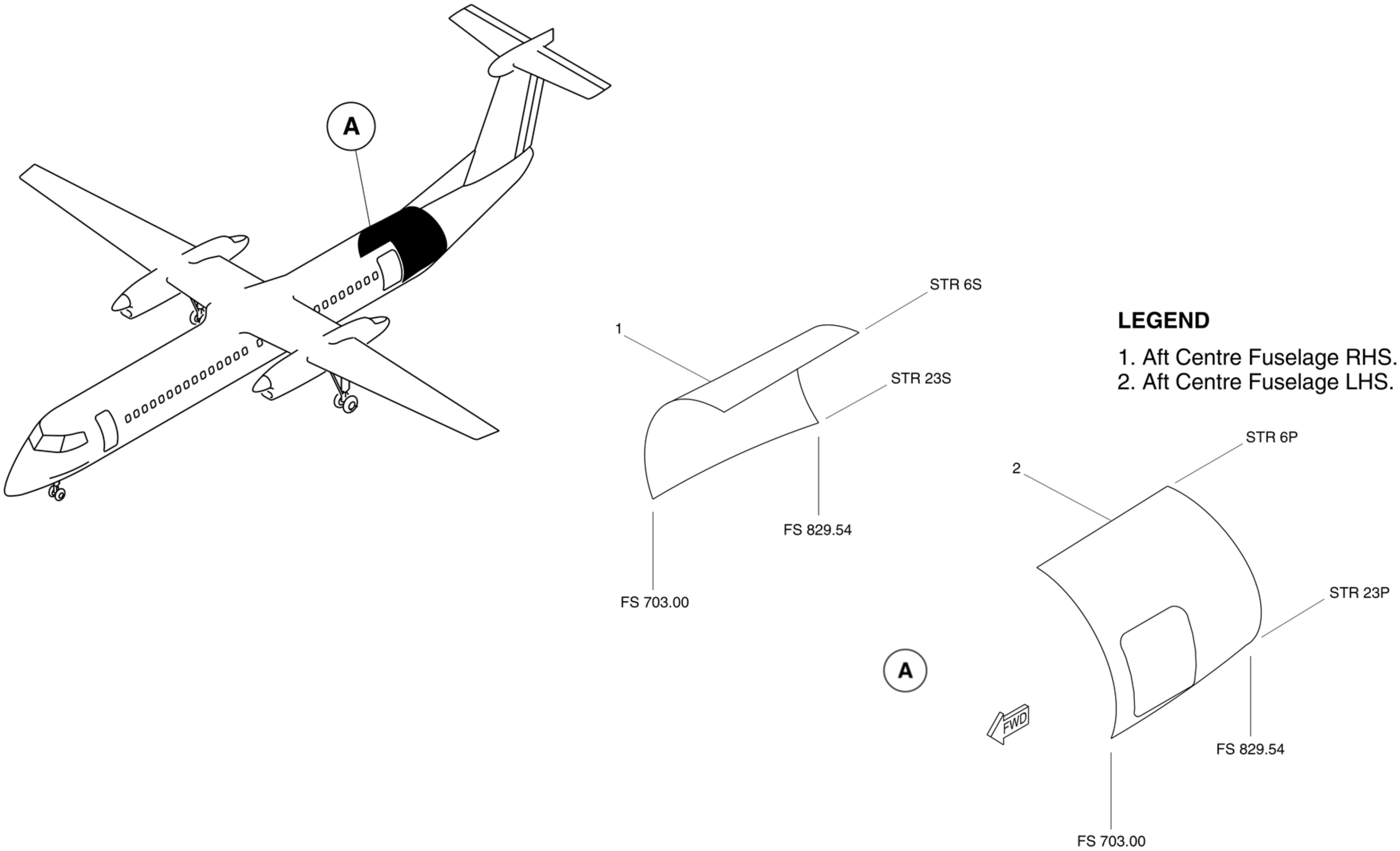


Figure 25-57 Aft Fuselage Insulation